TimeMorph: Dual-view LLM-based Few-Shot Time Series Classification

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Abstract

Few-shot time series classification (TSC) is challenging due to scarce labeled sequences. Reliable predictions require capturing both temporal dynamics and waveform morphology, reflecting how humans interpret sequential data by tracking numeric trends and recognizing overall shape patterns. Pretrained LLMs offer few-shot generative interfaces, but directly applying them is challenging: continuous time-series features are not naturally aligned with language embeddings, and predictions must be restricted to a fixed set of target classes. Therefore, we propose TimeMorph, a dual-view framework that extracts temporal dynamics via a temporal branch and waveform morphology via a morphology branch. To bridge time-series features and language embeddings, we employ stage-wise pretraining to project the fused tokens into a pretrained causal LLM. Extensive few-shot experiments on the UCR benchmark and additional external datasets show that TimeMorph consistently outperforms strong task-specific, Transformer-style, and LLM-style baselines. Further analyses validate the contributions of dual-view modeling, stage-wise pretraining, and class-token scoring.

1 Introduction

Time series classification (TSC) is challenging when labeled sequences are scarce in healthcare monitoring, industrial diagnosis, sensor analytics, and related domains [Ismail Fawaz et al., 2019]. In many practical scenarios, collecting labels requires domain expertise, new categories may appear with only a few examples, and class distributions can be highly imbalanced. Reliable few-shot TSC therefore requires more than fitting a classifier on a small support set. A successful model should capture transferable evidence from both temporal dynamics and waveform morphology, reflecting how humans naturally interpret sequences by tracking numerical evolution while recognizing shape-level patterns.

Discriminative evidence in time series is inherently heterogeneous. Some classes are distinguished by chronological dynamics, such as local trends, amplitudes, phase shifts, and long-range evolution, while others depend more strongly on waveform morphology, such as recurring peaks, valleys, cycles, motifs, and discriminative subsequences [Tang et al., 2020, Ye and Keogh, 2009b, Dempster et al., 2020, Ismail Fawaz et al., 2020, Wang and Oates, 2015a]. This dual-view observation suggests that effective few-shot TSC models should extract and combine both temporal and morphological features. However, many existing methods emphasize one dominant form of evidence. CNNs and Transformer-based models can achieve strong performance when abundant labeled training data are available, but they may overfit to unstable cues and generalize poorly under few-shot supervision [Ismail Fawaz et al., 2020, Dempster et al., 2020, Yang et al., 2024]. Prototype-based methods are more data-efficient, yet they still do not fully resolve the need to jointly preserve temporal and morphological evidence [Snell et al., 2017, Zhang et al., 2020, Tang et al., 2020]. In particular, models that mainly operate on ordered numerical trends can overlook the intuitive visual structure of

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the waveform, while purely shape-oriented representations may miss chronological dependencies needed for fine-grained discrimination.

Recently, pretrained large language models (LLMs) have become appealing for few-shot learning because they provide reusable pretrained computation, conditional-generation objectives, and token-level scoring interfaces [Brown et al., 2020, Raffel et al., 2020]. Time-series-specific adaptation work has explored numerical serialization, reprogramming, embedding alignment, and semantic anchors, but many of these methods focus on forecasting, representation learning, or open-ended generation rather than closed-set classification [Gruver et al., 2023, Zhou et al., 2023, Jin et al., 2024, Sun et al., 2023, Chang et al., 2025, Zhang et al., 2024, Jiang et al., 2024, Langer et al., 2025, Tan et al., 2024, Zheng et al., 2024]. Few-shot time-series classification with LLMs has only recently started to emerge [Chen et al., 2025]. These properties make LLMs a promising backbone for few-shot TSC, where supervised labels are limited but transferable computation and flexible decision interfaces are valuable. However, directly applying LLMs to time series remains challenging. Time-series observations are continuous, domain-specific, and structurally different from the discrete token sequences on which language models are pretrained. Moreover, few-shot TSC requires reliable closed-set discrimination over dataset-specific labels.

To make pretrained LLMs effective for dual-view few-shot TSC, three core challenges must be addressed. First, the representation should preserve both chronological dynamics and morphology-oriented cues, since either view may contain the decisive class signal. Second, fused continuous time-series features must be aligned with the pretrained LLM input space before downstream adaptation. Third, prediction should be restricted to valid class candidates instead of relying on unconstrained text generation, which may produce invalid labels and introduce tokenization or label-length bias [Raffel et al., 2020, Hokamp and Liu, 2017, Koo et al., 2024]. Without such an interface, the language model may become only a loosely connected generative component rather than a reliable few-shot classifier.

We propose **TimeMorph**, a dual-view time-series-to-language framework for few-shot TSC. For each sequence, TimeMorph constructs two complementary views: a temporal branch that models ordered local relations and long-range dynamics, and a morphology branch that transforms the waveform into a two-dimensional morphology map so that shape-level structure can be extracted by a frozen vision encoder. The two views are fused into continuous tokens and projected into a pretrained causal LLM through stage-wise pretraining, reducing the mismatch between time-series representations and the language-model embedding space. For downstream classification, TimeMorph replaces unconstrained decoding with supervised class-token scoring over dataset-specific class tokens, turning the generative interface into a discriminative closed-set classifier. When class names carry meaningful semantics, label verbalizers further initialize and regularize the corresponding class-token embeddings without requiring multi-token phrase decoding.

Our contributions are summarized as follows. First, we formulate LLM-based few-shot TSC as an interface-design problem that jointly considers complementary time-series evidence, continuous-to-language alignment, and closed-set prediction. Second, we introduce a dual-view time-series-to-language encoder that preserves chronological and morphology-oriented evidence before transferring continuous tokens into a pretrained causal LLM through stage-wise pretraining. Third, we propose supervised class-token scoring over valid labels, and extend it with label-verbalizer initialization and regularization when semantic label names are available. Across 128 UCR datasets, TimeMorph achieves the best average accuracy and average rank under 1-, 2-, 5-, and 10-shot settings, while additional external experiments and ablation studies validate the roles of dual-view modeling, stage-wise pretraining, and closed-set class-token scoring.

2 Related Work

LLMs and foundation models for time series. Recent surveys show that foundation-model research for time series has expanded from adapting pretrained language models to building native temporal foundation models Zhang et al. [2024], Jiang et al. [2024], Ye et al. [2024], Shi et al. [2025]. Representative adaptation-based methods include TEST Sun et al. [2023], which aligns time-series embeddings with the LLM embedding space through instance-, feature-, and text-prototype-aware contrastive learning; Time-LLM Jin et al. [2024], which reprograms time-series patches into a frozen LLM via text prototypes; and LLM4TS Chang et al. [2025], which uses a two-stage alignment-and-task fine-tuning strategy. OpenTSLM Langer et al. [2025] moves toward time-series language

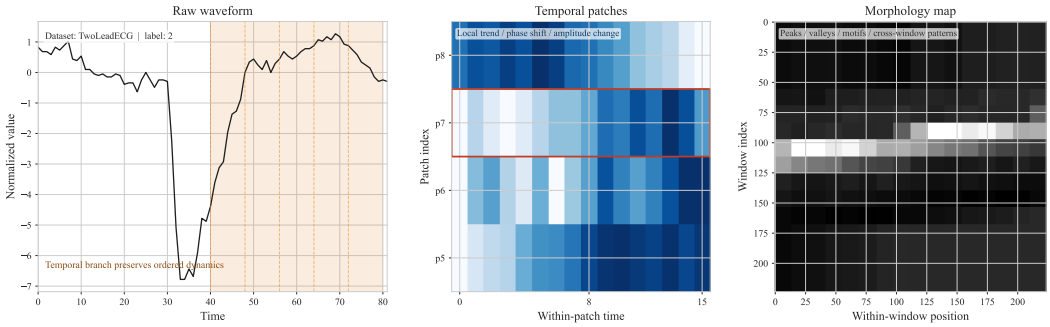

Figure 1: Illustrative example of the dual-view design. The same UCR sequence is represented as ordered temporal patches and as a windowed morphology map. The two views expose complementary evidence before TimeMorph transfers the continuous tokens to the language-model interface.

models that treat time series as a native modality through soft-prompt or cross-attention interfaces. At the same time, recent ablation studies caution that LLM components do not automatically improve time-series performance when the modality interface is poorly designed Tan et al. [2024], Zheng et al. [2024]. The closest few-shot classification line is LLMFew Chen et al. [2025], which introduces a patch-wise temporal convolution encoder and LoRA-tuned decoder for few-shot multivariate TSC. In contrast, our work focuses on dual-view temporal evidence, explicit time-series-to-language alignment, and class-token scoring rather than an external classifier head.

Temporal and morphological evidence in TSC. Classical and deep TSC methods support the view that discriminative evidence can appear at multiple structural levels. Shapelet methods identify local subsequences with high class-discriminative power Ye and Keogh [2009a], Tang et al. [2020], convolutional methods such as InceptionTime and ROCKET exploit local receptive fields and kernel responses at multiple scales Ismail Fawaz et al. [2020], Dempster et al. [2020], and patch-based Transformers preserve local subseries information while enabling longer-context attention Nie et al. [2022]. These methods primarily operate in the chronological-dependency space. In parallel, time-series imaging methods transform one-dimensional signals into two-dimensional structures such as Gramian angular fields and Markov transition fields, enabling vision models to capture global shape and transition patterns Wang and Oates [2015b]. Recent work further shows that visualization can improve LLM reasoning over time series and that frozen vision transformers applied to imaged time series can produce strong classification representations Liu et al. [2025], Roschmann et al. [2025]. These two lines motivate our dual-space hypothesis: few-shot TSC labels may depend on both chronological dependencies and morphological structures.

Generative classification and semantic-prior label spaces. Casting classification as generation is widely used in text-to-text learning, where labels are generated as target tokens under the same sequence objective as other tasks Raffel et al. [2020]. However, unconstrained decoding is mismatched with closed-set classification because many generated strings are irrelevant to the valid label set. Prior work on constrained decoding motivates restricting generation to valid lexical or structural outputs Hokamp and Liu [2017], Koo et al. [2024]. TimeMorph adopts the same closed-set principle but implements the final decision as supervised class-token scoring: the model learns dataset-specific class-token rows and compares their next-token logits rather than performing unconstrained generation. For datasets with meaningful label names, we further use label verbalizers as semantic priors over the same class-token rows. This keeps the decision aligned with the autoregressive interface while avoiding the length bias, shared-prefix artifacts, and language-prior sensitivity of direct phrase-likelihood classification.

3 Method

We propose **TimeMorph**, a dual-view time-series-to-language framework that adapts a pretrained causal language model to few-shot time series classification. The key design is to avoid translating a time series into raw text. Instead, TimeMorph transforms each sequence into a small set of continuous

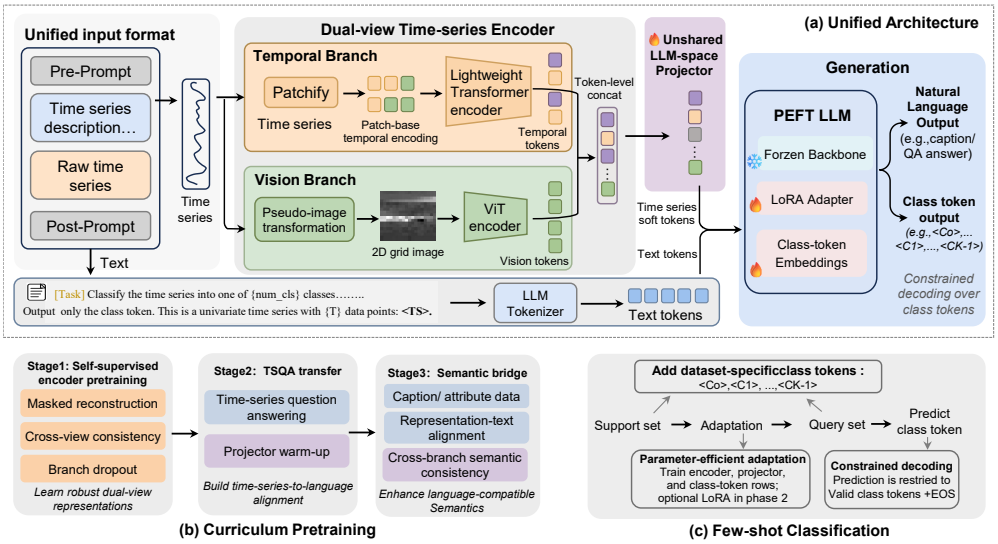

Figure 2: Overview of TimeMorph. The temporal branch extracts ordered patch-level dynamics from the raw waveform, while the morphology branch converts the same waveform into a windowed morphology map and extracts visual patch tokens with a frozen vision backbone. The two token streams are fused, projected into the input embedding space of a pretrained causal LLM, and used for closed-set prediction by class-token scoring over dataset-specific class tokens. Stage I learns dual-view time-series representations, Stage II aligns continuous time-series tokens with the language-model interface, and the optional semantic stage uses label verbalizers when meaningful class names are available.

tokens, places these tokens in the language-model embedding stream, and performs classification with class-token scoring over dataset-specific labels. This gives the language model a closed-set decision interface while preserving the temporal and morphological evidence needed for time series classification.

3.1 Problem Setup and Architecture

Let D be a target classification dataset with label set $\mathcal{Y}_D = \{0, \ldots, C-1\}$. In the S-shot setting, the support set $\mathcal{S}_D$ contains at most S labeled sequences per class. Given a query sequence $x \in \mathbb{R}^T$, the goal is to predict its label $y \in \mathcal{Y}_D$ after adapting to $\mathcal{S}_D$.

TimeMorph separates this problem into three modules, as shown in Figure 2. First, a dual-view encoder G_θ maps the input sequence into time-series tokens by combining a chronological temporal view and a morphology-oriented view. Second, a projection module P_ψ maps these tokens to the input dimension of a pretrained causal language model. Third, a closed-set class-token scoring interface converts the language model's next-token logits into a classifier over $\mathcal{Y}_D$. The language-model backbone provides the shared conditional computation, while dataset-specific class-token rows and lightweight adapters supply the final decision rule for each target dataset.

3.2 Dual-View Time-Series Encoder

Time series labels can depend on different types of evidence. Some classes are separated by ordered temporal dynamics, such as local trends, phase changes, or long-range evolution. Others are better characterized by waveform morphology, such as peaks, valleys, motifs, cycles, or discriminative subsequences. TimeMorph therefore constructs two complementary token sequences before interacting with the language model.

Temporal view. The temporal branch preserves the chronological structure of the raw sequence. Let x^{pad} be the sequence after right replicate padding. Given patch length ℓ and stride s, we extract

overlapping temporal patches

$$R_i = x^{\text{pad}}_{(i-1)s+1:(i-1)s+\ell}, \qquad i = 1, \ldots, N_t, \qquad N_t = \left\lfloor \frac{T + p_t - \ell}{s} \right\rfloor + 1, \tag{1}$$

where p_t is the padding length needed to form complete patches. Each patch is linearly embedded, combined with a learnable positional embedding, and passed through a temporal Transformer:

$$Z^t(x) = G_t\left([W_t R_1 + \pi_1; \ldots; W_t R_{N_t} + \pi_{N_t}]\right). \tag{2}$$

The output $Z^t(x)$ is a sequence of temporal tokens that encode ordered patch-level dynamics of the original waveform.

Morphology view. The morphology branch exposes shape-level structure by converting the same one-dimensional sequence into a two-dimensional windowed map. We first normalize x with robust per-sequence statistics:

$$\bar{x}_t = \frac{x_t - \text{median}(x)}{\text{IQR}(x) + \epsilon}. \tag{3}$$

The normalized sequence is then unfolded into overlapping local windows. Let $q = \max(1, \lfloor \sqrt{T} \rfloor)$ be the window length and $\delta = \max(1, \lfloor \rho q \rfloor)$ be the stride. We construct the morphology map by normalizing the unfolded matrix, resizing it to the input resolution of the vision backbone, and replicating it across RGB channels:

$$I(x) = \text{RGB}\left(\text{Resize}\left(\Gamma_{\text{img}}\left(\text{Unfold}_{q,\delta}(\bar{x})\right)\right)\right), \tag{4}$$

where $\Gamma_{\text{img}}(\cdot)$ denotes per-map normalization and contrast adjustment. Each row of $I(x)$ corresponds to a local waveform window, so the map preserves local subsequence shapes while making cross-window morphology visible to a vision encoder. A frozen vision backbone G_m extracts morphology tokens:

$$Z^m(x) = G_m(I(x)). \tag{5}$$

We discard the vision class token and use only the patch tokens. The vision backbone remains frozen; trainable parameters are introduced in the time-series encoder, projection modules, class-token rows, and language-side adapters.

Fusion. The temporal and morphology tokens are projected to a common dimension and concatenated:

$$\tilde{Z}^t(x) = Q_t(Z^t(x)), \qquad \tilde{Z}^m(x) = Q_m(Z^m(x)), \qquad Z_x = \left[\tilde{Z}^t(x); \tilde{Z}^m(x)\right]. \tag{6}$$

The fused sequence $Z_x \in \mathbb{R}^{N_x \times d_e}$ is the time-series token sequence used by the language-model interface.

3.3 Time-Series-to-Language Interface

The dual-view encoder produces continuous time-series tokens, whereas a causal language model expects embeddings drawn from its token embedding space. TimeMorph bridges this mismatch by reprogramming Z_x into the language-model embedding stream rather than serializing the numeric sequence as text.

Let P_ψ be the projector from encoder dimension d_e to the language-model hidden dimension, $\tau(\cdot)$ be the tokenizer, and E_{lm} be the language-model input embedding matrix. For a prompt $p = (p^{\text{pre}}, p^{\text{post}})$, the conditioned input embeddings are

$$H(x, p) = \left[E_{\text{lm}}(\tau(p^{\text{pre}})); P_\psi(Z_x); E_{\text{lm}}(\tau(p^{\text{post}}))\right]. \tag{7}$$

The projected time-series tokens therefore act as continuous prefix tokens inside the same embedding stream as text. Optional textual descriptions, when available, are appended to the prompt and encoded by the same tokenizer.

During stage-wise pretraining, the model is optimized on time-series-conditioned generation examples. For an answer string a with token sequence $u_{1:M} = \tau(a)$, we apply the language-modeling loss only to the answer tokens:

$$\mathcal{L}_{\text{LM}}(a \mid x, p) = -\sum_{m=1}^{M} \log p_\Theta\left(u_m \mid H(x, p), E_{\text{lm}}(u_{<m})\right). \tag{8}$$

This stage calibrates the continuous time-series tokens to the language-model input space. Downstream few-shot classification uses the same conditioned representation, but replaces free-form decoding with the closed-set rule below.

3.4 Closed-Set Class-Token Prediction

Few-shot TSC requires choosing among a fixed set of valid labels. Directly decoding natural-language label strings is undesirable because generated text may leave the label set and because phrase likelihoods are affected by tokenization, label length, and shared prefixes. TimeMorph instead assigns each target dataset a compact set of single-token class identifiers:

$$\mathcal{C}_D = \{\langle c_0 \rangle, \ldots, \langle c_{C-1} \rangle\}, \qquad \mathcal{V}_D = \{v_0, \ldots, v_{C-1}\}, \tag{9}$$

where v_c is the vocabulary index of class token $\langle c_c \rangle$.

Let p_D denote the classification prompt for dataset D. After the language model conditions on $H(x, p_D)$, let $h_\Theta(x, p_D)$ be the final hidden state at the last prompt position. With output embedding matrix W_{out}, the score for class c is the next-token logit assigned to its class token:

$$s_c(x) = \left[W_{\text{out}} h_\Theta(x, p_D)\right]_{v_c}. \tag{10}$$

Prediction is restricted to the valid class-token set:

$$\hat{y} = \arg \max_{c \in \{0,\ldots,C-1\}} s_c(x). \tag{11}$$

The supervised objective is the cross entropy over only the valid class tokens:

$$\mathcal{L}_{\text{ct}} = -\log \frac{\exp s_y(x)}{\sum_{c=0}^{C-1} \exp s_c(x)}. \tag{12}$$

Thus, the autoregressive vocabulary head is retained, but the decision is made by one forward pass and one restricted comparison over C valid class tokens. Within the vocabulary parameters, only the input embeddings and output rows of the new class tokens are trainable; all other vocabulary rows remain fixed.

Semantic initialization for named labels. For anonymous-label benchmarks, class tokens are learned as dataset-specific identifiers. When class names are meaningful, TimeMorph initializes each class-token row from the average embedding of template-consistent label verbalizers and regularizes its input and output rows toward this semantic prototype during adaptation. The resulting loss is

$$\mathcal{L}_{\text{adapt}} = \mathcal{L}_{\text{ct}} + \lambda_{\text{sem}} \mathcal{L}_{\text{sem}}. \tag{13}$$

For datasets without meaningful label names, we set $\lambda_{\text{sem}} = 0$; prototype construction and row regularization details are in Appendix B.

3.5 Stage-wise Pretraining and Few-Shot Adaptation

TimeMorph is trained in stages. Stage I learns time-series representations without class labels by combining masked temporal patch reconstruction with an invariance–variance–covariance regularizer [Bardes et al., 2022] over the temporal, morphology, and fused streams:

$$\mathcal{L}_{\text{I}} = \lambda_{\text{rec}} \mathcal{L}_{\text{rec}} + \lambda_t \mathcal{L}^t_{\text{ivc}} + \lambda_m \mathcal{L}^m_{\text{ivc}} + \lambda_f \mathcal{L}^f_{\text{ivc}}. \tag{14}$$

We also use branch dropout to prevent the fused representation from collapsing onto a single view. Stage II attaches P_ψ and the frozen causal language model, then calibrates the interface with an answer-token language-modeling loss on time-series question-answering examples:

$$\mathcal{L}_{\text{II}} = \mathbb{E}_{(x,p,a)\sim\mathcal{D}_{\text{TSQA}}} \left[\mathcal{L}_{\text{LM}}(a \mid x, p)\right]. \tag{15}$$

We first update the projector with the time-series encoder fixed, and then jointly optimize the encoder and projector; the language-model backbone remains frozen. An optional semantic stage further aligns outputs with label-related language when meaningful class names or descriptions are available, but it is not used for the anonymous-label UCR benchmark.

For each downstream dataset, adaptation first updates the dual-view encoder, the projection module, and the dataset-specific class-token rows. A second phase additionally updates LoRA adapters while keeping the pretrained language-model weights frozen. At inference time, TimeMorph performs one forward pass per query and ranks only the valid class tokens using Eq. (10). Implementation details and stage-wise data sources are provided in Appendix B.

4 Experiments

4.1 Experimental Setup

Protocol. We evaluate on all 128 datasets in the UCR Archive 2018 with a fixed full-label-space few-shot protocol. For each dataset D and support size $S \in \{1, 2, 5, 10\}$, all classes are included; we sample up to S labeled examples per class from the official training split and evaluate on all examples in the official test split. All methods use the same sampled support sets within each of five independent runs.

Pretraining data and default initialization. The main benchmark uses the Stage II checkpoint: Stage I is trained on raw TSQA-train series, raw M4-train series, and unlabeled official TRAIN-split sequences from a fixed 98-dataset UCR pool; Stage II is trained on the TSQA training split. Stage III is not used for Table 1. No pretraining stage uses any official UCR TEST split or UCR class label. For UCR datasets included in the 98-dataset pool, TimeMorph may have seen unlabeled official TRAIN sequences before downstream adaptation; after this pretraining, each target run uses only the sampled support labels for supervised adaptation.

Baselines and metrics. We compare against CNN/prototype baselines (RESNET, TAPNET, COSCO-RESNET), Transformer-style or decomposition-based baselines (PATCHTST, DLINEAR, TIMESNET, AUTOFORMER, CROSSFORMER, FEDFORMER, INFORMER), and the LLM-style GPT4TS baseline. Accuracy is averaged over five independent support-set draws and then macro-averaged over datasets. `Avg` averages the four shot settings, and `AvgRank` averages dataset-level ranks across shots; lower rank is better. Appendix tables report the sample standard deviation over these runs; we do not assume normality or perform formal hypothesis testing.

Fairness and training details. All methods share the same official TRAIN support pool, TEST query split, and sampled support indices. We do not add a validation split by default; each method follows its runner's checkpoint-selection rule, and some baselines select checkpoints using support-set performance. Appendix B.8 lists the training recipes and hyperparameters.

Implementation details. TimeMorph uses `meta-llama/Llama-3.2-1B`, `facebook/dinov2-base`, temporal patch length 16, stride 8, hidden size 128, and the Stage II checkpoint. Downstream adaptation uses 5 warm-up epochs, 55 joint-training epochs, and LoRA with rank 16 and alpha 32. Additional details are in Appendix B.

4.2 Main Results

Benchmark-level comparison. Table 1 shows that TimeMorph obtains the highest mean accuracy under all four support sizes: 48.31, 55.55, 64.14, and 69.16 for 1-, 2-, 5-, and 10-shot. The strongest non-TimeMorph baseline is COSCO-RESNET, with 47.88, 54.95, 63.55, and 67.18. Thus the improvement over the strongest specialized CNN/prototype baseline is modest in accuracy, especially in the lowest-shot regimes, but it is consistent across support sizes and is also reflected in the best `AvgRank` (4.24 vs. 4.70 for COSCO-RESNET). The gap is more substantial against Transformer-style and LLM-style baselines: TimeMorph improves `Avg` by 4.53 points over INFORMER and 18.36 points over GPT4TS. These results suggest that the proposed interface is most clearly beneficial relative to generic sequence and LLM-style baselines, while remaining competitive with strong specialized CNN/prototype methods.

Dataset-level robustness. Table 2 reports Win/Tie/Loss counts against representative baselines. TimeMorph wins substantially more often than it loses against PATCHTST, TIMESNET, and GPT4TS. Against the strongest CNN/prototype baselines, the picture is closer: TimeMorph is nearly balanced with COSCO-RESNET and RESNET at several shot levels, which is consistent with the modest average-accuracy margins in Table 1.

Additional external datasets. We also evaluate MIT-BIH and CinC2017 under full-label-space 10-, 20-, and 30-shot protocols, using 10 runs for each external setting. Table 3 reports the average over these shot settings, with the full table in Appendix B.5. TimeMorph obtains the best average

Table 1: Few-shot classification accuracy (%) on 128 UCR datasets, grouped by model family. We report the mean over five runs. `Avg` and `AvgRank` average the four shot-specific mean accuracies and average ranks, respectively. Best is bold and second-best is underlined.

Family	Model	1-shot	2-shot	5-shot	10-shot	Avg	AvgRank
CNN/prototype	ResNet	47.79	53.98	61.27	66.24	57.32	4.89
	TapNet	45.40	50.79	59.47	64.93	55.15	5.45
	COSCO-ResNet	<u>47.88</u>	<u>54.95</u>	<u>63.55</u>	<u>67.18</u>	<u>58.39</u>	<u>4.70</u>
Transformer/TS	DLinear	44.21	48.01	52.30	55.43	49.99	7.24
	PatchTST	42.39	48.43	56.94	62.72	52.62	6.54
	TimesNet	45.16	49.95	57.86	62.79	53.94	5.90
	Autoformer	36.78	39.45	43.20	46.04	41.37	9.47
	Crossformer	44.31	49.45	57.20	61.94	53.23	6.04
	FEDformer	37.71	41.71	47.30	51.50	44.56	8.68
	Informer	46.42	51.14	58.35	63.13	54.76	5.59
LLM/foundation	GPT4TS	35.61	38.30	42.94	46.88	40.93	9.26
Ours	TimeMorph	**48.31**	**55.55**	**64.14**	**69.16**	**59.29**	**4.24**

Table 2: Win/Tie/Loss of TimeMorph against selected baselines on 128 UCR datasets.

Baseline	1-shot	2-shot	5-shot	10-shot
COSCO-ResNet	68 / 0 / 60	68 / 1 / 59	58 / 0 / 70	64 / 0 / 64
PatchTST	97 / 0 / 31	97 / 0 / 31	99 / 0 / 29	94 / 2 / 32
TimesNet	88 / 0 / 40	91 / 1 / 36	86 / 0 / 42	93 / 1 / 34
GPT4TS	106 / 2 / 20	110 / 0 / 18	114 / 1 / 13	121 / 1 / 6
ResNet	64 / 1 / 63	66 / 1 / 61	67 / 0 / 61	69 / 0 / 59
TapNet	75 / 0 / 53	75 / 3 / 50	76 / 1 / 51	72 / 3 / 53

accuracy on both datasets. We therefore treat these experiments as evidence of transfer beyond UCR, not as a claim of uniform dominance across all time-series domains.

4.3 Computational Efficiency and Adaptation Cost

Because TimeMorph reuses a pretrained causal LLM, total parameters and adapted parameters must be reported separately. Table 4 shows the main trade-off: TimeMorph updates only 12.3M parameters per target dataset, but the frozen LLM backbone dominates memory and inference cost. TimeMorph is therefore most appropriate when transfer accuracy and reusable adaptation matter more than compact deployment. The complete efficiency table, including throughput and additional baselines, is in Appendix B.7.

4.4 Ablation Studies

Table 5 summarizes the main top-level UCR ablations on all 128 datasets. Each ablation follows the same five-run full-label-space few-shot protocol and modifies only the component under study. Focused compact views with the corresponding drops are provided in Tables 19–21. Because UCR labels are anonymous, the decision ablation tests closed-set class-token scoring rather than semantic label priors. The no-LLM ablation is reported separately in Table 22, with the full per-dataset comparison in Appendix Table 23.

UCR ablations. Removing curriculum pretraining lowers `Avg` by 3.35 points, indicating that Stage I+II initialization helps when downstream labels are too scarce to learn both the representation and LLM interface from scratch. Single-view variants also underperform the full model: removing the morphology branch drops `Avg` by 5.08 points, while removing the temporal branch drops it by 4.20 points, supporting the complementarity of chronological and morphological evidence. Replacing

Table 3: Average few-shot accuracy (%) on two external datasets, computed over 10-, 20-, and 30-shot settings with 10 runs per setting. Best is bold and second-best is underlined.

Model	MIT-BIH	CinC2017	Avg.
TimeMorph	**61.04**	**34.63**	**47.84**
ResNet	50.23	<u>33.63</u>	<u>41.93</u>
TapNet	38.73	31.29	35.01
COSCO-ResNet	36.19	32.48	34.34
PatchTST	50.46	29.21	39.84
GPT4TS	43.42	29.19	36.31
TimesNet	53.79	28.36	41.08
DLinear	<u>57.29</u>	24.87	41.08
Autoformer	41.64	27.45	34.55
Crossformer	56.76	25.62	41.19
FEDformer	56.93	25.05	40.99
Informer	53.25	28.39	40.82

Table 4: Compact efficiency comparison under the 10-shot UCR protocol. Runtime and memory are measured on the representative UCR runtime subset. Total parameters count the deployed model; updated parameters count downstream-adapted parameters only.

Model	Total params	Updated params	Peak GB	Infer ms/ex.
TimeMorph	1.28B	12.3M	13.75	156.85
COSCO-ResNet	662.1K	662.1K	0.58	0.25
ResNet	629.1K	629.1K	0.33	0.26
PatchTST	665.4K	598.3K	1.54	0.75
GPT4TS	82.2M	1.1M	0.56	0.86

class-token scoring with unconstrained generative adaptation lowers `Avg` by 4.15 points, suggesting that the closed-set interface is more stable for few-shot classification.

Semantic priors and LLM backbone. For external datasets with meaningful label names, initializing and regularizing class-token rows from label verbalizers improves the averaged result by 2.39 points on the compact MIT-BIH/CinC2017 summary, although the gain is not uniform at every shot; the full semantic-prior table is in Appendix B.6.

The no-LLM variant in Appendix Table 22 replaces the language pathway with a lightweight linear classifier on the same dual-view features and reduces `Avg` by 7.83 points. This suggests that the language pathway contributes beyond the dual-view encoder alone, while not isolating whether the gain comes from pretrained knowledge, additional capacity, or token scoring.

5 Limitations

TimeMorph improves average accuracy and average rank, but its gains over the strongest CNN/prototype baselines are modest in extremely low-shot settings. The main empirical message is therefore not that LLM-based TSC now universally dominates specialized CNN-style few-shot classifiers, but that a carefully designed time-series-to-language interface can substantially outperform previous Transformer-style and LLM-style baselines while remaining competitive with strong task-specific methods. TimeMorph also has a larger inference footprint than compact CNN baselines because it reuses a pretrained causal LLM, even though only a small fraction of parameters is updated during downstream adaptation. In our efficiency probe, this makes TimeMorph two to three orders of magnitude slower at inference than compact CNN/prototype baselines. The current UCR pretraining setting also uses unlabeled official TRAIN sequences from a fixed subset of UCR datasets, so it does not establish strict leave-dataset-out transfer for every UCR target dataset. The external semantic-prior results improve the averaged performance on the two reported semantic-label datasets, but individual shot-level effects still vary, so label semantics should be viewed as a prior over the class-token rows rather than as a replacement for supervised adaptation. TimeMorph may benefit applications where

Table 5: Summary of top-level ablations on the 128 UCR datasets under the same five-run full-label-space few-shot protocol as the main results. Each value is mean accuracy (%) over five runs. Best is bold, and the best ablated variant in each column is underlined. Since UCR labels are anonymous, the label-space ablation isolates closed-set class-token scoring rather than semantic label priors.

Variant	1-shot	2-shot	5-shot	10-shot	Avg
TimeMorph	**48.31**	**55.55**	**64.14**	**69.16**	**59.29**
w/o Curriculum Pretraining	44.70	<u>51.81</u>	60.65	<u>66.60</u>	<u>55.94</u>
Temporal only	43.51	50.55	58.40	64.39	54.21
Visual only	<u>45.88</u>	50.02	<u>60.86</u>	63.58	55.09
w/o Class-Token Label-Space Scoring	44.46	50.44	60.08	65.59	55.14

labeled time-series data are scarce, such as physiological monitoring or industrial inspection, but the present study is not a deployment-ready clinical or safety-critical system. Any deployment in such domains would require domain-specific validation, privacy review, and failure analysis beyond the scope of this paper.

References

A. Bardes, J. Ponce, and Y. LeCun. VICReg: Variance-invariance-covariance regularization for self-supervised learning. In *International Conference on Learning Representations*, 2022.

T. B. Brown, B. Mann, B. Ryder, M. Subbiah, J. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, et al. Language models are few-shot learners. In *Advances in Neural Information Processing Systems*, volume 33, pages 1877–1901, 2020.

C. Chang, W.-Y. Wang, W.-C. Peng, and T.-F. Chen. Llm4ts: Aligning pre-trained llms as data-efficient time-series forecasters. *ACM Transactions on Intelligent Systems and Technology*, 16(3): 1–20, 2025.

Y. Chen, Z. Li, C. Yang, X. Wang, and G. Xu. Large language models are few-shot multivariate time series classifiers. *Data Mining and Knowledge Discovery*, 39(5):66, 2025.

A. Dempster, F. Petitjean, and G. I. Webb. ROCKET: exceptionally fast and accurate time series classification using random convolutional kernels. *Data Mining and Knowledge Discovery*, 34(5): 1454–1495, 2020.

N. Gruver, M. Finzi, S. Qiu, and A. G. Wilson. Large language models are zero-shot time series forecasters. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

C. Hokamp and Q. Liu. Lexically constrained decoding for sequence generation using grid beam search. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1535–1546, 2017.

H. Ismail Fawaz, G. Forestier, J. Weber, L. Idoumghar, and P.-A. Muller. Deep learning for time series classification: a review. *Data mining and knowledge discovery*, 33(4):917–963, 2019.

H. Ismail Fawaz, B. Lucas, G. Forestier, C. Pelletier, D. F. Schmidt, J. Weber, G. I. Webb, L. Idoumghar, P.-A. Muller, and F. Petitjean. Inceptiontime: Finding alexnet for time series classification. *Data Mining and Knowledge Discovery*, 34(6):1936–1962, 2020.

Y. Jiang, Z. Pan, X. Zhang, S. Garg, A. Schneider, Y. Nevmyvaka, and D. Song. Empowering time series analysis with large language models: A survey. *arXiv preprint arXiv:2402.03182*, 2024.

M. Jin, S. Wang, L. Ma, Z. Chu, J. Zhang, X. Shi, P.-Y. Chen, Y. Liang, Y.-f. Li, S. Pan, et al. Time-llm: Time series forecasting by reprogramming large language models. In *International Conference on Learning Representations*, 2024.

T. Koo, F. Liu, and L. He. Automata-based constraints for language model decoding. *arXiv preprint arXiv:2407.08103*, 2024.

P. Langer, T. Kaar, M. Rosenblattl, M. A. Xu, W. Chow, M. Maritsch, R. Jakob, N. Wang, J. Liu, A. Verma, et al. Opentslm: Time-series language models for reasoning over multivariate medical text-and time-series data. *arXiv preprint arXiv:2510.02410*, 2025.

H. Liu, C. Liu, and B. A. Prakash. A picture is worth a thousand numbers: Enabling llms reason about time series via visualization. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 7486–7518, 2025.

Y. Nie, N. H. Nguyen, P. Sinthong, and J. Kalagnanam. A time series is worth 64 words: Long-term forecasting with transformers. In *The Eleventh International Conference on Learning Representations*, 2022.

C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li, and P. J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of machine learning research*, 21(140):1–67, 2020.

S. Roschmann, Q. Bouniot, V. Feofanov, I. Redko, and Z. Akata. Time series representations for classification lie hidden in pretrained vision transformers. *arXiv preprint arXiv:2506.08641*, 2025.

F. Shi, X. Yin, K. Wang, W. Tu, Q. Sun, and H. Ning. Large language models for time series analysis: Techniques, applications, and challenges. *arXiv preprint arXiv:2506.11040*, 2025.

J. Snell, K. Swersky, and R. Zemel. Prototypical networks for few-shot learning. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

C. Sun, H. Li, Y. Li, and S. Hong. Test: Text prototype aligned embedding to activate llm's ability for time series. *arXiv preprint arXiv:2308.08241*, 2023.

M. Tan, M. A. Merrill, V. Gupta, T. Althoff, and T. Hartvigsen. Are language models actually useful for time series forecasting? *Advances in Neural Information Processing Systems*, 37:60162–60191, 2024.

W. Tang, L. Liu, and G. Long. Interpretable time-series classification on few-shot samples. In *2020 International joint conference on neural networks (IJCNN)*, pages 1–8. IEEE, 2020.

Z. Wang and T. Oates. Encoding time series as images for visual inspection and classification using tiled convolutional neural networks. In *Workshops at the Twenty-Ninth AAAI Conference on Artificial Intelligence*, 2015a.

Z. Wang and T. Oates. Imaging time-series to improve classification and imputation. In *Proceedings of the Twenty-Fourth International Joint Conference on Artificial Intelligence*, pages 3939–3945, 2015b.

C. Yang, X. Wang, L. Yao, G. Long, and G. Xu. Dyformer: A dynamic transformer-based architecture for multivariate time series classification. *Information Sciences*, 656:119881, 2024.

J. Ye, W. Zhang, K. Yi, Y. Yu, Z. Li, J. Li, and F. Tsung. A survey of time series foundation models: Generalizing time series representation with large language model. *arXiv e-prints*, pages arXiv–2405, 2024.

L. Ye and E. Keogh. Time series shapelets: a new primitive for data mining. In *Proceedings of the 15th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 947–956, 2009a.

L. Ye and E. Keogh. Time series shapelets: A new primitive for data mining. In *Proceedings of the 15th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 947–956, 2009b.

X. Zhang, Y. Gao, J. Lin, and C.-T. Lu. Tapnet: Multivariate time series classification with attentional prototypical network. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pages 6845–6852, 2020.

X. Zhang, R. R. Chowdhury, R. K. Gupta, and J. Shang. Large language models for time series: A survey. *arXiv preprint arXiv:2402.01801*, 2024.

L. N. Zheng, C. G. Dong, W. E. Zhang, L. Yue, M. Xu, O. Maennel, and W. Chen. Revisited large language model for time series analysis through modality alignment. *arXiv preprint arXiv:2410.12326*, 2024.

T. Zhou, P. Niu, X. Wang, L. Sun, and R. Jin. One fits all: Power general time series analysis by pretrained lm. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

A Full UCR Benchmark Results

This appendix reports the complete per-dataset few-shot benchmark results for all 128 UCR datasets. Each cell is shown as mean $\pm$ standard deviation over five runs, and the summary rows at the end of each table report average accuracy, average rank, and the number of datasets attaining the best mean accuracy (counting ties).

Table 6: Per-dataset 1-shot classification results on 128 shared UCR datasets. Each value is reported as mean ± standard deviation over 5 runs.

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	TimeMorph
ACSF1	**53.80 ± 4.97**	42.00 ± 4.95	48.40 ± 7.30	24.00 ± 4.24	35.60 ± 3.21	27.67 ± 7.02	24.33 ± 9.71	28.33 ± 2.52	27.80 ± 7.09	31.67 ± 9.81	18.60 ± 4.51	44.00 ± 5.20
Adiac	33.81 ± 1.23	12.84 ± 3.09	33.81 ± 1.23	8.85 ± 2.36	11.97 ± 1.69	21.40 ± 4.24	34.53 ± 1.77	8.10 ± 1.92	40.82 ± 3.61	24.21 ± 3.24	3.02 ± 1.05	**45.17 ± 5.28**
AllGestureWiimoteX	25.71 ± 4.01	**26.86 ± 2.52**	24.43 ± 2.55	15.57 ± 2.95	21.86 ± 4.59	20.71 ± 2.87	11.33 ± 1.19	16.76 ± 2.52	13.43 ± 1.74	18.90 ± 1.08	21.86 ± 4.59	25.11 ± 4.65
AllGestureWiimoteY	**26.37 ± 2.89**	26.17 ± 2.88	22.71 ± 4.62	17.49 ± 2.02	25.17 ± 5.50	21.81 ± 2.78	12.29 ± 1.27	20.29 ± 1.98	13.06 ± 1.34	21.33 ± 2.38	25.17 ± 5.50	25.54 ± 4.42
AllGestureWiimoteZ	**27.11 ± 1.95**	22.66 ± 4.13	25.49 ± 2.57	18.26 ± 2.01	19.71 ± 2.87	19.24 ± 1.57	13.29 ± 0.43	18.76 ± 2.79	14.14 ± 1.72	19.71 ± 1.79	19.71 ± 2.87	22.26 ± 3.96
ArrowHead	47.66 ± 8.94	46.63 ± 9.45	48.00 ± 7.74	52.00 ± 6.38	49.71 ± 1.40	47.62 ± 9.46	38.29 ± 4.00	50.86 ± 1.71	35.89 ± 2.60	50.86 ± 6.05	41.49 ± 12.34	**52.57 ± 3.43**
BME	49.60 ± 2.85	50.80 ± 2.13	49.73 ± 3.42	56.67 ± 3.43	62.93 ± 5.35	55.33 ± 7.86	51.56 ± 3.15	61.33 ± 4.06	55.87 ± 7.71	58.89 ± 6.19	35.73 ± 3.22	**66.40 ± 3.88**
Beef	36.00 ± 6.41	31.33 ± 5.06	27.33 ± 7.96	38.67 ± 9.89	32.00 ± 7.67	40.00 ± 8.82	34.44 ± 6.94	**41.11 ± 1.92**	30.67 ± 6.83	40.00 ± 6.67	31.33 ± 9.31	34.00 ± 2.79
BeetleFly	55.00 ± 24.24	60.00 ± 25.00	61.00 ± 24.85	67.00 ± 4.47	61.00 ± 9.62	60.00 ± 13.23	55.00 ± 5.00	61.67 ± 11.55	48.00 ± 2.74	60.00 ± 13.23	**70.00 ± 11.18**	**70.00 ± 11.73**
BirdChicken	62.00 ± 8.37	**71.00 ± 9.62**	70.00 ± 17.68	56.00 ± 4.18	56.00 ± 4.18	51.67 ± 2.89	51.67 ± 12.58	53.33 ± 5.77	55.00 ± 5.00	51.67 ± 7.64	59.00 ± 9.62	68.00 ± 14.83
CBF	81.67 ± 6.47	**87.62 ± 2.73**	71.71 ± 7.91	54.02 ± 3.44	66.38 ± 7.40	55.85 ± 2.16	33.63 ± 1.20	58.00 ± 5.61	36.89 ± 2.37	54.78 ± 1.68	35.20 ± 3.75	72.48 ± 4.85
Car	37.00 ± 4.62	39.00 ± 4.80	41.67 ± 9.28	38.67 ± 11.63	27.33 ± 9.90	42.22 ± 5.09	24.44 ± 3.47	43.89 ± 5.85	27.00 ± 4.15	40.00 ± 7.26	30.00 ± 9.65	**55.67 ± 8.94**
Chinatown	66.53 ± 19.14	63.73 ± 26.98	62.51 ± 27.20	67.41 ± 23.86	67.35 ± 11.32	77.07 ± 16.93	81.83 ± 2.19	77.94 ± 9.20	77.61 ± 8.88	75.41 ± 7.92	56.68 ± 20.52	**93.59 ± 3.92**
ChlorineConcentration	34.61 ± 2.26	35.41 ± 4.36	34.04 ± 2.01	33.26 ± 2.96	35.54 ± 3.20	33.64 ± 3.96	32.92 ± 4.26	33.42 ± 5.09	33.78 ± 2.74	34.11 ± 3.50	**39.93 ± 15.64**	32.61 ± 3.30
CinCECGTorso	37.20 ± 9.73	36.03 ± 6.56	36.51 ± 8.82	40.06 ± 3.86	35.72 ± 7.09	44.86 ± 6.03	35.36 ± 5.77	**50.53 ± 4.66**	34.97 ± 4.68	45.68 ± 5.50	42.49 ± 6.76	48.74 ± 6.47
Coffee	78.57 ± 9.78	**92.86 ± 8.38**	84.29 ± 13.02	72.86 ± 14.42	74.29 ± 18.80	63.10 ± 10.91	65.48 ± 7.43	71.43 ± 16.37	76.43 ± 4.07	64.29 ± 7.14	51.43 ± 4.07	75.71 ± 13.69
Computers	57.52 ± 7.64	**60.40 ± 6.13**	59.12 ± 4.47	47.12 ± 5.87	52.64 ± 10.26	50.93 ± 1.29	48.80 ± 0.69	51.87 ± 2.57	48.96 ± 2.79	52.27 ± 3.23	45.20 ± 5.79	56.40 ± 5.77
CricketX	26.15 ± 3.29	26.10 ± 3.67	20.56 ± 4.67	17.28 ± 1.96	18.15 ± 4.50	20.34 ± 0.39	8.89 ± 0.82	17.52 ± 1.26	11.38 ± 2.35	20.68 ± 0.74	7.28 ± 1.05	**26.26 ± 5.29**
CricketY	**26.97 ± 3.03**	24.46 ± 4.25	22.72 ± 1.74	18.56 ± 4.13	17.90 ± 3.61	22.82 ± 1.56	10.34 ± 1.89	20.94 ± 3.72	11.23 ± 0.80	21.20 ± 3.98	9.54 ± 2.33	23.33 ± 5.19
CricketZ	**32.51 ± 2.32**	28.72 ± 2.15	26.56 ± 4.19	19.64 ± 2.73	22.26 ± 3.91	20.34 ± 4.11	11.03 ± 1.54	20.00 ± 2.31	12.26 ± 1.25	20.34 ± 2.57	7.64 ± 1.05	29.28 ± 4.82
Crop	25.10 ± 4.94	28.14 ± 2.90	25.10 ± 4.94	10.23 ± 2.66	17.27 ± 1.43	27.73 ± 2.89	24.10 ± 4.30	26.19 ± 2.96	27.27 ± 3.58	30.31 ± 5.00	3.80 ± 0.58	**36.48 ± 3.38**
DiatomSizeReduction	79.67 ± 5.51	55.36 ± 19.14	84.12 ± 4.82	**89.22 ± 8.91**	49.61 ± 24.86	75.71 ± 7.74	55.01 ± 11.77	56.43 ± 20.40	50.07 ± 12.66	76.47 ± 11.70	41.44 ± 13.25	85.82 ± 8.55
DistalPhalanxOutlineAgeGroup	43.02 ± 17.16	43.02 ± 18.34	43.60 ± 14.92	42.73 ± 19.01	46.76 ± 16.49	45.80 ± 19.81	46.28 ± 21.27	**56.59 ± 9.36**	50.94 ± 19.66	46.28 ± 20.78	35.68 ± 14.61	42.01 ± 16.44
DistalPhalanxOutlineCorrect	62.10 ± 1.32	57.32 ± 9.22	**63.04 ± 2.44**	57.54 ± 11.13	53.55 ± 11.63	53.99 ± 14.78	53.26 ± 12.27	53.62 ± 11.97	55.14 ± 10.52	53.26 ± 12.97	56.74 ± 7.30	60.58 ± 2.98
DistalPhalanxTW	51.22 ± 5.15	44.03 ± 11.70	53.38 ± 4.02	43.74 ± 5.17	51.22 ± 6.58	45.32 ± 9.89	42.69 ± 4.40	38.13 ± 7.51	48.20 ± 5.84	47.48 ± 5.62	18.56 ± 14.51	**54.53 ± 1.72**
DodgerLoopDay	32.50 ± 7.34	25.00 ± 4.05	29.75 ± 7.68	**41.50 ± 2.05**	26.00 ± 6.58	40.42 ± 4.73	20.42 ± 1.91	32.92 ± 3.15	21.50 ± 4.45	40.42 ± 7.94	26.00 ± 6.58	34.50 ± 4.38
DodgerLoopGame	52.17 ± 6.30	55.07 ± 17.38	49.13 ± 8.90	**60.87 ± 12.35**	59.86 ± 22.05	51.45 ± 9.96	51.21 ± 5.14	56.52 ± 10.82	50.43 ± 2.44	51.45 ± 9.59	59.86 ± 22.05	56.96 ± 11.69
DodgerLoopWeekend	84.49 ± 8.37	74.78 ± 12.27	83.19 ± 10.24	93.62 ± 3.81	**96.23 ± 1.19**	95.65 ± 1.26	63.77 ± 4.41	94.44 ± 0.42	68.41 ± 4.57	93.72 ± 5.86	**96.23 ± 1.19**	85.36 ± 4.30
ECG200	63.00 ± 12.61	**72.00 ± 6.75**	66.20 ± 8.32	71.80 ± 4.44	61.60 ± 12.82	61.33 ± 10.97	65.00 ± 11.79	63.00 ± 20.66	64.20 ± 14.29	68.67 ± 7.09	51.40 ± 12.93	60.60 ± 6.91
ECG5000	54.22 ± 21.00	50.05 ± 20.59	61.53 ± 13.66	62.80 ± 19.19	53.35 ± 20.14	**69.07 ± 12.52**	54.44 ± 9.79	57.20 ± 16.09	62.35 ± 12.99	68.84 ± 13.99	19.70 ± 25.30	67.18 ± 10.57
ECGFiveDays	58.65 ± 7.69	57.24 ± 6.51	54.80 ± 13.00	57.63 ± 2.63	**62.21 ± 8.35**	57.99 ± 4.84	54.86 ± 3.17	55.87 ± 1.56	54.08 ± 6.73	58.46 ± 5.08	51.54 ± 6.38	61.77 ± 6.61
EOGHorizontalSignal	20.11 ± 2.01	22.38 ± 1.86	22.60 ± 2.77	22.43 ± 4.01	21.22 ± 3.15	20.63 ± 4.63	15.29 ± 2.63	23.57 ± 5.45	10.39 ± 2.26	22.01 ± 4.37	27.24 ± 2.99	**29.06 ± 4.16**
EOGVerticalSignal	11.27 ± 3.67	14.09 ± 3.23	13.26 ± 3.09	24.20 ± 3.27	23.76 ± 3.73	23.39 ± 2.79	11.23 ± 0.70	24.86 ± 4.17	11.27 ± 3.83	23.11 ± 4.07	19.06 ± 6.08	**27.35 ± 1.90**
Earthquakes	50.07 ± 8.55	51.08 ± 6.33	50.94 ± 6.46	49.35 ± 5.05	45.61 ± 20.94	48.68 ± 19.16	49.40 ± 1.10	49.40 ± 14.03	47.05 ± 5.90	47.00 ± 24.65	46.04 ± 13.71	**55.54 ± 11.57**
ElectricDevices	28.83 ± 9.01	32.82 ± 12.31	28.35 ± 12.24	17.59 ± 1.67	27.21 ± 7.38	21.93 ± 4.54	17.11 ± 1.87	15.12 ± 2.81	17.04 ± 2.05	15.61 ± 7.10	17.06 ± 3.10	**33.14 ± 10.96**
EthanolLevel	24.44 ± 1.30	25.40 ± 0.85	25.28 ± 1.57	24.92 ± 1.50	23.80 ± 1.65	26.13 ± 0.58	26.53 ± 1.55	26.67 ± 1.29	25.32 ± 1.50	25.80 ± 1.39	24.76 ± 1.40	**26.68 ± 1.86**
FaceAll	**42.17 ± 4.79**	32.36 ± 9.47	**42.17 ± 4.79**	28.66 ± 9.09	19.36 ± 3.07	29.11 ± 4.19	33.12 ± 5.46	25.94 ± 1.75	34.05 ± 3.40	32.92 ± 2.10	8.64 ± 3.46	32.84 ± 6.86
FaceFour	68.64 ± 10.56	47.50 ± 11.12	56.36 ± 4.30	64.32 ± 4.73	54.09 ± 8.34	70.08 ± 4.30	52.65 ± 3.99	67.05 ± 7.10	51.36 ± 4.28	**73.11 ± 3.99**	42.95 ± 6.60	54.77 ± 8.25
FacesUCR	**54.83 ± 5.21**	37.63 ± 5.89	43.65 ± 3.88	35.69 ± 7.47	20.81 ± 1.95	36.24 ± 3.26	41.98 ± 2.95	40.05 ± 2.80	41.87 ± 3.09	42.63 ± 2.58	16.92 ± 7.54	50.37 ± 2.98
FiftyWords	13.27 ± 1.89	12.79 ± 3.45	13.27 ± 1.89	32.70 ± 4.33	19.60 ± 4.90	35.90 ± 3.31	3.44 ± 1.59	**38.97 ± 6.44**	11.30 ± 2.86	36.70 ± 4.41	3.74 ± 2.40	36.48 ± 2.66
Fish	**49.03 ± 5.96**	21.26 ± 3.60	45.71 ± 1.71	36.00 ± 3.98	23.20 ± 3.80	32.57 ± 4.57	16.95 ± 2.93	30.29 ± 2.62	21.71 ± 2.77	31.81 ± 4.76	20.80 ± 5.10	34.74 ± 4.91
FordA	59.23 ± 9.68	57.68 ± 10.00	**59.83 ± 8.76**	50.73 ± 1.01	56.70 ± 8.04	50.30 ± 0.33	49.57 ± 1.42	49.72 ± 0.63	49.68 ± 1.11	49.92 ± 0.80	50.61 ± 1.68	50.71 ± 0.98
FordB	52.54 ± 6.44	52.64 ± 6.88	52.40 ± 4.53	50.12 ± 1.83	**53.60 ± 4.09**	49.96 ± 0.94	50.53 ± 0.72	50.21 ± 0.80	50.81 ± 0.65	49.88 ± 0.98	50.05 ± 1.16	49.75 ± 2.56
FreezerRegularTrain	61.33 ± 21.55	58.22 ± 20.21	56.76 ± 19.80	61.31 ± 21.84	57.07 ± 18.69	59.91 ± 27.91	48.78 ± 1.68	53.81 ± 23.62	49.55 ± 8.37	60.27 ± 28.68	61.45 ± 8.09	**61.90 ± 22.25**
FreezerSmallTrain	67.85 ± 12.31	66.48 ± 12.45	67.42 ± 12.38	67.47 ± 12.29	63.19 ± 10.91	**73.32 ± 3.65**	55.33 ± 0.97	69.60 ± 5.96	57.00 ± 2.79	73.03 ± 4.44	65.92 ± 7.33	66.53 ± 10.64
Fungi	93.23 ± 4.00	35.81 ± 4.97	64.62 ± 2.56	68.39 ± 5.78	67.20 ± 19.03	**95.70 ± 0.93**	73.66 ± 2.69	86.74 ± 1.89	75.16 ± 1.03	**95.70 ± 0.93**	14.09 ± 7.48	79.35 ± 5.30
GestureMidAirD1	17.08 ± 4.26	19.38 ± 4.43	17.08 ± 4.26	34.31 ± 3.87	19.08 ± 6.45	31.54 ± 2.04	17.44 ± 0.89	34.62 ± 4.07	21.85 ± 4.63	**34.87 ± 2.47**	19.08 ± 6.45	32.00 ± 3.94
GestureMidAirD2	13.23 ± 5.14	17.08 ± 3.90	13.23 ± 5.14	29.08 ± 4.19	20.46 ± 5.77	29.74 ± 3.20	11.79 ± 2.22	29.23 ± 5.38	17.38 ± 5.95	**36.41 ± 0.89**	20.46 ± 5.77	33.85 ± 3.03
GestureMidAirD3	8.62 ± 1.84	9.08 ± 2.52	8.62 ± 1.84	20.00 ± 3.17	10.92 ± 1.26	**24.87 ± 4.24**	9.49 ± 1.60	23.08 ± 0.77	14.77 ± 4.47	22.56 ± 2.91	10.92 ± 1.26	18.46 ± 2.61

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Table 6: Per-dataset 1-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	TimeMorph
GesturePebbleZ1	49.07 ± 5.03	38.37 ± 6.03	**53.26 ± 6.97**	46.16 ± 13.08	23.60 ± 6.95	40.50 ± 15.60	20.93 ± 3.54	42.44 ± 7.05	18.95 ± 2.08	43.99 ± 8.75	23.60 ± 6.95	35.81 ± 4.48
GesturePebbleZ2	55.19 ± 11.38	42.91 ± 15.92	**57.47 ± 8.89**	48.35 ± 13.33	21.01 ± 3.48	47.68 ± 13.93	18.35 ± 4.94	43.88 ± 16.17	16.20 ± 5.17	55.06 ± 14.60	21.01 ± 3.48	33.29 ± 5.85
GunPoint	55.47 ± 4.79	58.13 ± 9.13	55.87 ± 5.32	53.20 ± 9.11	56.40 ± 7.58	**64.00 ± 8.08**	58.89 ± 10.59	63.33 ± 4.81	58.93 ± 8.45	62.89 ± 3.01	52.40 ± 11.44	58.00 ± 7.79
GunPointAgeSpan	49.94 ± 8.02	48.67 ± 3.53	49.11 ± 3.99	60.82 ± 22.09	58.73 ± 19.72	59.70 ± 21.56	52.11 ± 7.60	59.07 ± 20.81	52.41 ± 6.32	**61.92 ± 19.97**	48.10 ± 2.61	60.51 ± 10.32
GunPointMaleVersusFemale	63.99 ± 11.97	63.29 ± 11.59	64.30 ± 12.20	54.56 ± 4.47	56.65 ± 3.27	55.91 ± 6.97	50.95 ± 7.17	55.59 ± 9.92	50.19 ± 6.98	56.75 ± 8.76	58.04 ± 11.38	**81.01 ± 10.12**
GunPointOldVersusYoung	**92.38 ± 10.44**	89.97 ± 13.74	90.35 ± 13.23	53.90 ± 25.42	53.33 ± 25.91	58.31 ± 30.15	54.81 ± 4.33	57.46 ± 29.14	54.79 ± 8.98	58.10 ± 28.87	78.41 ± 20.62	57.97 ± 10.78
Ham	50.10 ± 9.32	51.24 ± 7.01	47.81 ± 11.28	55.24 ± 11.13	49.33 ± 2.81	48.89 ± 5.25	53.65 ± 3.06	51.43 ± 7.44	**55.81 ± 6.79**	50.16 ± 4.89	51.81 ± 10.55	52.76 ± 9.30
HandOutlines	46.81 ± 15.76	58.86 ± 12.82	55.84 ± 9.47	70.59 ± 7.20	75.68 ± 8.02	71.17 ± 7.55	51.08 ± 2.34	**76.76 ± 10.41**	52.81 ± 15.40	74.86 ± 11.22	61.19 ± 14.73	61.73 ± 6.31
Haptics	23.96 ± 5.22	22.66 ± 2.12	22.86 ± 4.94	**26.49 ± 6.85**	22.66 ± 4.38	25.11 ± 9.19	19.16 ± 1.81	23.16 ± 7.32	23.44 ± 3.46	25.43 ± 9.51	25.65 ± 5.26	25.65 ± 4.99
Herring	49.69 ± 6.39	49.69 ± 10.50	50.94 ± 11.51	48.12 ± 8.44	49.69 ± 7.44	47.40 ± 10.40	**51.04 ± 2.39**	49.48 ± 9.55	48.75 ± 8.44	48.44 ± 9.50	48.12 ± 10.39	50.00 ± 10.00
HouseTwenty	**80.84 ± 7.67**	74.62 ± 5.52	68.40 ± 16.13	50.25 ± 3.68	57.65 ± 8.25	68.63 ± 3.50	49.30 ± 7.15	61.62 ± 2.57	46.05 ± 6.81	66.67 ± 1.28	53.61 ± 8.52	62.18 ± 8.04
InlineSkate	16.40 ± 1.60	15.64 ± 0.76	18.00 ± 0.73	15.89 ± 3.32	14.69 ± 2.92	18.06 ± 2.11	15.15 ± 0.73	18.00 ± 2.36	15.02 ± 3.34	**18.67 ± 3.56**	14.87 ± 3.22	17.13 ± 2.24
InsectEPGRegularTrain	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	43.53 ± 11.83	50.44 ± 13.06	48.19 ± 5.92	42.03 ± 6.13	45.25 ± 3.22	46.27 ± 8.59	49.26 ± 6.69	87.15 ± 20.64	55.34 ± 14.47
InsectEPGSmallTrain	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	47.95 ± 6.69	55.10 ± 5.49	51.14 ± 8.16	39.36 ± 1.75	52.88 ± 6.50	37.67 ± 14.87	52.48 ± 11.47	85.78 ± 19.79	50.76 ± 7.86
InsectWingbeatSound	16.18 ± 3.00	14.23 ± 3.87	14.35 ± 2.82	39.88 ± 4.03	22.79 ± 2.65	43.42 ± 0.26	17.56 ± 4.92	43.16 ± 3.28	21.14 ± 2.99	**45.61 ± 5.13**	12.52 ± 3.37	25.27 ± 2.20
ItalyPowerDemand	61.83 ± 17.15	62.29 ± 17.84	64.74 ± 16.58	55.32 ± 7.10	60.35 ± 16.10	69.81 ± 17.24	68.35 ± 9.65	**78.33 ± 24.61**	68.98 ± 15.67	71.62 ± 21.13	47.31 ± 2.37	58.52 ± 15.27
LargeKitchenAppliances	48.48 ± 12.88	52.85 ± 11.74	46.72 ± 15.63	33.39 ± 3.38	**54.88 ± 8.28**	40.09 ± 11.55	34.93 ± 5.70	32.53 ± 3.23	31.47 ± 3.69	41.69 ± 12.56	33.81 ± 2.87	45.44 ± 5.60
Lightning2	50.49 ± 6.80	55.08 ± 10.15	51.80 ± 4.27	56.72 ± 11.39	51.80 ± 12.52	58.47 ± 11.16	48.63 ± 3.41	57.38 ± 11.36	51.48 ± 5.26	59.02 ± 8.67	55.08 ± 13.25	**59.67 ± 8.87**
Lightning7	44.93 ± 4.48	44.38 ± 4.60	40.82 ± 7.47	43.01 ± 4.18	42.19 ± 7.34	53.42 ± 10.70	15.07 ± 6.28	45.66 ± 6.76	18.63 ± 4.40	**56.16 ± 8.33**	24.11 ± 7.72	41.64 ± 7.60
Mallat	64.03 ± 14.52	50.42 ± 9.84	64.49 ± 9.43	80.37 ± 4.76	56.96 ± 10.09	**82.32 ± 3.73**	54.67 ± 2.15	75.96 ± 4.82	64.55 ± 4.53	75.72 ± 2.60	37.30 ± 5.46	68.24 ± 7.54
Meat	42.00 ± 9.96	44.00 ± 10.18	58.33 ± 10.99	**77.67 ± 17.26**	52.67 ± 11.94	46.11 ± 9.18	38.33 ± 8.66	33.33 ± 0.00	36.00 ± 2.53	66.11 ± 20.97	33.67 ± 0.75	55.67 ± 9.62
MedicalImages	27.92 ± 5.84	28.45 ± 4.75	24.95 ± 5.95	26.34 ± 8.67	24.16 ± 4.26	22.32 ± 5.35	26.84 ± 12.59	23.86 ± 6.12	**28.53 ± 8.23**	23.16 ± 6.10	4.21 ± 2.27	27.66 ± 4.41
MelbournePedestrian	**51.29 ± 5.58**	44.27 ± 4.81	47.47 ± 6.50	9.85 ± 0.23	31.99 ± 4.95	21.74 ± 7.37	46.44 ± 8.26	40.62 ± 4.65	43.40 ± 7.11	43.73 ± 5.33	31.99 ± 4.95	39.16 ± 5.22
MiddlePhalanxOutlineAgeGroup	38.83 ± 7.74	37.53 ± 10.57	38.05 ± 7.83	34.03 ± 9.03	35.97 ± 8.21	37.45 ± 11.78	29.87 ± 4.55	38.53 ± 15.64	33.90 ± 7.76	32.90 ± 7.86	**41.69 ± 18.94**	39.48 ± 7.40
MiddlePhalanxOutlineCorrect	53.68 ± 13.63	52.10 ± 14.21	54.43 ± 13.33	53.75 ± 17.89	56.49 ± 9.58	**58.42 ± 17.26**	55.56 ± 15.16	54.07 ± 23.81	54.50 ± 10.22	57.50 ± 21.09	52.85 ± 12.08	51.68 ± 5.00
MiddlePhalanxTW	34.03 ± 16.27	44.68 ± 7.76	32.47 ± 17.97	30.52 ± 10.75	32.34 ± 10.78	29.00 ± 15.96	38.96 ± 3.37	**46.10 ± 3.62**	35.45 ± 13.57	42.21 ± 5.07	15.19 ± 11.64	37.14 ± 16.84
MixedShapesRegularTrain	45.36 ± 5.29	40.59 ± 8.80	43.58 ± 10.39	47.60 ± 10.74	39.59 ± 5.62	43.05 ± 15.67	17.31 ± 1.15	45.53 ± 11.76	25.71 ± 3.77	42.76 ± 16.81	45.03 ± 12.47	**56.06 ± 5.34**
MixedShapesSmallTrain	52.78 ± 8.02	39.55 ± 6.53	48.84 ± 9.40	53.90 ± 7.66	45.57 ± 7.91	52.81 ± 6.78	17.32 ± 0.61	53.84 ± 4.25	24.17 ± 3.24	53.31 ± 5.69	48.73 ± 8.44	**63.30 ± 3.34**
MoteStrain	62.99 ± 13.96	60.67 ± 12.58	64.35 ± 11.70	63.40 ± 15.50	63.27 ± 13.76	68.45 ± 11.94	56.26 ± 12.55	63.82 ± 16.98	61.65 ± 9.57	**68.66 ± 11.89**	57.99 ± 9.98	54.36 ± 12.77
NonInvasiveFetalECGThorax1	19.05 ± 4.42	8.04 ± 2.52	19.05 ± 4.42	38.23 ± 1.81	12.07 ± 7.41	36.76 ± 3.30	12.57 ± 2.38	28.62 ± 3.97	14.44 ± 2.07	**42.58 ± 4.62**	5.58 ± 2.48	39.16 ± 2.28
NonInvasiveFetalECGThorax2	20.69 ± 5.30	9.71 ± 2.65	20.69 ± 5.30	47.48 ± 2.72	14.81 ± 5.33	46.67 ± 0.18	20.97 ± 3.22	34.59 ± 7.78	23.69 ± 1.73	**49.89 ± 4.96**	8.30 ± 1.72	47.48 ± 2.06
OSULeaf	**45.62 ± 7.45**	30.41 ± 8.35	38.35 ± 12.00	24.05 ± 2.14	22.15 ± 5.63	27.82 ± 3.13	17.77 ± 1.80	26.03 ± 1.89	20.00 ± 5.09	26.17 ± 1.56	17.60 ± 2.44	29.83 ± 5.82
OliveOil	28.00 ± 23.64	26.67 ± 11.06	47.33 ± 13.21	**65.33 ± 15.20**	36.00 ± 22.66	32.22 ± 16.44	44.44 ± 10.72	28.89 ± 11.71	32.00 ± 9.60	61.11 ± 15.40	27.33 ± 11.88	28.00 ± 14.26
PLAID	15.46 ± 6.16	13.97 ± 4.45	16.95 ± 4.17	17.77 ± 3.34	21.79 ± 0.97	21.54 ± 5.77	13.78 ± 3.08	19.93 ± 5.41	14.41 ± 2.63	20.30 ± 4.75	21.79 ± 0.97	**32.10 ± 5.72**
PhalangesOutlinesCorrect	**54.90 ± 6.98**	51.28 ± 10.93	54.41 ± 7.38	53.40 ± 8.33	52.54 ± 10.38	48.87 ± 4.48	51.01 ± 10.34	53.38 ± 8.94	52.73 ± 9.70	52.49 ± 9.30	54.83 ± 6.32	51.93 ± 9.54
Phoneme	**10.43 ± 1.42**	8.46 ± 1.00	**10.43 ± 1.42**	4.21 ± 0.54	5.61 ± 2.38	6.14 ± 1.81	2.39 ± 1.01	5.87 ± 1.54	3.20 ± 2.06	4.98 ± 3.20	3.42 ± 0.41	6.87 ± 1.15
PickupGestureWiimoteZ	35.60 ± 7.80	44.80 ± 4.82	42.00 ± 5.10	40.80 ± 5.40	36.00 ± 3.16	46.67 ± 9.87	17.33 ± 5.03	39.33 ± 6.43	20.40 ± 4.98	**48.67 ± 8.33**	36.00 ± 3.16	47.20 ± 7.29
PigAirwayPressure	23.94 ± 3.76	9.13 ± 2.66	**38.65 ± 2.92**	3.65 ± 1.47	3.75 ± 1.33	5.29 ± 0.83	4.01 ± 0.73	4.01 ± 1.11	3.27 ± 0.92	4.81 ± 0.96	6.15 ± 1.64	5.48 ± 2.19
PigArtPressure	40.58 ± 5.14	10.10 ± 2.36	**60.10 ± 2.28**	9.90 ± 2.42	2.21 ± 0.80	10.58 ± 2.10	8.01 ± 2.00	10.58 ± 1.73	4.81 ± 0.76	9.94 ± 2.27	15.58 ± 3.05	15.38 ± 4.07
PigCVP	22.50 ± 3.47	14.33 ± 2.79	**50.29 ± 2.90**	5.00 ± 1.25	3.27 ± 0.71	5.93 ± 0.73	3.53 ± 1.39	6.89 ± 0.56	2.02 ± 0.22	4.33 ± 0.83	8.94 ± 1.94	11.44 ± 2.08
Plane	**98.86 ± 2.56**	91.62 ± 6.26	95.43 ± 3.65	87.24 ± 8.67	85.14 ± 9.49	82.86 ± 4.15	78.73 ± 5.25	88.57 ± 9.09	86.48 ± 4.39	88.57 ± 6.67	26.10 ± 13.02	93.90 ± 6.92
PowerCons	76.89 ± 9.68	**80.22 ± 9.41**	78.11 ± 13.88	60.22 ± 11.82	61.00 ± 7.74	62.96 ± 3.70	49.44 ± 7.47	58.33 ± 2.78	51.33 ± 4.56	62.78 ± 6.41	55.22 ± 21.33	60.89 ± 11.09
ProximalPhalanxOutlineAgeGroup	48.49 ± 34.51	48.20 ± 31.13	48.68 ± 30.82	39.61 ± 19.82	39.12 ± 18.60	43.74 ± 32.89	49.59 ± 25.02	**60.81 ± 40.85**	38.15 ± 22.87	46.50 ± 27.14	32.68 ± 16.57	48.98 ± 23.27
ProximalPhalanxOutlineCorrect	54.43 ± 13.05	53.40 ± 15.85	51.20 ± 16.57	56.22 ± 5.47	53.26 ± 10.68	44.67 ± 13.09	46.96 ± 12.63	**58.76 ± 6.63**	52.78 ± 12.27	43.99 ± 6.30	57.59 ± 13.02	52.23 ± 12.03
ProximalPhalanxTW	54.63 ± 10.31	57.56 ± 9.33	58.15 ± 6.64	54.34 ± 11.16	49.76 ± 11.67	**60.81 ± 11.71**	55.93 ± 9.23	38.37 ± 0.75	55.80 ± 7.24	57.07 ± 10.70	17.37 ± 16.23	55.22 ± 6.63
RefrigerationDevices	36.05 ± 6.49	35.68 ± 5.16	37.44 ± 5.37	34.99 ± 2.77	33.97 ± 4.76	**41.16 ± 5.10**	34.58 ± 1.47	34.93 ± 0.92	34.72 ± 0.99	38.84 ± 5.04	32.75 ± 2.24	29.76 ± 11.02
Rock	29.20 ± 8.07	34.40 ± 7.92	28.40 ± 6.54	41.60 ± 16.52	26.40 ± 8.53	38.67 ± 25.32	**57.33 ± 5.03**	44.67 ± 18.90	36.80 ± 8.07	40.00 ± 24.25	36.40 ± 14.93	39.20 ± 13.75
ScreenType	33.92 ± 4.73	34.19 ± 7.39	**34.29 ± 6.61**	33.87 ± 2.72	33.87 ± 4.61	32.44 ± 5.27	32.18 ± 3.99	28.00 ± 3.40	30.99 ± 3.01	33.60 ± 4.19	30.61 ± 2.66	32.53 ± 4.84
SemgHandGenderCh2	50.87 ± 2.35	55.37 ± 5.38	53.17 ± 4.35	50.17 ± 11.45	**61.00 ± 15.57**	42.78 ± 13.47	52.78 ± 1.84	39.89 ± 8.47	44.67 ± 12.14	35.61 ± 1.06	55.73 ± 4.74	51.90 ± 14.92
SemgHandMovementCh2	28.89 ± 4.22	**29.87 ± 4.86**	29.60 ± 5.24	25.29 ± 1.13	24.89 ± 3.57	20.37 ± 0.13	21.70 ± 2.31	22.52 ± 2.76	18.71 ± 1.91	20.07 ± 3.46	27.02 ± 5.10	28.13 ± 4.95

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Table 6: Per-dataset 1-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	TimeMorph
SemgHandSubjectCh2	28.89 ± 5.77	29.07 ± 4.60	28.00 ± 2.65	35.60 ± 4.19	29.91 ± 10.77	26.37 ± 1.89	23.11 ± 2.89	26.67 ± 4.87	21.56 ± 1.94	26.96 ± 3.68	27.91 ± 3.11	**37.42 ± 8.82**
ShakeGestureWiimoteZ	63.20 ± 7.69	58.00 ± 11.83	**67.60 ± 11.61**	40.40 ± 11.26	43.20 ± 9.55	45.33 ± 3.06	16.00 ± 7.21	40.00 ± 3.46	24.40 ± 4.98	46.00 ± 6.00	43.20 ± 9.55	60.40 ± 7.92
ShapeletSim	66.33 ± 12.25	**87.00 ± 16.91**	73.89 ± 23.85	50.11 ± 2.27	75.22 ± 8.30	53.89 ± 1.67	52.96 ± 1.40	48.89 ± 5.64	50.44 ± 1.82	51.11 ± 0.96	47.78 ± 3.07	62.44 ± 7.44
ShapesAll	23.63 ± 4.60	15.60 ± 1.98	23.63 ± 4.60	45.70 ± 2.59	19.27 ± 4.58	44.72 ± 2.58	6.00 ± 0.58	43.94 ± 2.14	8.07 ± 3.90	44.06 ± 1.11	8.13 ± 2.75	**50.20 ± 3.03**
SmallKitchenAppliances	40.37 ± 15.51	**46.24 ± 15.42**	44.32 ± 15.50	42.19 ± 3.16	38.13 ± 22.42	32.98 ± 5.23	32.80 ± 3.67	35.82 ± 4.27	35.41 ± 3.13	36.89 ± 7.74	38.88 ± 7.80	42.88 ± 15.27
SmoothSubspace	**74.93 ± 5.59**	74.40 ± 5.13	74.80 ± 8.48	31.73 ± 3.96	50.00 ± 10.52	46.67 ± 1.76	53.33 ± 9.45	50.22 ± 10.96	62.27 ± 6.56	60.67 ± 10.35	56.67 ± 5.19	68.53 ± 8.95
SonyAIBORobotSurface1	77.57 ± 10.30	**80.80 ± 8.99**	80.37 ± 5.89	59.80 ± 6.73	64.53 ± 18.83	52.86 ± 10.38	56.96 ± 5.82	55.35 ± 9.37	56.07 ± 4.43	57.68 ± 9.32	55.61 ± 9.09	57.20 ± 8.62
SonyAIBORobotSurface2	80.29 ± 6.43	**84.09 ± 5.18**	80.90 ± 6.99	67.39 ± 19.08	73.03 ± 13.78	66.18 ± 11.61	60.37 ± 27.05	61.35 ± 27.10	69.49 ± 15.13	68.07 ± 16.06	57.31 ± 15.03	66.09 ± 10.78
StarLightCurves	63.75 ± 3.45	62.30 ± 9.88	57.59 ± 9.08	57.39 ± 13.40	57.55 ± 15.28	**67.54 ± 17.11**	37.78 ± 3.53	57.92 ± 15.55	27.99 ± 0.00	67.36 ± 17.40	64.81 ± 14.61	63.03 ± 14.23
Strawberry	52.27 ± 4.26	51.84 ± 3.86	54.43 ± 4.33	54.86 ± 4.34	53.62 ± 3.45	58.29 ± 0.95	54.86 ± 13.21	**60.18 ± 0.68**	56.11 ± 9.26	57.93 ± 0.83	45.89 ± 11.15	51.46 ± 5.21
SwedishLeaf	**65.95 ± 6.01**	46.08 ± 6.36	**65.95 ± 6.01**	27.42 ± 3.44	40.83 ± 4.40	36.69 ± 1.99	27.20 ± 2.42	24.53 ± 4.08	31.65 ± 2.48	38.19 ± 0.81	8.10 ± 1.23	57.98 ± 2.72
Symbols	**85.77 ± 4.83**	70.17 ± 4.91	76.68 ± 7.53	79.60 ± 7.65	72.50 ± 6.12	81.54 ± 3.92	37.15 ± 0.97	82.18 ± 0.42	46.13 ± 9.06	80.57 ± 0.95	51.54 ± 6.72	80.94 ± 7.54
SyntheticControl	72.53 ± 6.61	76.40 ± 5.24	68.67 ± 10.59	38.47 ± 6.28	80.33 ± 9.12	47.78 ± 5.87	44.89 ± 3.20	48.22 ± 0.38	42.13 ± 2.87	50.00 ± 4.48	24.53 ± 7.68	**84.20 ± 6.54**
ToeSegmentation1	**69.82 ± 7.39**	69.65 ± 4.27	65.09 ± 8.05	51.67 ± 1.47	55.79 ± 6.85	49.27 ± 1.10	49.71 ± 3.29	46.64 ± 4.73	52.46 ± 2.82	48.39 ± 3.23	49.47 ± 2.47	58.16 ± 3.70
ToeSegmentation2	68.92 ± 8.63	70.31 ± 3.38	70.92 ± 10.76	50.31 ± 5.37	65.54 ± 9.84	50.26 ± 2.91	35.90 ± 10.81	50.51 ± 4.24	33.23 ± 16.47	52.05 ± 6.54	41.54 ± 14.95	**75.23 ± 12.13**
Trace	61.00 ± 4.64	65.20 ± 9.09	65.80 ± 6.61	49.60 ± 3.65	74.40 ± 6.15	51.33 ± 5.51	42.33 ± 4.51	47.67 ± 0.58	44.80 ± 7.69	51.33 ± 4.93	37.80 ± 6.42	**99.20 ± 1.79**
TwoLeadECG	**69.48 ± 12.74**	64.97 ± 10.26	66.51 ± 13.23	54.03 ± 2.33	57.82 ± 5.04	59.32 ± 3.05	53.32 ± 7.30	52.62 ± 3.32	54.47 ± 5.41	60.32 ± 1.06	50.73 ± 1.02	64.88 ± 11.28
TwoPatterns	48.77 ± 2.41	**61.51 ± 6.85**	46.17 ± 11.65	31.86 ± 3.24	36.86 ± 6.10	29.76 ± 2.56	25.52 ± 1.93	28.23 ± 0.74	26.06 ± 1.27	29.87 ± 1.68	26.32 ± 2.53	45.74 ± 2.64
UMD	43.06 ± 4.02	**54.58 ± 9.53**	45.56 ± 6.14	46.81 ± 7.70	50.14 ± 6.28	46.30 ± 5.02	45.60 ± 3.21	47.69 ± 4.19	48.61 ± 2.55	47.45 ± 6.56	31.39 ± 4.35	45.47 ± 8.73
UWaveGestureLibraryAll	29.36 ± 4.20	24.46 ± 5.19	28.34 ± 3.82	66.77 ± 5.89	41.63 ± 5.40	69.19 ± 5.01	25.78 ± 1.68	67.23 ± 2.91	26.67 ± 3.53	**69.21 ± 6.24**	50.70 ± 5.42	33.37 ± 0.26
UWaveGestureLibraryX	26.45 ± 4.19	26.60 ± 3.44	24.71 ± 1.83	47.27 ± 6.01	35.36 ± 5.04	**49.30 ± 7.17**	22.37 ± 0.14	47.22 ± 8.02	24.34 ± 5.30	49.17 ± 7.61	17.28 ± 8.20	30.33 ± 0.23
UWaveGestureLibraryY	25.37 ± 4.63	26.13 ± 4.45	24.62 ± 4.93	43.95 ± 4.51	38.35 ± 7.26	47.05 ± 4.02	24.58 ± 1.89	46.37 ± 4.93	26.81 ± 2.91	**47.18 ± 5.20**	20.37 ± 4.71	40.36 ± 0.25
UWaveGestureLibraryZ	24.77 ± 4.74	27.03 ± 3.18	24.45 ± 3.15	42.90 ± 1.96	36.37 ± 3.72	44.61 ± 1.78	22.47 ± 2.82	44.70 ± 2.64	25.10 ± 2.80	44.75 ± 1.63	22.77 ± 3.47	**47.44 ± 2.61**
Wafer	50.39 ± 4.56	44.94 ± 10.31	49.91 ± 13.57	**67.87 ± 4.81**	51.56 ± 17.59	63.90 ± 3.73	64.15 ± 3.76	63.99 ± 9.60	63.82 ± 9.82	65.48 ± 1.03	43.02 ± 26.21	46.27 ± 7.11
Wine	**57.41 ± 14.70**	54.07 ± 4.61	54.81 ± 11.31	52.96 ± 8.34	50.74 ± 6.23	48.77 ± 3.85	50.00 ± 0.00	50.00 ± 0.00	52.22 ± 4.97	37.65 ± 19.80	51.48 ± 3.31	51.11 ± 7.48
WordSynonyms	14.92 ± 3.03	13.70 ± 5.79	14.92 ± 3.03	19.78 ± 2.70	16.96 ± 4.46	19.75 ± 1.90	6.79 ± 1.46	**21.16 ± 1.50**	10.06 ± 3.54	20.53 ± 2.41	4.98 ± 1.34	20.91 ± 2.73
Worms	**36.10 ± 9.87**	33.25 ± 6.66	35.06 ± 11.43	17.14 ± 4.04	23.90 ± 9.39	19.91 ± 6.54	23.38 ± 4.50	18.61 ± 5.86	17.66 ± 7.03	19.05 ± 5.25	17.66 ± 2.36	24.68 ± 8.95
WormsTwoClass	44.79 ± 3.48	49.09 ± 6.96	52.99 ± 6.71	49.09 ± 3.72	50.39 ± 3.23	49.35 ± 3.44	48.92 ± 10.50	52.38 ± 1.98	53.51 ± 4.81	**53.68 ± 3.75**	52.99 ± 2.82	47.53 ± 10.69
Yoga	**55.60 ± 4.57**	50.68 ± 5.18	53.54 ± 4.28	49.99 ± 1.44	49.75 ± 4.01	49.48 ± 0.91	51.11 ± 3.62	49.62 ± 0.72	51.12 ± 3.30	48.62 ± 0.52	49.35 ± 2.85	50.42 ± 4.51
Avg. Acc.	47.79	45.40	47.88	44.21	42.39	45.16	36.78	44.31	37.71	46.42	35.61	**48.31**
Avg. Rank	5.11	5.80	5.19	6.45	6.92	6.04	9.11	6.17	8.51	5.30	8.82	**4.58**
#Best	25	16	14	7	5	11	2	11	2	15	4	**26**

Table 7: Per-dataset 2-shot classification results on 128 shared UCR datasets. Each value is reported as mean ± standard deviation over 5 runs.

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	TimeMorph
ACSF1	**61.60 ± 7.86**	47.20 ± 6.30	59.20 ± 5.26	25.80 ± 4.38	42.00 ± 3.16	40.33 ± 2.31	30.67 ± 2.52	44.00 ± 6.24	35.80 ± 3.19	40.67 ± 5.69	24.80 ± 4.66	54.33 ± 4.16
Adiac	35.45 ± 2.60	14.27 ± 3.74	35.45 ± 2.60	18.87 ± 2.86	17.80 ± 5.08	27.79 ± 2.14	44.67 ± 0.39	16.62 ± 1.77	55.24 ± 4.67	32.91 ± 3.49	3.48 ± 1.27	**56.01 ± 1.25**
AllGestureWiimoteX	**32.43 ± 2.94**	27.69 ± 3.48	32.09 ± 3.58	19.46 ± 2.47	23.77 ± 3.63	24.43 ± 4.71	11.86 ± 1.00	22.67 ± 2.29	14.09 ± 1.28	21.10 ± 5.46	23.77 ± 3.63	26.51 ± 4.36
AllGestureWiimoteY	**35.11 ± 5.43**	28.57 ± 4.31	28.91 ± 5.81	20.54 ± 2.69	33.31 ± 2.65	24.33 ± 2.50	12.00 ± 1.38	21.90 ± 1.33	16.14 ± 1.67	22.57 ± 1.00	33.31 ± 2.65	29.43 ± 1.69
AllGestureWiimoteZ	**29.80 ± 2.53**	26.94 ± 2.84	29.31 ± 4.56	16.77 ± 2.02	22.97 ± 5.35	19.86 ± 4.12	12.76 ± 0.36	19.57 ± 2.48	14.97 ± 0.98	20.29 ± 3.48	22.97 ± 5.35	27.91 ± 5.21
ArrowHead	50.17 ± 5.13	44.80 ± 3.82	45.14 ± 5.25	**57.03 ± 4.92**	49.14 ± 13.02	53.14 ± 6.02	34.10 ± 6.27	50.67 ± 2.01	42.29 ± 4.92	55.43 ± 6.44	43.77 ± 8.76	**57.03 ± 2.81**
BME	48.40 ± 4.91	55.33 ± 7.02	54.40 ± 8.12	50.80 ± 5.11	67.33 ± 7.90	48.89 ± 2.14	64.00 ± 5.21	58.89 ± 3.91	67.33 ± 7.44	54.89 ± 9.20	50.80 ± 8.05	**79.47 ± 11.40**
Beef	42.00 ± 2.98	36.67 ± 8.50	36.70 ± 9.60	49.33 ± 7.23	39.33 ± 6.41	48.89 ± 5.09	40.00 ± 3.33	34.44 ± 5.09	42.67 ± 4.94	**51.11 ± 7.70**	32.67 ± 7.23	43.33 ± 5.27
BeetleFly	50.00 ± 9.35	67.00 ± 18.23	55.00 ± 12.75	52.00 ± 11.51	57.00 ± 16.43	45.00 ± 10.00	48.33 ± 12.58	38.33 ± 7.64	45.00 ± 9.35	56.67 ± 7.64	57.00 ± 4.47	**75.00 ± 18.03**
BirdChicken	71.00 ± 6.52	72.00 ± 4.47	**76.00 ± 7.42**	52.00 ± 14.40	65.00 ± 10.61	45.00 ± 8.66	51.67 ± 7.64	46.67 ± 5.77	53.00 ± 2.74	50.00 ± 13.23	49.00 ± 2.24	69.00 ± 22.19
CBF	85.76 ± 5.83	**90.62 ± 7.16**	82.18 ± 10.67	64.78 ± 6.67	82.69 ± 10.89	68.07 ± 6.00	34.89 ± 2.41	73.78 ± 6.88	36.78 ± 1.75	70.00 ± 6.65	42.84 ± 7.09	85.91 ± 4.23
Car	42.00 ± 15.43	41.67 ± 13.49	52.67 ± 7.87	55.33 ± 10.63	43.00 ± 11.45	54.44 ± 6.74	32.78 ± 0.96	52.78 ± 10.84	37.00 ± 7.76	55.56 ± 4.19	31.00 ± 10.71	**63.33 ± 7.45**
Chinatown	95.74 ± 1.26	**96.73 ± 1.19**	**96.73 ± 0.81**	54.52 ± 24.75	91.08 ± 5.39	85.33 ± 8.50	86.10 ± 2.71	84.55 ± 3.36	80.70 ± 5.41	82.80 ± 4.37	45.01 ± 24.65	91.84 ± 4.85
ChlorineConcentration	34.61 ± 4.95	35.24 ± 5.79	35.19 ± 5.79	34.54 ± 2.93	34.28 ± 2.22	31.22 ± 6.21	33.24 ± 2.95	33.84 ± 2.69	33.91 ± 2.07	33.69 ± 2.59	**39.08 ± 9.47**	32.47 ± 3.03
CinCECGTorso	46.52 ± 12.75	48.35 ± 9.58	51.06 ± 10.06	39.77 ± 4.46	54.91 ± 12.73	48.14 ± 2.63	36.18 ± 3.15	53.94 ± 2.51	50.30 ± 12.26	48.89 ± 2.81	49.00 ± 5.40	**62.95 ± 6.66**
Coffee	**94.29 ± 7.41**	88.57 ± 12.73	**94.29 ± 3.19**	92.14 ± 3.91	90.71 ± 10.89	73.81 ± 8.99	63.10 ± 7.43	85.71 ± 3.57	84.29 ± 10.29	77.38 ± 4.12	65.71 ± 15.07	88.57 ± 4.66
Computers	52.24 ± 10.15	54.40 ± 10.81	52.72 ± 12.14	48.08 ± 7.70	53.28 ± 11.25	48.00 ± 10.51	48.67 ± 4.60	48.00 ± 11.45	50.40 ± 1.72	46.67 ± 11.22	50.16 ± 7.53	**57.73 ± 17.44**
CricketX	**36.51 ± 3.27**	34.10 ± 3.98	33.03 ± 3.86	20.62 ± 2.07	30.56 ± 3.87	27.35 ± 2.06	8.12 ± 1.41	26.15 ± 1.85	12.15 ± 1.24	27.44 ± 2.96	11.54 ± 1.86	35.98 ± 3.13
CricketY	**37.54 ± 5.48**	30.62 ± 2.69	33.28 ± 3.86	22.56 ± 4.44	24.21 ± 2.69	26.50 ± 7.27	9.06 ± 0.39	25.90 ± 6.29	12.67 ± 0.94	27.52 ± 6.80	10.00 ± 3.24	30.43 ± 5.23
CricketZ	**40.41 ± 4.15**	35.49 ± 2.72	33.69 ± 5.21	20.36 ± 1.13	24.92 ± 3.09	26.75 ± 2.72	8.12 ± 1.50	24.44 ± 0.78	12.97 ± 2.47	26.50 ± 2.74	11.18 ± 2.42	33.33 ± 3.02
Crop	31.31 ± 3.87	37.82 ± 2.09	31.31 ± 3.87	14.67 ± 0.98	24.66 ± 2.54	36.33 ± 2.63	35.06 ± 3.15	37.89 ± 1.28	35.83 ± 2.39	38.66 ± 2.99	4.63 ± 0.75	**43.94 ± 2.11**
DiatomSizeReduction	78.43 ± 9.98	84.44 ± 4.38	89.74 ± 8.40	83.73 ± 5.41	63.01 ± 13.25	84.42 ± 6.89	59.91 ± 2.78	84.97 ± 5.38	59.28 ± 18.44	82.35 ± 7.47	35.10 ± 12.79	**90.39 ± 5.62**
DistalPhalanxOutlineAgeGroup	66.91 ± 6.08	**67.77 ± 3.31**	65.76 ± 6.53	55.54 ± 12.17	58.42 ± 11.15	57.07 ± 19.57	50.84 ± 14.56	56.12 ± 13.32	56.26 ± 12.92	57.07 ± 14.47	38.99 ± 8.80	**67.77 ± 3.39**
DistalPhalanxOutlineCorrect	52.32 ± 12.15	51.59 ± 11.49	52.54 ± 12.44	49.57 ± 11.83	51.45 ± 15.36	**57.49 ± 13.09**	56.76 ± 11.56	55.80 ± 11.93	51.38 ± 12.34	56.64 ± 12.97	50.65 ± 9.31	55.80 ± 14.95
DistalPhalanxTW	49.50 ± 8.83	**55.54 ± 4.21**	54.10 ± 4.92	39.28 ± 8.61	53.96 ± 7.56	41.97 ± 11.56	42.45 ± 10.74	39.81 ± 18.49	47.05 ± 8.79	45.80 ± 16.17	22.45 ± 8.85	51.37 ± 8.97
DodgerLoopDay	28.75 ± 2.34	23.75 ± 2.50	28.00 ± 5.70	**44.00 ± 4.09**	38.50 ± 7.42	39.58 ± 4.02	23.75 ± 2.50	42.50 ± 7.60	19.75 ± 4.09	42.50 ± 3.31	38.50 ± 7.42	35.50 ± 8.78
DodgerLoopGame	54.93 ± 6.45	63.04 ± 8.02	58.41 ± 11.09	53.62 ± 8.51	65.65 ± 18.08	67.87 ± 7.26	48.55 ± 4.53	**71.50 ± 0.84**	50.14 ± 3.81	68.36 ± 3.42	65.65 ± 18.08	61.01 ± 13.03
DodgerLoopWeekend	92.75 ± 2.40	87.68 ± 5.59	92.03 ± 3.69	94.64 ± 2.74	96.09 ± 1.41	96.38 ± 1.26	64.98 ± 4.71	**97.34 ± 0.42**	65.22 ± 8.45	96.14 ± 0.84	96.09 ± 1.41	84.64 ± 2.73
ECG200	69.40 ± 7.77	75.20 ± 7.05	71.80 ± 5.54	76.80 ± 3.56	71.80 ± 4.38	**78.33 ± 1.53**	74.33 ± 5.69	75.67 ± 3.21	73.60 ± 3.65	75.33 ± 0.58	52.20 ± 18.20	73.00 ± 3.46
ECG5000	61.90 ± 13.09	69.05 ± 11.38	61.86 ± 10.65	**74.23 ± 10.96**	58.61 ± 14.82	72.42 ± 4.03	65.61 ± 11.76	69.40 ± 4.84	67.93 ± 11.62	66.36 ± 9.81	30.14 ± 19.40	67.68 ± 9.76
ECGFiveDays	68.78 ± 9.46	56.79 ± 4.91	71.73 ± 8.76	57.12 ± 2.23	67.29 ± 3.58	66.55 ± 6.20	74.64 ± 13.99	**75.18 ± 2.37**	68.76 ± 10.24	68.52 ± 6.54	57.79 ± 7.24	69.12 ± 9.82
EOGHorizontalSignal	20.66 ± 1.70	23.43 ± 5.48	21.38 ± 4.60	25.86 ± 4.54	28.07 ± 2.90	30.66 ± 2.91	12.06 ± 1.96	31.68 ± 2.61	9.56 ± 1.50	29.01 ± 2.49	29.83 ± 3.23	**37.35 ± 4.07**
EOGVerticalSignal	13.37 ± 1.32	16.57 ± 3.25	14.86 ± 2.77	26.80 ± 1.98	29.56 ± 4.11	26.98 ± 4.28	11.88 ± 1.54	26.34 ± 4.16	10.83 ± 1.45	26.61 ± 6.09	20.88 ± 2.94	**31.82 ± 3.47**
Earthquakes	52.95 ± 7.17	57.84 ± 11.11	49.35 ± 10.38	53.24 ± 4.93	**63.88 ± 6.42**	63.31 ± 10.45	49.16 ± 1.10	54.92 ± 17.17	56.55 ± 8.34	60.43 ± 21.88	57.12 ± 15.34	55.25 ± 14.33
ElectricDevices	37.63 ± 9.44	41.80 ± 2.08	34.70 ± 5.95	21.27 ± 5.73	37.69 ± 10.00	28.63 ± 3.21	22.56 ± 2.12	20.63 ± 2.87	22.76 ± 4.51	29.88 ± 3.14	14.36 ± 4.97	**47.14 ± 3.99**
EthanolLevel	25.32 ± 0.64	25.52 ± 1.56	25.12 ± 2.09	25.76 ± 1.71	26.96 ± 0.82	26.87 ± 2.61	**29.20 ± 1.91**	25.67 ± 3.18	25.24 ± 0.98	26.47 ± 2.08	25.12 ± 0.81	28.00 ± 3.11
FaceAll	**55.98 ± 6.19**	45.22 ± 8.18	**55.98 ± 6.19**	40.27 ± 5.76	24.12 ± 4.85	38.56 ± 3.11	38.84 ± 3.81	36.71 ± 2.55	38.89 ± 3.64	41.40 ± 3.79	11.35 ± 5.01	39.09 ± 1.28
FaceFour	77.73 ± 4.99	52.05 ± 10.37	81.82 ± 7.14	74.09 ± 5.98	59.32 ± 5.76	79.55 ± 9.02	57.58 ± 7.31	80.68 ± 3.41	62.50 ± 7.14	**84.85 ± 2.37**	40.45 ± 14.16	68.55 ± 4.91
FacesUCR	**67.20 ± 6.42**	51.90 ± 6.85	62.04 ± 8.47	41.67 ± 3.36	25.42 ± 6.93	44.55 ± 2.52	45.95 ± 4.27	49.07 ± 5.45	48.56 ± 3.79	49.27 ± 6.47	10.96 ± 5.26	55.08 ± 2.32
FiftyWords	19.03 ± 1.51	13.49 ± 1.95	19.03 ± 1.51	40.09 ± 3.22	28.35 ± 3.05	44.69 ± 2.97	5.05 ± 0.44	46.74 ± 2.43	16.53 ± 1.75	44.18 ± 3.70	15.16 ± 3.75	**47.30 ± 2.89**
Fish	**50.97 ± 10.77**	26.97 ± 9.88	50.51 ± 16.65	50.17 ± 6.36	31.54 ± 5.18	45.90 ± 8.88	20.00 ± 4.12	39.05 ± 5.64	22.97 ± 3.62	45.71 ± 10.47	22.86 ± 2.83	49.91 ± 10.25
FordA	**64.98 ± 8.59**	62.27 ± 9.82	62.82 ± 9.33	49.92 ± 1.84	60.48 ± 7.29	50.30 ± 1.31	50.66 ± 0.90	50.15 ± 0.79	50.27 ± 1.01	50.58 ± 1.43	50.76 ± 2.11	53.35 ± 2.84
FordB	**61.19 ± 3.67**	58.10 ± 1.40	58.67 ± 2.63	49.09 ± 1.35	56.89 ± 2.99	49.84 ± 0.63	50.99 ± 2.05	50.25 ± 1.29	49.95 ± 1.77	50.58 ± 0.68	50.77 ± 0.87	50.42 ± 2.72
FreezerRegularTrain	68.01 ± 9.57	67.73 ± 22.81	61.99 ± 17.00	67.56 ± 20.68	71.73 ± 5.42	59.79 ± 26.51	53.13 ± 0.84	60.39 ± 25.70	58.85 ± 2.14	60.74 ± 25.80	72.39 ± 6.35	**79.14 ± 7.94**
FreezerSmallTrain	66.47 ± 12.08	65.70 ± 14.41	66.00 ± 12.69	66.60 ± 13.71	64.67 ± 11.73	**71.47 ± 6.03**	56.34 ± 1.10	68.49 ± 7.01	56.88 ± 4.22	71.17 ± 6.80	65.57 ± 16.06	64.30 ± 13.28
Fungi	97.10 ± 2.45	34.84 ± 7.53	66.24 ± 2.06	68.17 ± 5.25	68.17 ± 13.75	**97.13 ± 0.82**	75.99 ± 1.64	82.62 ± 2.42	74.73 ± 0.85	96.95 ± 0.31	8.06 ± 3.93	81.97 ± 3.24
GestureMidAirD1	24.62 ± 1.44	29.08 ± 3.94	20.62 ± 1.44	38.31 ± 3.66	29.38 ± 2.74	37.69 ± 3.35	17.95 ± 2.70	36.67 ± 1.78	35.69 ± 2.53	36.92 ± 2.77	29.38 ± 2.74	**44.92 ± 6.31**
GestureMidAirD2	16.46 ± 2.22	21.85 ± 4.16	16.46 ± 2.22	35.69 ± 3.47	33.69 ± 3.66	42.05 ± 0.44	12.05 ± 1.60	44.36 ± 2.47	30.92 ± 4.26	**45.64 ± 3.20**	33.69 ± 3.66	43.85 ± 4.03
GestureMidAirD3	10.15 ± 1.99	11.85 ± 3.91	10.15 ± 1.99	25.08 ± 3.59	18.00 ± 2.64	26.41 ± 1.18	14.62 ± 4.07	**26.92 ± 1.54**	22.00 ± 3.63	**26.92 ± 2.77**	18.00 ± 2.64	21.08 ± 1.85

Continued on next page

Table 7: Per-dataset 2-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	TimeMorph
GesturePebbleZ1	62.44 ± 5.24	41.40 ± 5.40	**64.88 ± 2.42**	53.26 ± 4.18	29.53 ± 10.46	58.91 ± 3.41	16.86 ± 3.54	57.17 ± 0.89	24.19 ± 4.00	61.82 ± 3.41	29.53 ± 10.46	37.91 ± 6.35
GesturePebbleZ2	**66.58 ± 5.17**	44.05 ± 14.70	61.70 ± 6.13	51.90 ± 9.19	35.32 ± 9.94	55.06 ± 7.46	15.82 ± 0.63	46.84 ± 8.79	20.76 ± 3.15	56.75 ± 4.12	35.32 ± 9.94	41.39 ± 4.24
GunPoint	67.07 ± 8.15	65.20 ± 5.95	66.00 ± 5.89	60.13 ± 7.74	65.47 ± 7.68	55.33 ± 6.57	64.44 ± 5.18	58.44 ± 9.46	68.13 ± 5.80	57.11 ± 7.95	54.00 ± 6.53	**79.33 ± 16.25**
GunPointAgeSpan	65.38 ± 2.37	63.73 ± 4.38	**68.61 ± 1.37**	67.41 ± 13.19	61.08 ± 13.41	61.08 ± 8.21	52.22 ± 8.74	55.49 ± 10.60	52.41 ± 8.13	58.76 ± 12.22	47.41 ± 14.13	64.49 ± 7.90
GunPointMaleVersusFemale	79.30 ± 6.81	84.62 ± 8.74	81.65 ± 8.32	57.15 ± 7.06	65.95 ± 9.57	60.65 ± 4.86	50.11 ± 5.57	60.76 ± 10.27	52.59 ± 5.35	65.82 ± 3.12	65.00 ± 5.79	**87.66 ± 5.66**
GunPointOldVersusYoung	**91.94 ± 11.58**	89.97 ± 13.74	91.37 ± 12.08	72.89 ± 4.20	73.14 ± 5.24	73.54 ± 7.07	56.51 ± 2.29	72.49 ± 6.74	59.30 ± 3.02	75.87 ± 9.07	78.03 ± 17.26	70.92 ± 5.05
Ham	45.71 ± 9.69	50.86 ± 8.18	48.00 ± 10.27	54.86 ± 11.40	53.33 ± 7.28	54.92 ± 1.98	**58.73 ± 5.25**	50.48 ± 13.83	55.24 ± 4.62	57.78 ± 9.77	52.57 ± 10.93	50.86 ± 8.07
HandOutlines	57.19 ± 12.29	63.30 ± 2.05	60.65 ± 3.52	69.19 ± 14.14	63.57 ± 17.16	67.21 ± 16.82	52.16 ± 12.39	68.11 ± 20.52	41.57 ± 12.57	67.48 ± 15.69	**69.89 ± 13.52**	64.59 ± 10.14
Haptics	29.16 ± 3.65	24.35 ± 4.42	25.32 ± 2.75	28.31 ± 7.72	28.25 ± 7.08	**29.33 ± 7.26**	23.81 ± 0.68	29.22 ± 2.66	20.97 ± 3.07	29.00 ± 7.31	26.95 ± 8.77	26.62 ± 4.64
Herring	52.81 ± 11.18	**56.88 ± 7.13**	52.81 ± 11.23	51.25 ± 5.89	52.19 ± 12.13	47.92 ± 6.51	46.35 ± 3.61	45.83 ± 3.25	50.00 ± 4.69	48.96 ± 4.51	45.00 ± 9.84	55.00 ± 4.34
HouseTwenty	**76.30 ± 9.04**	69.24 ± 11.83	67.39 ± 13.31	60.00 ± 7.10	63.03 ± 13.56	68.91 ± 7.47	47.34 ± 6.31	59.94 ± 1.75	52.10 ± 8.21	71.15 ± 7.81	60.34 ± 11.20	71.93 ± 10.30
InlineSkate	18.58 ± 2.99	19.56 ± 1.81	**19.93 ± 2.98**	18.29 ± 2.68	17.05 ± 3.64	16.85 ± 1.00	15.76 ± 0.93	15.82 ± 1.62	16.29 ± 1.31	17.03 ± 0.46	16.36 ± 2.46	18.98 ± 2.54
InsectEPGRegularTrain	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	51.97 ± 7.58	58.39 ± 6.76	58.37 ± 4.44	37.08 ± 10.70	54.89 ± 2.35	37.99 ± 13.89	59.71 ± 4.91	81.69 ± 17.66	68.76 ± 9.60
InsectEPGSmallTrain	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	51.33 ± 4.40	58.47 ± 6.25	57.56 ± 4.71	41.37 ± 7.24	58.23 ± 4.08	42.65 ± 10.05	58.10 ± 4.04	80.40 ± 17.17	67.79 ± 2.45
InsectWingbeatSound	20.38 ± 3.08	16.62 ± 4.40	18.70 ± 3.47	47.87 ± 2.61	33.18 ± 5.98	50.44 ± 4.17	20.69 ± 3.83	49.41 ± 1.90	27.56 ± 2.55	**51.03 ± 4.83**	16.68 ± 7.51	31.49 ± 3.72
ItalyPowerDemand	68.88 ± 12.96	75.63 ± 13.47	73.61 ± 12.06	50.03 ± 0.16	79.18 ± 15.47	54.49 ± 7.77	75.87 ± 16.00	76.35 ± 22.01	**83.15 ± 15.39**	73.57 ± 21.06	50.07 ± 0.34	65.03 ± 13.70
LargeKitchenAppliances	45.65 ± 8.51	**54.19 ± 10.41**	50.93 ± 13.42	33.60 ± 7.89	44.85 ± 7.66	37.96 ± 12.60	32.00 ± 3.23	34.40 ± 7.77	32.96 ± 7.01	36.62 ± 11.05	32.00 ± 6.67	42.56 ± 5.00
Lightning2	57.05 ± 8.63	53.77 ± 7.90	59.34 ± 7.64	54.10 ± 10.50	51.80 ± 7.11	57.38 ± 4.34	49.73 ± 9.88	51.91 ± 4.13	52.13 ± 11.90	55.19 ± 6.21	50.16 ± 8.80	**63.28 ± 8.25**
Lightning7	45.48 ± 6.31	43.01 ± 5.94	42.74 ± 6.38	38.90 ± 8.91	47.95 ± 8.16	**59.36 ± 11.07**	19.63 ± 3.45	50.23 ± 12.43	27.40 ± 4.94	56.16 ± 10.34	20.82 ± 11.60	46.03 ± 6.17
Mallat	73.35 ± 15.31	47.53 ± 14.01	80.84 ± 4.13	86.79 ± 1.54	65.16 ± 19.15	**91.78 ± 0.99**	63.31 ± 2.66	78.79 ± 4.89	70.90 ± 5.59	87.65 ± 2.43	49.21 ± 6.11	80.67 ± 4.06
Meat	54.33 ± 7.42	42.00 ± 7.76	73.00 ± 16.39	**89.67 ± 7.85**	52.00 ± 12.16	53.33 ± 6.01	43.33 ± 8.33	41.11 ± 7.52	51.33 ± 18.61	57.22 ± 7.88	36.33 ± 5.82	57.67 ± 17.30
MedicalImages	33.45 ± 8.28	25.74 ± 5.92	33.61 ± 5.62	24.45 ± 6.71	27.45 ± 4.51	28.46 ± 3.05	27.85 ± 6.61	28.55 ± 2.80	**34.53 ± 1.25**	30.00 ± 1.71	29.82 ± 16.29	31.63 ± 5.59
MelbournePedestrian	**63.62 ± 4.54**	57.50 ± 8.49	63.00 ± 2.07	14.05 ± 5.17	42.42 ± 4.22	40.29 ± 3.90	51.77 ± 5.29	50.24 ± 2.74	54.32 ± 5.40	46.37 ± 2.70	42.42 ± 4.22	49.78 ± 4.68
MiddlePhalanxOutlineAgeGroup	47.01 ± 9.52	44.55 ± 12.64	45.45 ± 9.45	**50.39 ± 7.58**	44.29 ± 10.12	37.45 ± 10.11	46.75 ± 13.26	45.24 ± 16.76	48.57 ± 12.10	48.92 ± 14.26	22.86 ± 4.12	42.99 ± 12.95
MiddlePhalanxOutlineCorrect	47.01 ± 7.12	51.13 ± 7.83	49.42 ± 8.07	53.75 ± 12.47	44.54 ± 7.47	52.92 ± 7.99	47.54 ± 10.52	**54.07 ± 11.30**	48.59 ± 8.66	53.26 ± 15.07	50.72 ± 11.13	48.87 ± 3.56
MiddlePhalanxTW	37.92 ± 11.48	44.68 ± 3.99	42.73 ± 9.27	38.96 ± 14.59	39.22 ± 13.51	45.45 ± 5.95	43.07 ± 3.27	36.36 ± 13.61	**46.36 ± 2.32**	45.24 ± 3.58	16.36 ± 7.24	44.81 ± 7.78
MixedShapesRegularTrain	55.18 ± 4.21	33.76 ± 13.30	55.82 ± 8.68	58.38 ± 6.27	54.48 ± 12.34	62.60 ± 3.23	19.42 ± 0.39	63.00 ± 3.88	22.36 ± 6.85	63.64 ± 2.51	54.05 ± 4.44	**67.98 ± 4.06**
MixedShapesSmallTrain	45.73 ± 9.51	37.76 ± 12.10	51.77 ± 7.21	61.69 ± 3.51	54.28 ± 10.55	63.00 ± 2.46	19.05 ± 0.77	64.04 ± 1.32	24.01 ± 6.26	62.64 ± 3.39	57.53 ± 4.77	**68.35 ± 3.26**
MoteStrain	81.95 ± 4.07	75.34 ± 9.74	78.95 ± 3.25	70.97 ± 10.57	75.59 ± 8.13	78.41 ± 9.87	68.53 ± 3.31	82.48 ± 1.96	68.63 ± 10.07	**83.65 ± 4.68**	55.75 ± 11.79	58.56 ± 15.12
NonInvasiveFetalECGThorax1	25.55 ± 2.02	14.44 ± 3.90	25.55 ± 2.02	**55.96 ± 3.30**	18.44 ± 9.38	50.77 ± 4.84	18.37 ± 2.16	51.65 ± 2.21	21.42 ± 2.01	53.89 ± 1.99	13.55 ± 0.51	50.58 ± 2.13
NonInvasiveFetalECGThorax2	27.07 ± 5.56	22.30 ± 6.32	27.07 ± 5.56	**62.09 ± 2.37**	28.27 ± 8.71	55.69 ± 4.28	27.23 ± 2.03	56.05 ± 0.94	30.73 ± 3.23	58.63 ± 2.42	18.29 ± 1.75	59.72 ± 2.76
OSULeaf	55.70 ± 9.28	38.35 ± 6.31	**56.03 ± 5.89**	25.79 ± 2.77	25.29 ± 5.19	29.89 ± 3.34	18.18 ± 1.43	28.10 ± 2.07	21.90 ± 3.33	28.51 ± 4.37	23.31 ± 3.85	36.28 ± 6.97
OliveOil	64.67 ± 10.70	32.67 ± 5.48	71.33 ± 10.70	**80.67 ± 3.65**	17.33 ± 10.90	32.22 ± 11.71	38.89 ± 7.70	25.56 ± 12.62	24.00 ± 10.90	58.89 ± 22.69	22.00 ± 13.04	23.33 ± 9.72
PLAID	19.29 ± 3.99	16.65 ± 2.54	20.89 ± 4.30	22.98 ± 3.80	35.53 ± 7.89	27.56 ± 4.82	17.44 ± 2.97	25.88 ± 4.40	19.81 ± 3.24	28.68 ± 3.91	35.53 ± 7.89	**40.19 ± 5.06**
PhalangesOutlinesCorrect	52.91 ± 9.24	53.03 ± 10.79	52.61 ± 9.16	56.64 ± 11.15	54.13 ± 11.04	**60.64 ± 3.81**	58.70 ± 2.16	60.37 ± 4.20	52.26 ± 8.88	60.06 ± 3.62	49.04 ± 14.25	50.84 ± 2.93
Phoneme	**12.22 ± 1.88**	10.75 ± 2.12	11.22 ± 1.88	5.41 ± 0.40	7.31 ± 0.69	7.77 ± 0.84	4.13 ± 1.74	7.00 ± 0.45	6.19 ± 2.34	5.29 ± 0.61	5.20 ± 0.75	10.02 ± 0.59
PickupGestureWiimoteZ	43.20 ± 11.37	53.20 ± 9.55	**62.40 ± 1.67**	43.20 ± 6.72	40.40 ± 8.29	52.67 ± 10.07	21.33 ± 1.15	53.33 ± 4.16	27.60 ± 3.58	51.33 ± 6.43	40.40 ± 8.29	58.80 ± 3.03
PigAirwayPressure	28.37 ± 3.10	13.08 ± 1.90	**49.52 ± 0.76**	5.00 ± 0.26	3.65 ± 0.87	4.49 ± 0.28	3.21 ± 1.21	7.05 ± 0.73	4.62 ± 1.05	6.09 ± 0.73	5.10 ± 0.80	9.81 ± 0.43
PigArtPressure	50.58 ± 2.62	23.65 ± 3.56	**69.42 ± 3.86**	7.88 ± 0.26	5.29 ± 1.63	10.10 ± 0.48	8.01 ± 1.39	9.78 ± 1.11	7.12 ± 1.24	8.33 ± 0.56	19.13 ± 1.04	30.67 ± 1.38
PigCVP	31.54 ± 0.94	18.08 ± 2.27	**61.92 ± 1.78**	7.02 ± 0.43	3.94 ± 1.10	6.73 ± 0.00	4.01 ± 0.56	7.85 ± 1.00	4.13 ± 0.80	5.29 ± 0.83	10.10 ± 1.40	19.33 ± 0.99
Plane	**100.00 ± 0.00**	94.29 ± 3.75	99.05 ± 1.17	86.10 ± 6.69	91.62 ± 3.39	86.98 ± 7.02	84.44 ± 7.02	81.90 ± 1.90	87.24 ± 4.98	86.98 ± 6.34	21.71 ± 8.89	99.24 ± 0.80
PowerCons	82.11 ± 6.61	**83.78 ± 3.63**	82.33 ± 7.91	72.11 ± 2.98	70.78 ± 14.65	69.81 ± 4.10	52.22 ± 1.47	70.93 ± 7.46	50.67 ± 4.38	70.00 ± 8.73	65.89 ± 16.73	66.78 ± 15.52
ProximalPhalanxOutlineAgeGroup	**68.10 ± 13.07**	64.29 ± 28.01	64.78 ± 22.07	58.44 ± 28.63	68.00 ± 16.32	53.17 ± 31.21	56.26 ± 31.37	56.91 ± 37.90	59.22 ± 23.54	51.38 ± 35.67	33.95 ± 22.63	66.24 ± 17.89
ProximalPhalanxOutlineCorrect	**65.43 ± 5.55**	60.07 ± 16.64	63.85 ± 6.55	63.37 ± 8.02	62.96 ± 6.50	54.75 ± 14.29	54.30 ± 16.96	60.94 ± 6.65	59.86 ± 17.09	55.21 ± 16.86	44.88 ± 18.17	59.93 ± 12.82
ProximalPhalanxTW	50.83 ± 4.70	56.00 ± 9.92	**67.12 ± 8.12**	49.76 ± 11.43	53.37 ± 7.43	58.70 ± 5.35	53.98 ± 16.76	42.28 ± 24.98	56.00 ± 8.06	48.29 ± 19.01	16.98 ± 12.37	57.76 ± 7.59
RefrigerationDevices	**41.39 ± 9.49**	40.37 ± 10.57	41.23 ± 9.97	33.92 ± 0.58	35.31 ± 3.72	36.71 ± 1.61	33.87 ± 4.03	35.64 ± 1.78	33.81 ± 1.50	33.60 ± 3.03	33.07 ± 2.39	33.39 ± 7.13
Rock	32.00 ± 5.48	38.40 ± 8.65	37.80 ± 13.31	67.60 ± 10.53	44.80 ± 18.25	58.67 ± 13.61	**70.00 ± 11.14**	66.67 ± 8.08	47.60 ± 12.28	60.00 ± 14.00	44.40 ± 14.24	64.00 ± 7.07
ScreenType	36.91 ± 4.95	**40.21 ± 6.72**	38.40 ± 6.31	35.25 ± 3.07	36.75 ± 2.52	39.56 ± 4.47	33.78 ± 3.74	38.67 ± 2.63	33.71 ± 2.21	39.11 ± 4.89	33.81 ± 3.79	35.63 ± 4.40
SemgHandGenderCh2	55.13 ± 5.29	58.23 ± 11.97	54.27 ± 6.20	53.27 ± 13.91	**62.87 ± 8.65**	49.22 ± 10.20	55.56 ± 1.55	62.44 ± 3.53	43.87 ± 10.45	56.22 ± 13.04	53.70 ± 4.90	59.73 ± 10.87
SemgHandMovementCh2	33.51 ± 7.12	31.29 ± 5.18	**36.00 ± 6.56**	29.78 ± 3.16	23.56 ± 5.26	24.44 ± 2.52	18.89 ± 0.59	24.44 ± 1.39	18.71 ± 1.13	24.07 ± 5.35	27.91 ± 3.59	31.16 ± 4.70

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Table 7: Per-dataset 2-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	TimeMorph
SemgHandSubjectCh2	32.00 ± 3.88	30.71 ± 4.13	33.56 ± 5.04	**45.91 ± 5.66**	40.67 ± 4.39	29.56 ± 4.02	23.56 ± 2.47	32.52 ± 7.62	24.09 ± 3.07	30.81 ± 0.84	29.47 ± 4.28	43.20 ± 6.72
ShakeGestureWiimoteZ	76.00 ± 6.63	76.80 ± 7.43	**81.60 ± 0.89**	47.20 ± 7.95	58.40 ± 5.18	54.00 ± 6.00	26.00 ± 4.00	54.67 ± 4.62	32.40 ± 2.61	56.00 ± 5.29	58.40 ± 5.18	**81.60 ± 2.61**
ShapeletSim	78.89 ± 14.16	**98.33 ± 2.08**	92.11 ± 5.26	50.11 ± 2.87	80.22 ± 16.23	47.59 ± 3.16	50.74 ± 3.06	48.15 ± 1.79	49.22 ± 2.93	46.48 ± 3.06	49.67 ± 1.50	67.44 ± 10.92
ShapesAll	36.80 ± 3.60	28.17 ± 3.75	36.80 ± 3.60	50.73 ± 1.71	32.23 ± 5.06	55.78 ± 1.50	6.11 ± 0.63	53.17 ± 1.00	18.33 ± 5.95	53.33 ± 2.62	27.63 ± 3.10	**63.03 ± 2.22**
SmallKitchenAppliances	48.85 ± 7.91	50.40 ± 8.87	50.19 ± 7.71	41.49 ± 3.33	47.15 ± 6.04	49.60 ± 6.54	35.20 ± 2.93	40.09 ± 2.14	42.03 ± 4.49	44.09 ± 8.36	34.61 ± 4.49	**55.68 ± 8.74**
SmoothSubspace	78.93 ± 10.17	77.87 ± 5.55	80.13 ± 5.80	33.33 ± 0.00	51.20 ± 5.65	58.22 ± 4.73	68.89 ± 7.31	59.11 ± 4.23	70.67 ± 4.88	81.33 ± 6.36	48.27 ± 9.05	**85.07 ± 5.20**
SonyAIBORobotSurface1	90.42 ± 4.34	**94.08 ± 1.77**	89.38 ± 5.35	64.83 ± 12.29	70.05 ± 3.77	62.90 ± 13.91	61.90 ± 7.89	66.61 ± 10.89	59.10 ± 5.18	62.73 ± 9.45	52.65 ± 8.42	71.85 ± 8.28
SonyAIBORobotSurface2	**88.00 ± 2.82**	84.87 ± 6.58	83.67 ± 9.41	74.59 ± 3.28	78.70 ± 2.68	69.60 ± 7.04	77.26 ± 1.79	78.49 ± 0.86	78.30 ± 1.49	79.29 ± 1.17	56.81 ± 17.39	68.21 ± 5.28
StarLightCurves	**76.49 ± 11.97**	66.10 ± 9.91	66.22 ± 10.70	67.92 ± 13.52	71.57 ± 12.97	63.90 ± 14.70	35.37 ± 11.93	65.31 ± 14.87	34.69 ± 14.98	63.78 ± 14.15	68.45 ± 12.08	74.47 ± 15.08
Strawberry	62.81 ± 6.83	57.14 ± 6.89	60.22 ± 9.39	51.84 ± 8.28	56.59 ± 2.41	53.33 ± 7.10	**66.58 ± 2.89**	52.97 ± 8.18	59.62 ± 11.05	54.59 ± 4.47	51.03 ± 11.13	59.41 ± 6.38
SwedishLeaf	**72.45 ± 3.37**	54.30 ± 6.89	**72.45 ± 3.37**	34.02 ± 3.00	43.26 ± 7.26	48.91 ± 2.97	33.28 ± 3.94	46.24 ± 1.25	38.85 ± 4.16	47.52 ± 2.78	7.94 ± 0.74	64.99 ± 3.83
Symbols	**90.49 ± 1.42**	75.88 ± 7.72	88.48 ± 4.41	81.23 ± 0.99	76.18 ± 6.87	81.37 ± 1.84	37.42 ± 3.12	82.04 ± 0.58	58.95 ± 6.86	80.00 ± 1.74	51.50 ± 7.93	82.69 ± 3.43
SyntheticControl	79.67 ± 4.05	85.07 ± 4.63	82.60 ± 6.01	41.40 ± 6.37	83.80 ± 5.24	55.00 ± 3.38	44.11 ± 1.39	59.89 ± 3.15	47.93 ± 4.58	62.89 ± 3.15	20.80 ± 4.95	**87.93 ± 3.95**
ToeSegmentation1	**68.42 ± 7.02**	60.79 ± 8.14	61.58 ± 7.54	53.95 ± 2.19	58.60 ± 10.97	51.90 ± 4.30	53.80 ± 2.21	53.95 ± 0.44	52.37 ± 4.44	51.61 ± 3.08	50.70 ± 2.53	60.26 ± 8.54
ToeSegmentation2	76.15 ± 9.44	**79.69 ± 9.24**	72.15 ± 7.68	48.92 ± 5.64	68.62 ± 12.68	66.41 ± 15.71	51.54 ± 5.38	63.08 ± 15.78	47.08 ± 8.33	63.59 ± 15.79	54.31 ± 14.96	76.15 ± 12.44
Trace	76.00 ± 3.00	76.40 ± 3.05	73.80 ± 2.95	52.40 ± 4.04	76.60 ± 4.39	47.33 ± 3.21	51.00 ± 4.00	49.00 ± 3.00	55.60 ± 4.62	49.00 ± 5.00	46.00 ± 6.78	**98.40 ± 3.05**
TwoLeadECG	**82.53 ± 6.47**	79.19 ± 7.45	70.63 ± 4.92	55.68 ± 3.84	59.95 ± 3.74	62.36 ± 5.86	56.60 ± 1.27	55.90 ± 1.72	55.49 ± 2.80	60.14 ± 7.03	48.48 ± 5.21	74.63 ± 8.09
TwoPatterns	52.71 ± 4.44	71.75 ± 5.17	65.30 ± 11.45	32.62 ± 1.84	44.61 ± 9.45	31.66 ± 1.59	24.88 ± 1.30	28.57 ± 1.14	25.14 ± 0.61	29.88 ± 2.34	27.30 ± 2.15	**78.23 ± 0.41**
UMD	54.86 ± 7.95	71.67 ± 9.36	68.75 ± 16.74	54.17 ± 4.55	62.50 ± 12.00	52.31 ± 6.22	47.69 ± 9.93	60.19 ± 1.45	53.06 ± 4.83	56.25 ± 6.05	47.50 ± 4.16	**78.86 ± 14.11**
UWaveGestureLibraryAll	32.50 ± 3.38	28.30 ± 5.01	31.51 ± 4.14	**69.44 ± 6.55**	43.65 ± 6.39	66.54 ± 6.96	26.72 ± 0.50	66.58 ± 7.48	30.82 ± 2.10	66.61 ± 7.59	54.22 ± 7.86	42.54 ± 0.28
UWaveGestureLibraryX	31.94 ± 2.72	28.97 ± 3.44	30.99 ± 3.77	50.42 ± 3.39	42.01 ± 2.19	**51.69 ± 1.72**	21.35 ± 1.20	51.05 ± 1.76	28.72 ± 2.91	50.93 ± 1.76	22.53 ± 5.80	42.54 ± 0.28
UWaveGestureLibraryY	28.00 ± 4.17	26.42 ± 2.97	27.96 ± 5.26	43.36 ± 3.42	39.43 ± 5.20	42.54 ± 7.86	19.88 ± 2.66	**43.61 ± 6.37**	28.02 ± 2.78	41.75 ± 7.40	28.22 ± 7.42	42.54 ± 0.28
UWaveGestureLibraryZ	31.19 ± 2.60	30.19 ± 2.10	28.42 ± 4.12	44.60 ± 3.60	41.59 ± 3.78	42.17 ± 5.87	23.93 ± 2.40	43.23 ± 7.30	29.59 ± 1.71	40.53 ± 6.44	23.38 ± 7.02	**59.88 ± 2.03**
Wafer	47.08 ± 18.17	57.95 ± 22.06	52.16 ± 16.40	56.77 ± 13.38	49.59 ± 17.01	67.22 ± 1.00	55.02 ± 6.93	66.32 ± 1.28	56.21 ± 9.73	**67.64 ± 1.29**	54.99 ± 20.43	55.01 ± 15.62
Wine	**60.74 ± 8.43**	50.74 ± 7.92	60.48 ± 12.52	53.70 ± 8.88	47.41 ± 11.54	53.70 ± 3.21	53.09 ± 5.35	50.00 ± 0.00	50.74 ± 1.66	54.94 ± 26.28	50.74 ± 1.66	48.89 ± 5.34
WordSynonyms	20.63 ± 4.03	16.05 ± 1.67	20.63 ± 4.03	21.35 ± 4.96	20.06 ± 3.25	29.05 ± 0.77	6.11 ± 1.90	29.73 ± 1.73	13.92 ± 2.48	29.57 ± 1.41	6.43 ± 4.20	**34.92 ± 2.64**
Worms	**54.03 ± 11.78**	44.94 ± 11.42	52.73 ± 7.32	22.60 ± 2.99	28.57 ± 6.43	19.48 ± 1.30	17.32 ± 5.41	17.75 ± 5.25	15.58 ± 4.00	20.35 ± 3.75	18.70 ± 5.00	38.70 ± 4.63
WormsTwoClass	55.32 ± 9.03	52.47 ± 8.50	**55.84 ± 3.31**	48.57 ± 2.69	49.09 ± 4.63	49.78 ± 2.70	51.52 ± 8.84	47.19 ± 3.97	47.27 ± 4.37	46.32 ± 1.50	48.83 ± 7.82	54.03 ± 7.72
Yoga	**52.99 ± 4.56**	50.68 ± 2.33	50.48 ± 3.16	50.86 ± 0.91	51.49 ± 3.75	50.12 ± 2.91	51.48 ± 2.93	50.70 ± 2.90	51.46 ± 3.58	50.58 ± 2.12	51.69 ± 4.15	50.64 ± 4.90
Avg. Acc.	53.98	50.79	54.95	48.01	48.43	49.95	39.45	49.45	41.71	51.14	38.30	**55.55**
Avg. Rank	4.84	5.63	4.83	6.78	6.42	6.13	9.36	6.29	8.63	5.64	9.17	**4.28**
#Best	**33**	13	19	10	2	9	4	6	3	7	2	32

Table 8: Per-dataset 5-shot classification results on 128 shared UCR datasets. Each value is reported as mean ± standard deviation over 5 runs.

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	TimeMorph
ACSF1	**76.00 ± 2.92**	64.60 ± 5.18	72.80 ± 1.64	36.20 ± 3.49	50.40 ± 2.61	60.00 ± 2.00	40.00 ± 1.00	62.67 ± 2.52	46.80 ± 3.70	58.67 ± 1.53	30.60 ± 4.83	68.33 ± 2.89
Adiac	56.21 ± 4.18	42.81 ± 4.46	56.21 ± 4.18	35.29 ± 1.32	34.58 ± 2.33	43.82 ± 1.54	46.97 ± 3.07	38.62 ± 1.68	64.50 ± 5.47	43.99 ± 3.86	7.11 ± 1.15	**69.51 ± 2.42**
AllGestureWiimoteX	**40.46 ± 4.50**	34.14 ± 3.58	38.71 ± 5.08	23.63 ± 2.19	35.40 ± 4.31	32.24 ± 3.43	13.57 ± 1.57	29.95 ± 4.86	18.17 ± 3.28	29.38 ± 3.89	35.40 ± 4.31	39.40 ± 2.86
AllGestureWiimoteY	43.20 ± 1.70	40.60 ± 1.51	36.20 ± 3.42	27.89 ± 2.07	40.97 ± 3.06	36.76 ± 2.07	15.38 ± 0.81	34.43 ± 2.02	19.49 ± 2.37	35.43 ± 1.55	40.97 ± 3.06	**48.11 ± 2.89**
AllGestureWiimoteZ	**41.46 ± 4.33**	35.71 ± 5.12	36.46 ± 5.30	20.20 ± 1.22	27.11 ± 2.51	26.57 ± 0.25	13.86 ± 1.71	26.67 ± 2.65	17.37 ± 2.23	26.86 ± 0.99	27.11 ± 2.51	36.14 ± 0.97
ArrowHead	66.40 ± 5.44	61.03 ± 5.46	64.23 ± 7.62	59.43 ± 5.27	58.86 ± 12.66	60.19 ± 7.46	42.67 ± 0.87	60.76 ± 1.19	48.11 ± 2.87	63.43 ± 8.42	54.06 ± 9.79	**73.60 ± 3.71**
BME	55.60 ± 7.80	56.93 ± 3.90	68.00 ± 9.90	58.53 ± 7.20	89.60 ± 10.65	58.89 ± 5.55	73.33 ± 8.11	76.89 ± 7.34	80.00 ± 6.70	73.33 ± 5.93	53.07 ± 5.63	**97.73 ± 1.53**
Beef	54.67 ± 4.47	47.33 ± 2.79	55.33 ± 2.98	52.67 ± 7.60	43.33 ± 4.71	64.44 ± 3.85	50.00 ± 3.33	52.22 ± 8.39	62.00 ± 8.03	**84.44 ± 1.92**	35.33 ± 4.47	51.33 ± 6.91
BeetleFly	72.00 ± 7.58	60.00 ± 11.73	79.00 ± 10.84	71.00 ± 14.32	57.00 ± 8.37	70.00 ± 8.66	51.67 ± 2.89	66.67 ± 16.07	49.00 ± 6.52	70.00 ± 10.00	59.00 ± 18.51	**91.67 ± 2.89**
BirdChicken	69.00 ± 12.94	77.00 ± 9.75	**90.00 ± 3.54**	57.00 ± 5.70	61.00 ± 16.73	63.33 ± 7.64	36.67 ± 2.89	65.00 ± 10.00	54.00 ± 8.94	68.33 ± 15.28	54.00 ± 4.18	85.00 ± 11.73
CBF	95.87 ± 2.59	95.58 ± 3.67	**95.89 ± 3.27**	61.47 ± 7.78	95.67 ± 4.48	74.52 ± 5.66	39.04 ± 0.80	80.11 ± 4.97	42.29 ± 1.09	80.59 ± 4.09	41.20 ± 4.62	93.76 ± 3.90
Car	59.33 ± 9.02	51.00 ± 7.96	65.67 ± 5.73	52.33 ± 6.41	45.67 ± 2.79	68.89 ± 6.31	31.67 ± 5.00	48.89 ± 4.19	41.00 ± 12.56	57.78 ± 5.85	36.00 ± 4.50	**69.67 ± 3.98**
Chinatown	96.15 ± 2.16	96.97 ± 0.49	**97.43 ± 0.38**	50.96 ± 33.10	96.03 ± 2.24	84.94 ± 15.12	91.06 ± 1.94	93.00 ± 5.85	89.56 ± 2.00	87.95 ± 8.75	73.00 ± 0.76	93.46 ± 10.56
ChlorineConcentration	35.97 ± 2.09	35.63 ± 3.63	36.15 ± 2.53	35.57 ± 3.02	35.80 ± 3.16	35.02 ± 2.94	36.15 ± 1.24	35.49 ± 4.56	37.02 ± 1.05	35.96 ± 1.50	38.60 ± 5.65	**39.58 ± 3.20**
CinCECGTorso	49.41 ± 5.43	57.77 ± 5.88	56.17 ± 3.91	43.81 ± 1.84	69.26 ± 4.67	63.41 ± 2.73	37.80 ± 1.42	64.95 ± 2.40	64.83 ± 11.83	68.41 ± 5.18	54.58 ± 3.46	**82.56 ± 7.22**
Coffee	86.43 ± 14.81	97.86 ± 1.96	**100.00 ± 0.00**	92.86 ± 3.57	97.86 ± 1.96	86.90 ± 2.06	76.19 ± 2.06	82.14 ± 9.45	92.86 ± 10.41	85.71 ± 0.00	50.00 ± 6.19	94.29 ± 1.96
Computers	58.80 ± 7.81	61.60 ± 5.76	60.96 ± 6.58	46.64 ± 4.37	54.48 ± 7.54	49.87 ± 5.33	50.53 ± 1.29	48.00 ± 8.43	49.04 ± 4.09	48.67 ± 4.31	46.56 ± 5.23	**62.67 ± 4.74**
CricketX	45.90 ± 4.41	43.08 ± 3.97	43.08 ± 2.15	25.23 ± 1.80	36.10 ± 5.19	34.87 ± 4.89	11.79 ± 1.43	30.77 ± 0.51	16.92 ± 1.32	33.68 ± 3.44	11.33 ± 1.41	**49.06 ± 3.21**
CricketY	42.67 ± 4.49	43.08 ± 3.40	43.28 ± 6.01	30.41 ± 3.78	36.92 ± 6.92	37.86 ± 2.42	9.06 ± 1.21	36.58 ± 2.06	15.23 ± 1.85	37.26 ± 2.44	12.05 ± 2.18	**49.66 ± 1.41**
CricketZ	51.59 ± 3.02	49.64 ± 3.11	47.79 ± 6.26	26.31 ± 2.67	36.31 ± 4.35	37.18 ± 1.18	10.77 ± 0.51	34.87 ± 1.43	15.44 ± 1.54	37.18 ± 0.77	10.92 ± 1.59	**53.68 ± 2.06**
Crop	49.93 ± 1.45	46.93 ± 2.27	49.93 ± 1.45	31.91 ± 2.21	38.13 ± 2.76	46.92 ± 2.69	44.47 ± 0.97	47.82 ± 2.55	47.76 ± 1.51	48.69 ± 2.69	8.15 ± 3.22	**52.18 ± 2.24**
DiatomSizeReduction	85.75 ± 6.98	70.26 ± 16.84	**95.95 ± 1.26**	61.57 ± 2.74	65.49 ± 10.17	87.69 ± 0.50	68.63 ± 2.55	66.34 ± 8.52	73.53 ± 9.37	87.25 ± 0.00	41.83 ± 13.83	93.66 ± 1.97
DistalPhalanxOutlineAgeGroup	65.76 ± 4.54	68.63 ± 3.96	67.19 ± 5.75	58.99 ± 14.81	63.31 ± 4.46	64.75 ± 3.74	67.63 ± 2.16	69.30 ± 3.24	68.20 ± 2.35	68.59 ± 2.31	53.67 ± 12.57	**69.35 ± 1.20**
DistalPhalanxOutlineCorrect	66.81 ± 3.92	61.96 ± 7.85	**68.62 ± 6.57**	56.23 ± 10.81	60.07 ± 6.07	62.44 ± 3.65	62.68 ± 3.32	58.94 ± 3.37	63.41 ± 7.40	59.30 ± 3.08	55.87 ± 8.58	63.77 ± 11.63
DistalPhalanxTW	58.99 ± 2.69	57.12 ± 1.40	58.56 ± 3.40	49.64 ± 7.01	**60.58 ± 2.61**	57.31 ± 0.42	51.32 ± 3.62	59.47 ± 4.40	54.10 ± 4.24	56.59 ± 5.05	31.94 ± 15.91	56.40 ± 4.87
DodgerLoopDay	37.75 ± 3.35	40.50 ± 4.11	37.00 ± 3.26	46.75 ± 4.01	44.25 ± 2.09	**52.50 ± 5.45**	22.08 ± 0.72	45.42 ± 2.60	24.25 ± 3.49	50.83 ± 4.73	44.25 ± 2.09	45.75 ± 3.26
DodgerLoopGame	66.52 ± 4.02	82.32 ± 3.38	72.17 ± 3.64	76.09 ± 5.15	**84.49 ± 1.89**	75.36 ± 5.02	50.00 ± 2.61	78.99 ± 7.63	53.33 ± 2.09	77.29 ± 4.82	**84.49 ± 1.89**	72.61 ± 10.25
DodgerLoopWeekend	95.51 ± 1.65	92.46 ± 3.46	94.78 ± 1.88	97.39 ± 0.40	97.25 ± 0.61	96.62 ± 0.84	66.43 ± 4.25	**97.83 ± 0.00**	72.90 ± 1.82	96.86 ± 0.42	97.25 ± 0.61	93.62 ± 2.37
ECG200	76.40 ± 3.58	77.80 ± 6.26	76.20 ± 4.60	75.00 ± 1.58	62.80 ± 16.36	77.67 ± 3.06	76.67 ± 3.79	78.00 ± 2.00	77.60 ± 3.21	**79.33 ± 4.93**	69.60 ± 4.51	67.40 ± 8.91
ECG5000	78.85 ± 9.03	75.51 ± 9.29	75.56 ± 6.88	**84.88 ± 3.51**	71.69 ± 7.99	82.01 ± 5.12	76.33 ± 7.90	83.48 ± 5.73	74.58 ± 8.80	82.23 ± 5.84	18.64 ± 16.84	73.89 ± 8.21
ECGFiveDays	71.54 ± 13.30	70.27 ± 8.41	70.22 ± 9.52	69.38 ± 6.83	77.54 ± 3.65	72.94 ± 9.66	**87.65 ± 8.42**	75.22 ± 2.60	83.30 ± 11.51	75.61 ± 4.26	60.49 ± 6.59	70.96 ± 3.91
EOGHorizontalSignal	24.97 ± 1.88	32.71 ± 1.62	27.35 ± 3.43	29.56 ± 3.95	37.73 ± 6.23	36.92 ± 1.62	10.96 ± 1.15	38.40 ± 2.19	9.94 ± 1.90	35.17 ± 4.76	31.99 ± 1.89	**46.35 ± 2.27**
EOGVerticalSignal	18.18 ± 3.62	22.10 ± 3.22	18.78 ± 4.22	29.01 ± 2.45	**38.23 ± 4.43**	31.77 ± 5.00	8.47 ± 1.80	31.95 ± 3.43	10.39 ± 1.59	31.22 ± 4.29	22.82 ± 2.72	36.74 ± 2.24
Earthquakes	50.94 ± 12.41	50.22 ± 8.85	52.23 ± 11.61	**56.83 ± 2.92**	52.09 ± 5.05	56.35 ± 11.87	51.32 ± 9.11	56.12 ± 3.14	53.38 ± 4.56	54.68 ± 7.61	52.81 ± 9.99	54.68 ± 9.31
ElectricDevices	36.69 ± 5.52	36.63 ± 6.97	36.08 ± 5.74	23.65 ± 1.86	38.77 ± 8.33	38.73 ± 7.16	22.89 ± 3.05	25.09 ± 5.42	31.29 ± 6.08	38.61 ± 5.70	11.47 ± 4.02	**47.06 ± 4.18**
EthanolLevel	27.52 ± 1.92	26.36 ± 0.86	26.60 ± 1.66	27.88 ± 1.25	24.88 ± 0.18	28.40 ± 1.73	28.60 ± 2.78	26.53 ± 0.12	25.40 ± 0.75	**29.53 ± 0.64**	26.32 ± 1.07	25.92 ± 1.40
FaceAll	67.63 ± 4.66	**67.75 ± 7.28**	67.63 ± 4.66	53.96 ± 2.14	41.22 ± 3.29	55.76 ± 2.36	53.79 ± 5.19	55.44 ± 2.51	51.51 ± 5.08	54.30 ± 2.41	17.64 ± 1.73	54.54 ± 1.26
FaceFour	88.18 ± 2.85	61.82 ± 13.74	**89.77 ± 4.62**	78.86 ± 3.17	57.50 ± 8.82	86.36 ± 1.14	64.77 ± 2.27	87.88 ± 1.74	71.82 ± 1.24	85.61 ± 1.74	41.14 ± 6.88	75.68 ± 4.37
FacesUCR	**78.67 ± 2.69**	73.00 ± 4.70	76.25 ± 3.16	50.94 ± 2.11	43.00 ± 7.53	68.33 ± 3.31	58.46 ± 2.45	68.99 ± 1.59	60.62 ± 1.49	67.61 ± 2.78	27.20 ± 3.34	68.58 ± 2.57
FiftyWords	30.07 ± 3.14	27.91 ± 3.50	30.07 ± 3.14	47.03 ± 2.26	37.71 ± 3.50	52.23 ± 1.46	9.45 ± 3.91	56.19 ± 3.63	25.10 ± 2.19	49.38 ± 1.41	40.84 ± 2.35	**61.58 ± 2.55**
Fish	61.26 ± 15.63	48.69 ± 6.91	67.66 ± 6.55	56.80 ± 6.22	42.97 ± 5.32	**68.76 ± 3.49**	25.52 ± 2.93	56.57 ± 5.51	40.80 ± 3.78	67.05 ± 1.44	26.40 ± 8.24	64.57 ± 5.92
FordA	71.41 ± 10.22	**71.94 ± 9.09**	70.55 ± 10.29	50.33 ± 1.09	63.62 ± 6.82	50.28 ± 2.11	51.67 ± 1.61	49.67 ± 1.72	51.11 ± 2.04	49.82 ± 1.42	50.91 ± 1.25	54.00 ± 2.22
FordB	**61.26 ± 2.75**	58.96 ± 1.47	59.04 ± 3.16	50.00 ± 1.41	55.88 ± 4.39	50.86 ± 0.77	49.51 ± 0.81	48.81 ± 1.26	49.56 ± 0.80	50.21 ± 1.36	50.27 ± 2.02	52.32 ± 2.78
FreezerRegularTrain	71.93 ± 7.16	**86.24 ± 2.60**	79.17 ± 4.44	76.42 ± 0.69	77.88 ± 9.36	76.43 ± 0.19	57.27 ± 4.31	76.76 ± 0.69	62.60 ± 6.21	71.04 ± 6.76	59.41 ± 11.73	85.11 ± 3.83
FreezerSmallTrain	**77.10 ± 3.40**	75.64 ± 1.31	75.24 ± 0.61	75.24 ± 1.24	71.21 ± 6.86	75.59 ± 1.06	60.63 ± 1.62	76.02 ± 0.68	66.32 ± 5.59	73.06 ± 2.85	63.24 ± 10.73	75.05 ± 3.10
Fungi	96.34 ± 4.07	34.19 ± 4.54	66.45 ± 5.80	74.09 ± 4.22	74.73 ± 9.52	**97.13 ± 0.31**	75.63 ± 1.35	90.14 ± 2.24	75.27 ± 0.54	96.06 ± 0.82	10.75 ± 6.51	81.18 ± 3.16
GestureMidAirD1	37.54 ± 4.63	52.92 ± 3.41	37.54 ± 4.63	46.00 ± 0.64	43.38 ± 3.51	53.85 ± 0.77	21.03 ± 3.11	52.05 ± 3.11	40.46 ± 4.85	52.82 ± 1.94	43.38 ± 3.51	**60.31 ± 2.35**
GestureMidAirD2	28.15 ± 3.51	42.15 ± 7.00	28.15 ± 3.51	48.31 ± 1.58	46.46 ± 3.63	50.00 ± 1.54	12.82 ± 3.55	49.49 ± 1.60	33.08 ± 9.04	50.00 ± 0.77	46.46 ± 3.63	**54.62 ± 1.96**
GestureMidAirD3	14.00 ± 2.69	19.85 ± 2.33	14.00 ± 2.69	30.92 ± 2.20	26.92 ± 3.31	25.13 ± 1.18	14.10 ± 3.11	31.79 ± 1.60	26.31 ± 4.82	27.18 ± 1.78	26.92 ± 3.31	**32.15 ± 3.33**

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Table 8: Per-dataset 5-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	TimeMorph
GesturePebbleZ1	71.74 ± 3.41	64.53 ± 3.05	73.60 ± 2.12	65.35 ± 3.15	63.95 ± 8.94	65.89 ± 3.50	23.45 ± 1.21	73.64 ± 2.20	28.95 ± 4.06	**76.16 ± 6.07**	63.95 ± 8.94	63.84 ± 8.99
GesturePebbleZ2	67.34 ± 3.97	52.53 ± 17.40	**73.04 ± 4.07**	58.35 ± 6.18	48.10 ± 7.12	62.03 ± 7.30	19.41 ± 4.45	64.98 ± 9.05	24.81 ± 3.02	60.34 ± 8.06	48.10 ± 7.12	60.89 ± 8.06
GunPoint	82.53 ± 12.46	77.33 ± 12.19	79.87 ± 13.86	59.60 ± 4.87	73.07 ± 2.03	66.00 ± 6.36	55.11 ± 6.19	70.22 ± 2.52	65.87 ± 7.58	66.00 ± 2.91	62.40 ± 5.79	**90.13 ± 10.49**
GunPointAgeSpan	75.70 ± 7.16	69.81 ± 6.17	76.46 ± 8.05	75.51 ± 2.81	73.67 ± 12.74	76.79 ± 5.43	49.26 ± 6.34	75.74 ± 4.37	60.38 ± 3.16	69.20 ± 7.84	65.70 ± 10.14	**81.52 ± 9.45**
GunPointMaleVersusFemale	91.20 ± 8.64	**96.65 ± 4.17**	90.32 ± 9.07	61.14 ± 7.95	87.53 ± 12.91	74.37 ± 13.69	55.17 ± 0.48	75.00 ± 13.22	61.20 ± 4.73	77.95 ± 14.78	63.92 ± 13.57	94.62 ± 6.24
GunPointOldVersusYoung	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	75.87 ± 2.49	70.29 ± 8.33	70.79 ± 4.48	56.93 ± 3.97	73.23 ± 4.31	58.98 ± 5.43	70.48 ± 8.59	92.38 ± 8.67	80.44 ± 8.63
Ham	60.95 ± 6.21	60.00 ± 6.77	62.10 ± 4.59	62.86 ± 11.45	58.86 ± 7.48	60.00 ± 11.70	63.49 ± 0.55	56.51 ± 8.59	63.62 ± 3.26	**66.35 ± 16.28**	56.76 ± 9.56	60.57 ± 9.91
HandOutlines	46.86 ± 11.04	52.86 ± 13.07	67.24 ± 3.85	80.76 ± 4.40	74.49 ± 7.28	**83.06 ± 3.92**	47.84 ± 8.25	81.62 ± 4.95	35.95 ± 0.00	82.34 ± 4.91	75.57 ± 5.75	68.54 ± 6.21
Haptics	32.14 ± 2.79	28.25 ± 2.44	31.49 ± 1.92	35.39 ± 2.49	28.18 ± 3.44	36.58 ± 0.19	21.32 ± 3.65	**37.12 ± 1.14**	24.68 ± 3.46	**37.12 ± 0.94**	31.82 ± 7.39	26.95 ± 4.88
Herring	50.94 ± 5.70	55.00 ± 8.93	52.50 ± 7.46	51.88 ± 6.76	50.94 ± 6.68	48.96 ± 5.02	**57.29 ± 4.51**	51.56 ± 7.16	46.88 ± 13.58	49.48 ± 2.39	49.06 ± 7.13	45.00 ± 4.87
HouseTwenty	83.70 ± 6.31	81.18 ± 9.08	81.85 ± 9.95	58.99 ± 5.13	76.81 ± 7.55	73.67 ± 7.34	54.90 ± 2.57	66.95 ± 4.78	59.66 ± 6.85	72.83 ± 4.23	63.03 ± 4.28	**84.37 ± 7.51**
InlineSkate	20.15 ± 4.70	22.95 ± 2.93	19.75 ± 2.66	21.05 ± 2.12	18.98 ± 2.35	24.85 ± 1.07	16.00 ± 1.31	24.06 ± 0.93	16.51 ± 0.15	24.06 ± 0.90	19.20 ± 1.41	**28.07 ± 1.19**
InsectEPGRegularTrain	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	56.47 ± 1.83	78.96 ± 9.81	59.57 ± 2.67	38.96 ± 2.13	57.30 ± 1.41	53.17 ± 6.32	61.31 ± 3.42	89.88 ± 9.24	80.24 ± 3.80
InsectEPGSmallTrain	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	53.98 ± 1.90	77.03 ± 8.38	59.44 ± 2.44	52.61 ± 2.78	60.51 ± 1.98	55.66 ± 1.65	62.12 ± 1.62	83.13 ± 0.00	83.05 ± 2.80
InsectWingbeatSound	25.39 ± 1.91	24.63 ± 1.27	26.10 ± 5.93	53.74 ± 0.54	42.18 ± 1.58	56.25 ± 3.65	25.84 ± 1.75	**58.96 ± 0.98**	38.75 ± 2.42	57.51 ± 1.69	20.02 ± 5.68	41.44 ± 2.25
ItalyPowerDemand	87.66 ± 4.88	89.29 ± 4.25	88.36 ± 4.24	62.66 ± 11.58	93.39 ± 3.05	58.50 ± 7.42	89.57 ± 5.31	93.78 ± 2.95	**95.72 ± 1.14**	85.46 ± 2.44	56.64 ± 14.68	84.26 ± 5.94
LargeKitchenAppliances	60.00 ± 4.27	58.51 ± 9.39	**62.67 ± 10.64**	37.55 ± 2.23	60.48 ± 4.88	44.09 ± 3.07	36.98 ± 2.46	40.44 ± 3.60	37.71 ± 3.74	44.09 ± 6.61	37.17 ± 1.99	50.13 ± 3.07
Lightning2	64.92 ± 6.82	62.30 ± 7.42	67.87 ± 5.51	60.33 ± 3.91	64.92 ± 4.86	**69.95 ± 2.50**	50.82 ± 5.68	68.85 ± 5.68	61.64 ± 7.91	68.31 ± 3.41	59.02 ± 6.86	69.51 ± 10.47
Lightning7	58.90 ± 7.12	54.25 ± 7.09	60.55 ± 3.80	51.23 ± 2.84	50.96 ± 8.07	67.58 ± 3.45	18.26 ± 3.95	61.64 ± 3.62	30.41 ± 0.61	**70.32 ± 2.85**	22.19 ± 2.63	66.30 ± 3.83
Mallat	71.03 ± 13.07	70.33 ± 5.20	93.72 ± 2.08	88.67 ± 2.40	77.56 ± 9.50	**94.66 ± 0.33**	65.67 ± 5.01	87.51 ± 2.78	79.72 ± 5.50	90.06 ± 1.99	56.21 ± 5.78	90.88 ± 1.15
Meat	64.33 ± 14.22	50.00 ± 9.93	81.00 ± 8.47	85.33 ± 10.70	49.33 ± 15.48	76.67 ± 10.41	47.78 ± 15.03	37.22 ± 5.36	50.00 ± 11.61	**88.89 ± 2.55**	34.00 ± 1.49	64.00 ± 11.82
MedicalImages	**45.61 ± 3.94**	43.95 ± 4.74	42.63 ± 4.22	27.26 ± 7.75	32.26 ± 2.93	37.41 ± 0.59	36.84 ± 3.23	38.03 ± 3.53	40.16 ± 6.60	38.20 ± 1.38	5.24 ± 3.29	42.03 ± 2.75
MelbournePedestrian	76.79 ± 2.31	74.96 ± 2.28	**77.84 ± 3.11**	11.65 ± 1.29	58.25 ± 3.82	62.66 ± 1.83	70.73 ± 4.35	71.76 ± 3.72	70.23 ± 2.95	67.90 ± 5.27	58.25 ± 3.82	65.81 ± 3.77
MiddlePhalanxOutlineAgeGroup	47.01 ± 8.99	54.03 ± 8.97	50.78 ± 7.01	52.08 ± 8.63	49.09 ± 6.85	50.22 ± 4.61	51.73 ± 7.75	53.90 ± 7.87	**55.06 ± 4.76**	54.11 ± 9.46	31.82 ± 16.75	49.74 ± 4.72
MiddlePhalanxOutlineCorrect	53.20 ± 7.83	52.71 ± 15.62	58.14 ± 12.45	46.94 ± 12.76	48.93 ± 10.44	49.60 ± 3.67	**58.65 ± 6.95**	55.44 ± 15.07	51.68 ± 10.94	49.60 ± 9.23	50.93 ± 8.48	48.45 ± 7.93
MiddlePhalanxTW	**45.84 ± 9.64**	**45.84 ± 3.45**	45.71 ± 4.74	**45.84 ± 4.10**	43.12 ± 4.67	44.37 ± 0.37	40.04 ± 2.62	43.72 ± 4.56	44.16 ± 4.95	44.59 ± 2.28	22.86 ± 6.13	43.64 ± 5.71
MixedShapesRegularTrain	63.93 ± 9.77	61.94 ± 4.99	68.21 ± 8.35	68.04 ± 1.34	68.63 ± 3.58	67.15 ± 2.55	20.78 ± 0.21	70.05 ± 0.70	30.67 ± 6.49	67.15 ± 1.60	64.11 ± 1.37	**76.68 ± 5.11**
MixedShapesSmallTrain	66.24 ± 12.23	57.21 ± 4.26	72.37 ± 7.91	66.37 ± 2.62	66.13 ± 3.66	68.30 ± 1.42	22.61 ± 0.88	67.60 ± 3.67	25.23 ± 4.04	66.78 ± 1.26	60.00 ± 5.54	**78.71 ± 1.70**
MoteStrain	84.74 ± 4.04	83.85 ± 4.63	85.10 ± 2.03	80.19 ± 8.46	77.96 ± 3.58	**85.73 ± 4.20**	72.68 ± 2.25	80.09 ± 3.68	77.11 ± 6.04	85.36 ± 1.87	60.30 ± 15.64	80.37 ± 4.50
NonInvasiveFetalECGThorax1	40.80 ± 1.72	31.63 ± 3.55	40.80 ± 1.72	**72.10 ± 1.96**	51.79 ± 9.05	70.09 ± 3.02	22.04 ± 0.40	70.72 ± 1.13	32.89 ± 2.08	69.19 ± 0.16	28.56 ± 4.30	67.42 ± 0.56
NonInvasiveFetalECGThorax2	41.15 ± 5.46	43.13 ± 8.61	41.15 ± 5.46	**78.55 ± 0.96**	60.71 ± 3.55	75.84 ± 1.88	37.61 ± 1.37	75.40 ± 1.50	50.66 ± 2.46	74.93 ± 2.49	37.30 ± 2.25	75.31 ± 0.82
OSULeaf	57.85 ± 5.97	50.74 ± 10.99	**72.23 ± 4.84**	32.40 ± 1.86	35.37 ± 3.13	38.84 ± 1.09	16.80 ± 1.04	32.64 ± 2.71	24.05 ± 2.60	36.91 ± 1.04	23.06 ± 4.07	49.92 ± 6.46
OliveOil	59.33 ± 13.21	28.67 ± 13.66	**74.00 ± 6.83**	31.33 ± 9.60	26.67 ± 10.00	42.22 ± 25.02	40.00 ± 10.00	32.22 ± 13.47	40.00 ± 21.73	32.22 ± 22.19	26.67 ± 10.00	28.67 ± 11.69
PLAID	24.02 ± 4.96	24.51 ± 3.44	22.87 ± 2.95	30.06 ± 4.34	47.93 ± 3.93	36.56 ± 2.69	26.88 ± 2.01	40.16 ± 2.24	27.26 ± 3.48	37.93 ± 4.09	47.93 ± 3.93	**52.96 ± 5.39**
PhalangesOutlinesCorrect	56.22 ± 7.89	55.22 ± 9.82	**56.85 ± 6.96**	52.75 ± 11.57	53.15 ± 7.53	51.83 ± 11.55	50.35 ± 9.47	51.20 ± 13.21	53.05 ± 9.33	50.85 ± 11.66	53.61 ± 9.97	53.61 ± 8.69
Phoneme	**19.05 ± 2.97**	18.15 ± 2.47	**19.05 ± 2.97**	6.61 ± 0.25	9.00 ± 1.08	10.58 ± 0.95	6.87 ± 0.83	9.00 ± 0.29	9.61 ± 1.88	9.30 ± 1.34	8.21 ± 0.75	15.63 ± 1.56
PickupGestureWiimoteZ	56.00 ± 10.20	66.80 ± 5.93	**73.60 ± 4.56**	52.40 ± 1.67	56.80 ± 9.55	57.33 ± 3.06	27.33 ± 1.15	58.00 ± 3.46	42.00 ± 2.00	58.67 ± 1.15	56.80 ± 9.55	68.00 ± 2.45
PigAirwayPressure	25.00 ± 2.61	13.37 ± 2.05	**48.17 ± 0.40**	5.10 ± 0.26	4.62 ± 0.55	4.81 ± 0.00	4.17 ± 2.22	6.73 ± 0.48	4.04 ± 0.87	5.93 ± 0.56	4.42 ± 0.86	9.81 ± 0.73
PigArtPressure	50.96 ± 5.81	20.77 ± 3.29	**68.17 ± 4.63**	8.46 ± 0.65	5.10 ± 1.82	8.97 ± 1.21	6.57 ± 2.00	8.97 ± 0.73	6.92 ± 1.65	9.29 ± 1.11	18.17 ± 2.24	33.08 ± 1.33
PigCVP	29.33 ± 2.90	18.75 ± 2.40	**62.98 ± 1.73**	6.83 ± 0.22	3.37 ± 1.32	7.53 ± 0.56	4.65 ± 1.11	7.53 ± 0.28	3.75 ± 1.24	4.81 ± 0.48	10.38 ± 1.21	19.52 ± 0.55
Plane	**100.00 ± 0.00**	98.10 ± 2.43	99.62 ± 0.52	88.95 ± 7.93	95.24 ± 3.16	97.78 ± 0.55	96.51 ± 1.98	95.87 ± 1.45	97.14 ± 1.51	95.87 ± 0.55	23.05 ± 11.32	99.24 ± 1.24
PowerCons	80.56 ± 3.36	**84.44 ± 5.02**	84.00 ± 6.30	70.11 ± 5.26	81.11 ± 10.74	75.37 ± 5.45	54.26 ± 3.90	80.93 ± 10.72	52.22 ± 4.39	78.52 ± 10.32	68.89 ± 15.97	76.11 ± 6.55
ProximalPhalanxOutlineAgeGroup	73.37 ± 8.19	81.76 ± 3.12	79.90 ± 3.95	65.56 ± 9.78	75.41 ± 5.61	79.02 ± 7.19	79.02 ± 8.46	**84.07 ± 1.71**	74.34 ± 7.73	80.00 ± 3.41	43.61 ± 12.69	76.88 ± 5.79
ProximalPhalanxOutlineCorrect	**69.35 ± 11.27**	69.00 ± 5.91	69.28 ± 9.35	64.47 ± 0.52	63.71 ± 8.48	65.18 ± 3.44	64.60 ± 1.57	63.69 ± 1.73	63.16 ± 5.61	66.67 ± 4.47	55.26 ± 17.89	65.84 ± 4.70
ProximalPhalanxTW	62.34 ± 7.16	66.44 ± 4.62	**69.37 ± 2.52**	53.37 ± 11.45	65.27 ± 5.74	58.05 ± 7.67	58.21 ± 7.39	55.45 ± 4.15	60.00 ± 4.28	55.28 ± 3.66	27.32 ± 18.90	66.24 ± 3.19
RefrigerationDevices	42.83 ± 1.65	43.20 ± 6.13	**44.00 ± 6.18**	33.33 ± 2.73	37.92 ± 6.42	38.93 ± 3.24	31.20 ± 3.23	32.98 ± 2.22	32.32 ± 2.29	38.04 ± 3.74	34.61 ± 1.83	40.96 ± 5.59
Rock	42.00 ± 5.29	36.80 ± 7.16	38.40 ± 4.34	80.40 ± 2.97	60.80 ± 5.93	80.67 ± 1.15	81.33 ± 3.06	**88.67 ± 2.31**	59.20 ± 3.63	85.33 ± 3.06	33.60 ± 5.90	76.00 ± 3.16
ScreenType	**42.56 ± 7.61**	41.33 ± 8.02	40.64 ± 8.17	35.41 ± 2.78	36.16 ± 4.77	35.56 ± 2.39	32.18 ± 2.24	36.09 ± 2.30	32.53 ± 1.41	35.02 ± 0.41	34.08 ± 2.12	34.51 ± 4.14
SemgHandGenderCh2	58.13 ± 2.49	66.47 ± 7.67	56.07 ± 2.82	65.13 ± 3.61	**67.13 ± 2.65**	50.28 ± 5.31	55.83 ± 13.67	55.06 ± 2.04	54.63 ± 4.30	47.94 ± 16.07	59.97 ± 7.08	66.80 ± 7.25
SemgHandMovementCh2	30.89 ± 3.99	33.91 ± 3.59	34.40 ± 6.06	32.84 ± 5.27	26.49 ± 1.89	27.70 ± 2.36	20.59 ± 1.85	29.48 ± 2.44	21.64 ± 2.79	23.41 ± 3.03	33.16 ± 2.38	**36.31 ± 5.92**

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Table 8: Per-dataset 5-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	TimeMorph
SemgHandSubjectCh2	37.11 ± 4.58	44.09 ± 1.20	35.60 ± 3.13	50.76 ± 1.90	49.02 ± 0.83	46.67 ± 2.12	27.78 ± 0.59	51.19 ± 2.74	30.62 ± 1.93	41.48 ± 6.03	36.62 ± 4.30	**54.31 ± 3.37**
ShakeGestureWiimoteZ	80.40 ± 4.34	91.60 ± 2.61	89.20 ± 1.10	56.40 ± 1.67	72.80 ± 5.22	62.00 ± 0.00	35.33 ± 1.15	60.67 ± 3.06	44.00 ± 2.45	55.33 ± 2.31	72.80 ± 5.22	**92.00 ± 0.00**
ShapeletSim	82.78 ± 10.96	**99.67 ± 0.75**	92.56 ± 6.55	49.44 ± 2.69	88.22 ± 10.06	55.93 ± 2.31	49.81 ± 2.80	49.63 ± 4.21	50.67 ± 1.82	49.44 ± 3.33	49.67 ± 3.98	92.00 ± 3.88
ShapesAll	57.57 ± 3.68	48.33 ± 5.58	57.57 ± 3.68	58.13 ± 1.68	51.50 ± 3.69	64.67 ± 1.64	9.22 ± 0.79	65.83 ± 3.88	13.83 ± 3.47	59.22 ± 2.28	56.47 ± 1.46	**76.20 ± 1.83**
SmallKitchenAppliances	59.47 ± 8.51	57.39 ± 6.38	**62.35 ± 6.85**	39.04 ± 4.11	57.55 ± 2.52	57.51 ± 6.62	38.67 ± 2.54	39.56 ± 5.26	46.67 ± 3.53	56.62 ± 3.96	44.00 ± 7.15	51.63 ± 12.25
SmoothSubspace	86.80 ± 2.13	82.00 ± 2.83	**88.67 ± 2.36**	33.87 ± 1.19	57.07 ± 4.58	65.78 ± 2.04	69.33 ± 3.71	61.11 ± 0.77	77.07 ± 5.51	84.00 ± 4.81	37.07 ± 6.74	88.00 ± 2.87
SonyAIBORobotSurface1	93.94 ± 2.22	**95.71 ± 1.41**	92.85 ± 2.46	61.83 ± 10.72	84.73 ± 2.02	71.21 ± 3.16	67.39 ± 1.20	71.60 ± 6.11	64.96 ± 2.40	64.89 ± 3.03	54.54 ± 12.34	84.86 ± 2.67
SonyAIBORobotSurface2	**92.51 ± 3.32**	91.14 ± 0.89	91.54 ± 4.19	78.15 ± 2.59	85.83 ± 4.10	79.96 ± 1.73	83.11 ± 2.73	81.78 ± 2.73	80.76 ± 4.57	83.21 ± 2.88	61.15 ± 14.84	77.21 ± 3.67
StarLightCurves	81.52 ± 8.85	60.67 ± 16.70	76.44 ± 1.22	74.70 ± 4.25	76.72 ± 3.78	75.63 ± 4.10	28.48 ± 9.45	75.33 ± 3.39	27.99 ± 0.00	75.69 ± 5.23	75.95 ± 2.54	**81.86 ± 7.93**
Strawberry	63.51 ± 2.99	64.22 ± 4.90	60.22 ± 9.83	58.49 ± 2.90	63.03 ± 7.09	61.89 ± 4.92	**67.21 ± 4.90**	59.82 ± 1.80	61.03 ± 3.64	65.77 ± 7.40	52.92 ± 6.34	58.05 ± 7.04
SwedishLeaf	**79.81 ± 1.70**	76.00 ± 4.97	**79.81 ± 1.70**	47.97 ± 1.68	65.66 ± 4.54	62.61 ± 4.53	49.23 ± 4.26	67.15 ± 4.00	56.93 ± 3.43	62.29 ± 1.68	15.90 ± 6.80	79.78 ± 2.56
Symbols	88.78 ± 7.99	88.74 ± 7.74	**91.24 ± 0.96**	83.44 ± 1.29	84.92 ± 3.31	84.29 ± 1.01	36.11 ± 0.23	86.40 ± 1.63	58.19 ± 7.40	83.75 ± 0.90	27.56 ± 4.43	88.04 ± 1.30
SyntheticControl	87.80 ± 2.38	**96.33 ± 0.33**	92.60 ± 1.01	48.53 ± 3.68	92.73 ± 2.94	73.22 ± 3.10	52.11 ± 3.20	66.22 ± 3.66	56.27 ± 3.45	72.78 ± 3.66	20.93 ± 4.87	95.07 ± 1.62
ToeSegmentation1	**85.96 ± 7.05**	77.37 ± 5.14	81.58 ± 5.14	54.74 ± 3.58	68.16 ± 3.79	55.56 ± 2.16	53.95 ± 2.74	51.61 ± 0.91	53.25 ± 3.38	54.82 ± 1.32	48.77 ± 4.31	70.53 ± 10.39
ToeSegmentation2	74.77 ± 12.53	77.85 ± 7.33	74.77 ± 8.79	50.77 ± 2.88	68.46 ± 6.25	58.97 ± 1.60	50.26 ± 0.44	59.49 ± 6.72	37.54 ± 3.14	64.10 ± 2.70	52.31 ± 9.12	**78.00 ± 6.36**
Trace	88.60 ± 7.37	99.00 ± 1.00	93.00 ± 8.00	53.00 ± 4.42	76.40 ± 5.77	51.67 ± 4.93	53.33 ± 2.08	49.67 ± 3.79	59.80 ± 2.59	55.67 ± 8.50	48.80 ± 10.71	**100.00 ± 0.00**
TwoLeadECG	**97.98 ± 4.17**	95.47 ± 5.89	75.84 ± 5.58	58.75 ± 2.38	82.55 ± 14.28	62.16 ± 1.52	60.81 ± 0.10	56.34 ± 2.59	61.88 ± 1.65	68.51 ± 9.61	52.10 ± 2.73	96.68 ± 5.58
TwoPatterns	64.62 ± 3.64	**74.98 ± 5.04**	74.94 ± 1.43	38.09 ± 3.07	51.91 ± 7.48	40.83 ± 3.89	26.00 ± 0.92	39.83 ± 3.80	26.50 ± 1.83	37.54 ± 1.63	27.50 ± 4.13	74.22 ± 2.81
UMD	87.08 ± 10.10	91.53 ± 5.28	**96.67 ± 2.05**	56.53 ± 3.69	74.44 ± 17.00	61.57 ± 1.45	62.96 ± 5.82	68.52 ± 10.79	71.81 ± 2.17	70.14 ± 10.69	48.47 ± 7.32	95.78 ± 2.33
UWaveGestureLibraryAll	42.53 ± 1.15	39.33 ± 5.11	44.53 ± 1.84	77.02 ± 5.21	57.17 ± 2.24	**78.18 ± 3.39**	26.42 ± 1.62	77.47 ± 3.61	35.67 ± 1.53	76.96 ± 4.32	65.94 ± 6.48	48.36 ± 0.30
UWaveGestureLibraryX	39.64 ± 3.12	46.31 ± 4.46	46.23 ± 4.40	56.88 ± 2.64	50.95 ± 2.22	58.42 ± 2.37	19.27 ± 3.18	**58.58 ± 2.34**	31.22 ± 3.55	57.86 ± 1.47	26.48 ± 5.59	47.36 ± 0.30
UWaveGestureLibraryY	37.00 ± 1.20	38.44 ± 2.57	40.23 ± 3.63	**52.69 ± 2.24**	48.41 ± 3.12	51.42 ± 3.10	21.22 ± 1.11	51.72 ± 2.98	30.92 ± 5.43	50.81 ± 4.53	19.79 ± 3.07	51.36 ± 0.31
UWaveGestureLibraryZ	38.74 ± 2.41	44.95 ± 4.51	40.85 ± 4.13	48.35 ± 3.41	47.76 ± 4.10	50.34 ± 2.37	20.23 ± 1.68	49.48 ± 1.66	31.23 ± 4.16	48.05 ± 3.36	25.20 ± 6.78	**55.54 ± 2.61**
Wafer	76.83 ± 9.15	69.99 ± 14.88	71.79 ± 15.46	55.60 ± 16.77	73.34 ± 19.66	80.42 ± 16.85	76.25 ± 9.34	**83.86 ± 16.37**	75.80 ± 12.54	83.26 ± 17.85	64.01 ± 22.29	71.65 ± 9.58
Wine	57.41 ± 13.03	48.15 ± 3.93	**61.11 ± 5.40**	55.93 ± 3.56	53.33 ± 5.62	47.53 ± 8.35	54.94 ± 8.55	50.00 ± 0.00	46.67 ± 6.47	37.65 ± 19.80	50.00 ± 0.00	51.48 ± 2.03
WordSynonyms	25.08 ± 1.54	21.22 ± 1.56	25.08 ± 1.54	29.56 ± 2.06	30.38 ± 2.74	45.87 ± 4.63	7.99 ± 1.24	44.67 ± 4.83	18.81 ± 2.07	41.33 ± 4.67	22.26 ± 2.64	**47.05 ± 3.87**
Worms	57.14 ± 9.76	55.58 ± 5.91	**62.86 ± 4.55**	24.68 ± 4.68	40.26 ± 5.11	22.94 ± 0.75	19.91 ± 3.27	25.97 ± 0.00	16.62 ± 2.32	22.08 ± 1.30	22.08 ± 5.11	44.42 ± 6.26
WormsTwoClass	**68.05 ± 7.21**	59.74 ± 6.69	62.08 ± 7.97	48.83 ± 3.13	51.17 ± 8.09	52.38 ± 2.70	42.86 ± 3.44	51.08 ± 5.25	48.83 ± 3.26	52.81 ± 3.27	48.83 ± 5.92	54.81 ± 7.59
Yoga	**55.71 ± 3.37**	53.57 ± 4.72	55.19 ± 2.57	50.13 ± 2.22	53.89 ± 3.78	49.79 ± 3.21	52.94 ± 1.57	47.50 ± 3.55	51.67 ± 1.99	49.68 ± 4.28	48.87 ± 2.05	51.31 ± 4.34
Avg. Acc.	61.27	59.47	63.55	52.30	56.94	57.86	43.20	57.20	47.30	58.35	42.94	**64.14**
Avg. Rank	4.77	5.34	4.23	7.73	6.60	5.69	9.55	5.94	8.70	5.79	9.48	**4.19**
#Best	21	13	30	6	4	8	4	7	2	8	1	**36**

Table 9: Per-dataset 10-shot classification results on 128 shared UCR datasets. Each value is reported as mean ± standard deviation over 5 runs.

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	TimeMorph
ACSF1	**85.40 ± 2.19**	76.80 ± 3.35	80.80 ± 0.84	38.60 ± 0.89	58.80 ± 2.59	64.67 ± 0.58	42.67 ± 1.53	67.00 ± 2.00	53.60 ± 2.51	67.00 ± 2.00	36.80 ± 3.27	81.33 ± 1.53
Adiac	62.46 ± 4.53	56.98 ± 3.84	62.46 ± 4.53	45.32 ± 1.06	47.72 ± 5.48	56.01 ± 2.77	48.68 ± 0.78	47.91 ± 6.68	71.71 ± 1.94	48.00 ± 3.58	12.84 ± 1.42	**76.42 ± 1.68**
AllGestureWiimoteX	49.37 ± 3.15	43.11 ± 2.57	39.26 ± 2.50	23.91 ± 1.60	43.54 ± 2.85	37.14 ± 0.52	15.00 ± 0.87	35.19 ± 1.33	25.80 ± 2.24	31.00 ± 1.57	43.54 ± 2.85	**52.31 ± 2.26**
AllGestureWiimoteY	54.89 ± 2.09	50.74 ± 4.62	40.17 ± 2.24	27.26 ± 1.11	48.03 ± 1.58	42.57 ± 3.19	16.95 ± 2.22	42.57 ± 3.81	30.37 ± 4.65	41.24 ± 2.84	48.03 ± 1.58	**56.34 ± 3.54**
AllGestureWiimoteZ	**49.34 ± 1.38**	46.03 ± 2.76	39.80 ± 2.97	22.43 ± 1.80	34.54 ± 4.20	31.29 ± 1.13	16.67 ± 1.15	29.67 ± 3.32	21.71 ± 1.69	32.19 ± 0.59	34.54 ± 4.20	44.43 ± 2.91
ArrowHead	72.23 ± 5.22	63.89 ± 7.06	65.49 ± 9.09	64.34 ± 0.65	62.86 ± 7.57	60.38 ± 1.19	48.19 ± 2.58	59.62 ± 2.88	57.83 ± 5.80	63.43 ± 3.48	45.83 ± 13.23	**81.49 ± 3.22**
BME	64.27 ± 9.29	69.33 ± 1.56	78.67 ± 4.64	64.53 ± 0.30	97.60 ± 3.29	71.11 ± 2.34	78.89 ± 1.02	88.44 ± 1.02	91.33 ± 2.31	86.67 ± 0.00	46.40 ± 7.46	**99.20 ± 0.30**
Beef	51.33 ± 6.91	44.00 ± 6.83	52.00 ± 2.98	61.33 ± 4.47	46.00 ± 6.41	70.00 ± 3.33	51.11 ± 5.09	54.44 ± 3.85	64.00 ± 7.23	**80.00 ± 0.00**	37.33 ± 4.35	68.00 ± 1.83
BeetleFly	71.00 ± 8.94	74.00 ± 6.52	75.00 ± 6.12	86.00 ± 4.18	86.00 ± 2.24	81.67 ± 2.89	48.33 ± 2.89	83.33 ± 2.89	49.00 ± 11.94	75.00 ± 5.00	70.00 ± 12.25	**96.67 ± 2.89**
BirdChicken	84.00 ± 12.45	92.00 ± 5.70	**94.00 ± 4.18**	68.00 ± 2.74	81.00 ± 9.62	71.67 ± 7.64	41.67 ± 7.64	68.33 ± 16.07	56.00 ± 5.48	56.67 ± 7.64	56.00 ± 4.18	89.00 ± 5.48
CBF	98.96 ± 0.23	98.27 ± 0.96	**99.24 ± 0.32**	67.49 ± 2.09	95.98 ± 4.64	90.07 ± 1.09	39.37 ± 1.51	91.44 ± 1.93	51.24 ± 7.49	90.11 ± 0.88	49.31 ± 9.05	97.71 ± 0.69
Car	57.67 ± 10.84	58.67 ± 3.21	65.33 ± 2.17	73.00 ± 2.98	62.67 ± 5.22	70.00 ± 2.89	27.22 ± 5.85	65.56 ± 0.96	37.33 ± 3.84	72.22 ± 1.92	50.33 ± 8.28	**76.33 ± 5.45**
Chinatown	98.19 ± 0.13	98.13 ± 0.16	**98.54 ± 0.21**	54.58 ± 24.80	97.73 ± 0.75	94.75 ± 2.10	95.63 ± 1.01	97.28 ± 0.61	96.62 ± 0.60	96.60 ± 1.10	63.56 ± 20.21	94.56 ± 3.09
ChlorineConcentration	**41.18 ± 3.66**	34.41 ± 8.63	37.86 ± 5.20	34.42 ± 4.40	35.51 ± 6.32	35.06 ± 1.39	38.85 ± 1.62	32.99 ± 4.01	39.05 ± 4.08	39.50 ± 4.07	33.01 ± 11.68	34.77 ± 6.93
CinCECGTorso	54.03 ± 3.03	65.94 ± 5.62	64.64 ± 5.11	45.36 ± 1.09	80.65 ± 7.24	81.59 ± 1.76	35.82 ± 2.41	84.42 ± 2.94	63.93 ± 1.27	85.19 ± 1.24	52.77 ± 4.47	**94.71 ± 4.65**
Coffee	97.14 ± 6.39	97.14 ± 1.60	**100.00 ± 0.00**	96.43 ± 4.37	97.86 ± 1.96	91.67 ± 2.06	86.90 ± 16.50	92.86 ± 0.00	98.57 ± 3.19	90.48 ± 2.06	56.43 ± 7.74	95.71 ± 2.99
Computers	66.48 ± 3.03	63.20 ± 6.81	**68.08 ± 4.62**	48.88 ± 2.52	56.40 ± 10.98	52.13 ± 9.60	52.13 ± 6.02	50.93 ± 12.73	49.68 ± 5.04	53.87 ± 12.94	51.52 ± 3.69	65.60 ± 6.66
CricketX	60.72 ± 2.35	55.18 ± 4.36	51.28 ± 3.40	30.15 ± 2.84	42.92 ± 1.99	46.50 ± 1.21	14.27 ± 1.71	45.38 ± 2.45	20.87 ± 2.94	45.38 ± 1.94	19.33 ± 2.95	**66.92 ± 2.71**
CricketY	56.92 ± 2.81	49.44 ± 1.94	47.69 ± 2.43	32.82 ± 4.21	42.15 ± 5.17	44.62 ± 0.44	9.32 ± 1.04	44.02 ± 2.09	17.90 ± 2.44	41.11 ± 0.53	23.95 ± 3.97	**57.95 ± 1.33**
CricketZ	63.18 ± 1.66	57.38 ± 2.58	48.82 ± 4.98	30.56 ± 3.52	43.18 ± 5.27	46.32 ± 1.71	12.91 ± 2.15	47.78 ± 1.41	18.26 ± 2.08	45.30 ± 1.50	21.95 ± 3.02	**65.30 ± 1.16**
Crop	57.92 ± 1.22	54.52 ± 1.67	57.92 ± 1.22	42.46 ± 2.10	44.96 ± 1.35	54.14 ± 2.07	52.84 ± 0.15	54.82 ± 1.79	54.74 ± 0.79	54.60 ± 1.45	25.02 ± 2.10	**57.97 ± 1.25**
DiatomSizeReduction	87.32 ± 3.50	71.05 ± 13.58	**96.80 ± 0.91**	44.77 ± 1.16	62.68 ± 4.64	87.04 ± 1.80	66.88 ± 2.86	51.96 ± 5.03	48.89 ± 18.63	87.47 ± 0.19	30.13 ± 0.15	92.61 ± 2.00
DistalPhalanxOutlineAgeGroup	68.20 ± 2.98	68.78 ± 4.24	70.50 ± 1.61	69.50 ± 1.66	69.50 ± 2.81	66.67 ± 1.10	65.71 ± 3.62	70.50 ± 1.25	68.78 ± 2.72	66.91 ± 1.25	48.06 ± 11.76	**70.65 ± 2.18**
DistalPhalanxOutlineCorrect	64.78 ± 12.49	65.07 ± 11.29	63.62 ± 12.87	50.00 ± 13.55	60.14 ± 9.92	58.94 ± 4.04	**67.15 ± 2.41**	55.43 ± 14.84	62.39 ± 9.46	61.71 ± 6.72	47.61 ± 8.80	60.51 ± 11.85
DistalPhalanxTW	55.83 ± 3.65	59.14 ± 2.12	**60.58 ± 3.58**	47.34 ± 6.42	57.55 ± 6.08	53.96 ± 3.14	53.00 ± 2.91	54.92 ± 0.83	52.52 ± 4.35	54.92 ± 2.91	32.37 ± 14.50	54.10 ± 3.11
DodgerLoopDay	49.25 ± 3.71	47.25 ± 4.28	39.75 ± 3.11	41.50 ± 2.24	47.00 ± 4.11	**49.58 ± 5.64**	28.75 ± 2.50	45.42 ± 1.91	29.00 ± 2.40	48.75 ± 1.25	47.00 ± 4.11	48.75 ± 1.77
DodgerLoopGame	68.70 ± 6.09	83.33 ± 5.28	75.07 ± 2.64	79.28 ± 0.97	**88.84 ± 1.67**	76.81 ± 1.26	48.79 ± 3.42	81.40 ± 1.51	53.48 ± 2.83	78.99 ± 1.45	**88.84 ± 1.67**	83.62 ± 0.65
DodgerLoopWeekend	94.06 ± 1.39	94.20 ± 2.11	96.81 ± 0.83	**97.83 ± 0.00**	96.81 ± 0.65	96.86 ± 0.42	72.71 ± 1.82	97.58 ± 0.42	76.38 ± 1.82	97.10 ± 0.00	96.81 ± 0.65	**97.83 ± 0.00**
ECG200	80.00 ± 4.00	**83.00 ± 2.00**	80.40 ± 4.28	77.60 ± 4.34	79.80 ± 6.53	79.33 ± 3.21	78.00 ± 3.61	81.00 ± 5.00	82.00 ± 2.92	82.00 ± 3.61	63.00 ± 15.26	77.40 ± 4.88
ECG5000	83.56 ± 5.09	78.23 ± 2.20	80.72 ± 4.37	**88.37 ± 1.22**	76.31 ± 5.83	79.59 ± 2.75	73.64 ± 2.21	84.75 ± 1.16	79.28 ± 5.11	82.19 ± 1.79	24.56 ± 15.15	73.78 ± 6.18
ECGFiveDays	95.40 ± 1.88	88.99 ± 4.44	**97.56 ± 1.20**	74.87 ± 4.61	94.61 ± 2.00	83.82 ± 7.05	96.44 ± 1.42	91.64 ± 1.90	97.51 ± 1.12	86.18 ± 8.29	57.47 ± 4.38	75.28 ± 1.71
EOGHorizontalSignal	30.61 ± 3.99	43.15 ± 5.84	30.94 ± 3.48	29.89 ± 1.60	38.78 ± 6.14	39.59 ± 1.67	8.01 ± 1.27	41.62 ± 1.42	10.61 ± 3.44	40.06 ± 3.72	39.56 ± 4.17	**49.17 ± 1.62**
EOGVerticalSignal	21.71 ± 2.53	28.23 ± 2.47	19.83 ± 3.08	31.55 ± 2.19	40.06 ± 1.98	36.00 ± 1.12	8.47 ± 0.84	37.75 ± 1.67	11.99 ± 1.33	37.02 ± 1.99	30.55 ± 3.17	**42.21 ± 2.26**
Earthquakes	53.81 ± 8.62	53.96 ± 9.50	51.37 ± 3.47	52.52 ± 6.49	54.82 ± 2.85	60.91 ± 4.63	49.16 ± 8.34	59.71 ± 6.86	54.68 ± 5.26	**61.87 ± 3.60**	47.91 ± 6.66	54.68 ± 7.63
ElectricDevices	40.12 ± 4.42	38.65 ± 3.96	36.00 ± 4.64	28.04 ± 1.63	39.08 ± 8.26	48.59 ± 6.21	30.07 ± 3.26	33.74 ± 3.43	37.44 ± 1.49	48.14 ± 2.51	16.08 ± 5.57	**49.42 ± 5.02**
EthanolLevel	26.56 ± 1.96	25.76 ± 1.05	26.16 ± 0.71	29.28 ± 1.78	25.92 ± 1.70	28.07 ± 2.55	25.20 ± 3.34	26.80 ± 0.69	25.60 ± 1.13	**29.87 ± 2.47**	25.56 ± 1.05	28.24 ± 0.50
FaceAll	77.73 ± 3.48	**80.30 ± 1.67**	77.73 ± 3.48	67.09 ± 2.40	54.49 ± 5.41	67.46 ± 1.82	59.68 ± 1.83	67.32 ± 2.82	60.69 ± 2.63	64.06 ± 2.32	32.89 ± 2.71	63.01 ± 1.34
FaceFour	87.95 ± 3.07	59.32 ± 7.51	**92.95 ± 2.03**	74.77 ± 0.95	66.36 ± 10.77	85.98 ± 1.31	65.91 ± 2.27	88.26 ± 0.66	69.55 ± 1.69	83.71 ± 1.31	34.55 ± 3.82	79.09 ± 1.90
FacesUCR	**87.80 ± 1.60**	84.26 ± 2.06	82.41 ± 2.48	63.80 ± 1.79	54.40 ± 3.05	80.54 ± 1.15	70.11 ± 1.05	83.69 ± 1.13	72.08 ± 1.16	79.51 ± 1.53	34.67 ± 3.28	79.08 ± 1.41
FiftyWords	45.36 ± 2.39	36.70 ± 1.89	45.36 ± 2.39	53.67 ± 1.47	50.46 ± 2.83	64.32 ± 1.87	16.85 ± 0.67	67.99 ± 1.84	37.05 ± 3.62	57.36 ± 1.54	56.40 ± 1.98	**71.34 ± 2.10**
Fish	**83.31 ± 3.78**	56.00 ± 4.28	71.54 ± 6.28	65.60 ± 3.49	62.74 ± 2.15	68.95 ± 4.29	24.38 ± 3.80	71.05 ± 1.44	57.14 ± 3.45	75.05 ± 2.31	38.17 ± 5.05	78.29 ± 3.92
FordA	78.26 ± 5.14	75.61 ± 0.71	**78.70 ± 3.26**	49.09 ± 0.93	67.88 ± 3.90	51.99 ± 2.52	52.22 ± 0.95	48.81 ± 1.31	53.05 ± 1.09	48.21 ± 1.67	50.06 ± 2.65	57.15 ± 1.85
FordB	**63.83 ± 5.74**	62.17 ± 3.33	61.70 ± 3.19	48.54 ± 0.72	59.06 ± 1.85	46.75 ± 0.61	49.79 ± 1.45	47.24 ± 1.24	50.47 ± 2.25	47.33 ± 0.14	49.33 ± 1.67	55.60 ± 1.91
FreezerRegularTrain	86.27 ± 5.59	**89.20 ± 7.73**	79.00 ± 5.98	76.33 ± 0.75	80.48 ± 6.98	76.36 ± 0.25	61.75 ± 7.11	76.22 ± 0.12	68.27 ± 6.23	74.90 ± 1.40	66.57 ± 12.21	**89.20 ± 5.65**
FreezerSmallTrain	78.20 ± 4.17	80.08 ± 7.30	74.46 ± 2.32	76.11 ± 0.45	68.74 ± 3.00	73.18 ± 1.45	65.15 ± 4.43	71.92 ± 2.00	68.25 ± 2.23	70.97 ± 2.25	65.53 ± 10.69	**82.86 ± 3.49**
Fungi	93.98 ± 3.37	38.06 ± 6.98	67.85 ± 1.59	67.74 ± 4.64	69.57 ± 7.88	**95.88 ± 1.64**	72.94 ± 1.89	91.22 ± 1.73	74.73 ± 0.76	**95.88 ± 0.31**	12.47 ± 9.15	79.57 ± 1.20
GestureMidAirD1	49.38 ± 3.66	60.92 ± 1.48	49.38 ± 3.66	49.69 ± 1.40	48.31 ± 2.52	54.36 ± 1.94	26.15 ± 2.31	58.21 ± 2.22	44.15 ± 6.68	57.44 ± 1.60	48.31 ± 2.52	**67.08 ± 1.00**
GestureMidAirD2	38.00 ± 5.29	47.23 ± 3.34	38.00 ± 5.29	50.46 ± 0.69	50.46 ± 2.75	53.08 ± 2.66	11.28 ± 2.47	55.13 ± 1.18	34.00 ± 4.05	51.79 ± 1.18	50.46 ± 2.75	**55.69 ± 1.85**
GestureMidAirD3	20.15 ± 2.40	25.23 ± 1.38	20.15 ± 2.40	32.92 ± 0.34	29.23 ± 3.17	28.46 ± 0.77	15.13 ± 2.70	33.08 ± 2.04	29.38 ± 4.79	31.28 ± 0.44	29.23 ± 3.17	**33.69 ± 2.63**

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Table 9: Per-dataset 10-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	TimeMorph
GesturePebbleZ1	76.86 ± 5.40	75.47 ± 8.13	77.67 ± 2.20	66.63 ± 2.78	68.84 ± 4.61	75.00 ± 1.16	22.29 ± 3.87	77.33 ± 2.10	38.95 ± 10.62	**81.40 ± 2.53**	68.84 ± 4.61	75.12 ± 5.19
GesturePebbleZ2	73.16 ± 4.51	63.42 ± 5.15	73.16 ± 2.82	60.38 ± 3.46	67.59 ± 4.18	69.62 ± 2.90	19.20 ± 0.73	74.26 ± 2.56	28.86 ± 9.85	73.00 ± 1.59	67.59 ± 4.18	**75.57 ± 4.64**
GunPoint	85.73 ± 9.32	82.13 ± 8.02	91.73 ± 11.16	68.27 ± 5.09	77.60 ± 4.39	71.78 ± 3.79	64.00 ± 6.57	72.00 ± 1.15	77.73 ± 11.90	76.67 ± 6.67	62.00 ± 16.53	**97.47 ± 2.23**
GunPointAgeSpan	89.56 ± 7.42	72.47 ± 9.28	74.87 ± 11.59	83.54 ± 2.13	82.22 ± 4.04	85.44 ± 1.67	52.22 ± 3.12	85.13 ± 2.39	71.08 ± 9.43	83.54 ± 3.02	53.61 ± 10.43	**92.72 ± 5.80**
GunPointMaleVersusFemale	92.78 ± 8.49	93.48 ± 5.65	92.22 ± 9.65	67.09 ± 4.97	95.57 ± 2.50	91.14 ± 4.91	60.97 ± 6.06	93.57 ± 3.96	74.05 ± 7.11	94.41 ± 2.22	67.22 ± 11.33	**98.54 ± 0.94**
GunPointOldVersusYoung	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	76.19 ± 1.96	79.17 ± 3.15	78.20 ± 2.30	57.46 ± 1.38	77.25 ± 5.77	61.90 ± 5.09	79.26 ± 2.47	85.78 ± 19.21	86.79 ± 4.71
Ham	60.76 ± 5.19	66.48 ± 2.64	65.52 ± 4.83	**71.24 ± 5.65**	62.86 ± 4.86	66.03 ± 7.40	63.49 ± 10.09	68.89 ± 6.69	61.71 ± 8.23	68.89 ± 7.02	61.52 ± 3.79	62.86 ± 3.37
HandOutlines	50.76 ± 12.61	50.92 ± 12.38	57.46 ± 2.17	81.57 ± 4.57	**82.65 ± 3.66**	81.62 ± 2.31	38.65 ± 4.09	78.47 ± 5.74	35.95 ± 0.00	81.62 ± 1.95	76.97 ± 8.74	71.51 ± 5.35
Haptics	29.68 ± 4.36	29.16 ± 4.62	33.31 ± 5.65	37.73 ± 3.74	31.95 ± 2.95	37.66 ± 2.89	23.81 ± 5.22	37.66 ± 2.77	25.45 ± 4.31	38.20 ± 2.76	35.32 ± 3.26	**39.22 ± 4.29**
Herring	55.00 ± 9.20	55.62 ± 5.91	55.00 ± 7.28	50.31 ± 6.57	**59.69 ± 3.39**	52.60 ± 3.93	47.92 ± 2.39	53.65 ± 8.02	49.06 ± 4.50	50.52 ± 1.80	57.19 ± 6.68	55.31 ± 5.01
HouseTwenty	**90.92 ± 1.10**	88.91 ± 1.82	89.58 ± 3.18	65.55 ± 5.48	81.51 ± 3.71	81.23 ± 2.43	60.50 ± 3.36	74.79 ± 1.46	64.20 ± 2.95	81.79 ± 4.63	70.76 ± 6.04	89.24 ± 2.81
InlineSkate	30.04 ± 2.77	24.04 ± 1.78	21.27 ± 3.03	25.16 ± 1.55	21.67 ± 3.27	28.42 ± 1.51	16.91 ± 0.91	28.73 ± 0.83	18.22 ± 1.24	29.09 ± 2.20	22.62 ± 1.07	**33.49 ± 2.77**
InsectEPGRegularTrain	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	54.62 ± 4.82	79.12 ± 3.30	61.31 ± 3.90	42.70 ± 1.23	60.37 ± 1.29	55.18 ± 4.83	64.39 ± 5.02	96.63 ± 7.54	86.59 ± 4.06
InsectEPGSmallTrain	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	60.00 ± 1.05	81.69 ± 6.46	69.08 ± 1.20	55.82 ± 1.06	69.88 ± 0.80	58.71 ± 4.87	66.40 ± 1.81	83.13 ± 0.00	85.06 ± 2.08
InsectWingbeatSound	34.13 ± 0.95	34.52 ± 3.87	30.58 ± 3.83	60.48 ± 0.59	47.36 ± 1.34	60.99 ± 1.05	39.46 ± 3.03	**61.67 ± 1.18**	49.66 ± 3.26	61.20 ± 0.34	46.64 ± 2.91	51.36 ± 2.05
ItalyPowerDemand	92.21 ± 1.02	92.56 ± 1.30	92.75 ± 1.07	53.43 ± 4.80	93.97 ± 0.64	87.66 ± 8.42	94.69 ± 1.03	95.59 ± 0.11	**95.82 ± 0.32**	94.33 ± 0.49	49.23 ± 14.84	92.19 ± 2.62
LargeKitchenAppliances	64.05 ± 4.75	66.13 ± 5.93	66.08 ± 5.44	40.85 ± 3.41	**67.95 ± 4.05**	46.22 ± 6.49	39.82 ± 1.74	43.11 ± 5.30	40.53 ± 2.82	45.33 ± 5.07	41.71 ± 0.52	55.09 ± 5.30
Lightning2	70.49 ± 3.84	64.92 ± 7.48	65.57 ± 4.34	58.03 ± 2.74	69.18 ± 5.84	71.04 ± 1.89	51.37 ± 4.13	63.93 ± 3.28	58.36 ± 5.26	68.85 ± 4.34	60.00 ± 3.77	**77.05 ± 4.18**
Lightning7	64.38 ± 5.97	66.03 ± 3.41	60.00 ± 4.37	60.27 ± 3.36	61.10 ± 9.76	74.43 ± 1.58	20.09 ± 4.81	65.75 ± 1.37	36.71 ± 3.41	**75.34 ± 2.37**	26.30 ± 5.76	66.85 ± 7.34
Mallat	68.22 ± 8.96	69.46 ± 5.32	**95.78 ± 0.57**	90.38 ± 0.87	87.49 ± 9.31	92.79 ± 0.83	66.92 ± 5.03	85.20 ± 1.21	79.12 ± 6.08	89.69 ± 0.98	25.09 ± 0.73	89.72 ± 1.50
Meat	53.00 ± 18.31	59.33 ± 6.08	85.00 ± 4.86	**87.00 ± 7.94**	60.00 ± 5.27	61.11 ± 4.19	68.89 ± 7.52	55.56 ± 19.25	55.33 ± 20.66	86.11 ± 7.88	38.67 ± 15.92	64.00 ± 6.08
MedicalImages	**56.61 ± 3.74**	54.08 ± 3.55	52.39 ± 5.17	34.84 ± 5.66	48.24 ± 3.52	46.27 ± 5.02	40.00 ± 3.41	45.09 ± 2.04	47.45 ± 1.06	47.72 ± 6.48	18.58 ± 8.91	54.68 ± 2.10
MelbournePedestrian	**84.60 ± 1.98**	81.55 ± 1.62	80.47 ± 2.59	30.74 ± 5.50	68.14 ± 1.82	73.08 ± 1.13	75.36 ± 1.75	78.01 ± 1.38	77.48 ± 0.93	76.77 ± 0.49	68.14 ± 1.82	70.88 ± 1.49
MiddlePhalanxOutlineAgeGroup	44.29 ± 12.74	54.68 ± 6.95	55.19 ± 4.97	54.16 ± 10.18	53.51 ± 5.82	**58.66 ± 1.35**	50.43 ± 7.47	54.11 ± 4.79	50.13 ± 8.45	56.49 ± 1.12	41.82 ± 18.66	51.04 ± 7.11
MiddlePhalanxOutlineCorrect	60.00 ± 5.84	**65.02 ± 2.98**	60.62 ± 1.86	54.23 ± 4.23	56.36 ± 2.69	50.86 ± 7.18	64.72 ± 6.49	54.98 ± 7.77	58.97 ± 7.04	52.23 ± 9.34	52.44 ± 10.59	59.18 ± 5.65
MiddlePhalanxTW	**49.74 ± 5.84**	48.05 ± 4.11	49.48 ± 2.12	45.71 ± 2.70	46.23 ± 6.05	46.10 ± 6.40	37.23 ± 5.86	42.64 ± 7.61	35.97 ± 2.62	41.34 ± 8.07	23.64 ± 8.52	44.42 ± 6.27
MixedShapesRegularTrain	67.65 ± 7.57	70.16 ± 5.08	72.53 ± 6.92	70.49 ± 2.59	73.35 ± 4.70	73.87 ± 1.26	24.40 ± 1.33	74.08 ± 2.13	28.76 ± 6.79	73.65 ± 2.23	63.69 ± 2.98	**79.60 ± 1.74**
MixedShapesSmallTrain	70.61 ± 10.30	69.46 ± 11.49	75.93 ± 3.55	73.56 ± 2.63	73.16 ± 8.70	78.08 ± 2.18	23.04 ± 1.80	77.03 ± 1.02	27.06 ± 6.78	77.29 ± 1.92	65.95 ± 4.04	**81.61 ± 2.47**
MoteStrain	89.38 ± 0.88	89.14 ± 3.04	90.18 ± 0.68	86.26 ± 0.20	86.58 ± 1.30	87.19 ± 1.69	77.40 ± 1.04	87.43 ± 0.48	85.32 ± 0.75	**90.65 ± 0.48**	67.41 ± 11.72	89.23 ± 1.53
NonInvasiveFetalECGThorax1	55.91 ± 3.81	50.28 ± 5.45	55.91 ± 3.81	**81.90 ± 0.59**	63.56 ± 4.40	77.13 ± 1.08	36.83 ± 0.46	78.18 ± 0.43	46.96 ± 3.44	75.71 ± 1.08	58.08 ± 1.54	76.84 ± 1.40
NonInvasiveFetalECGThorax2	62.50 ± 1.19	67.13 ± 2.71	62.50 ± 1.19	**86.79 ± 1.25**	75.21 ± 2.50	83.55 ± 0.48	57.79 ± 2.45	83.27 ± 1.00	69.23 ± 2.65	80.68 ± 0.66	70.22 ± 2.79	82.53 ± 1.33
OSULeaf	**72.64 ± 5.62**	69.26 ± 4.17	69.01 ± 3.75	34.88 ± 3.54	39.34 ± 3.85	46.42 ± 2.52	19.83 ± 4.60	40.50 ± 2.19	29.83 ± 2.82	37.19 ± 2.07	28.10 ± 5.12	59.42 ± 3.58
OliveOil	56.00 ± 14.79	40.00 ± 0.00	**66.67 ± 9.72**	40.00 ± 0.00	40.00 ± 0.00	40.00 ± 0.00	45.56 ± 9.62	40.00 ± 0.00	40.00 ± 0.00	40.00 ± 0.00	40.00 ± 0.00	40.00 ± 0.00
PLAID	28.04 ± 4.01	25.14 ± 3.42	22.46 ± 6.07	34.45 ± 1.13	54.53 ± 5.93	50.16 ± 2.34	32.59 ± 0.99	44.82 ± 3.36	37.77 ± 2.12	46.80 ± 0.84	54.53 ± 5.93	**59.89 ± 3.28**
PhalangesOutlinesCorrect	56.88 ± 6.98	56.74 ± 8.68	57.48 ± 7.09	54.27 ± 11.23	49.91 ± 6.62	**58.78 ± 4.01**	55.79 ± 3.74	55.36 ± 17.14	52.42 ± 4.71	54.23 ± 12.22	53.99 ± 12.03	55.73 ± 8.07
Phoneme	**28.10 ± 1.26**	25.96 ± 1.60	**28.10 ± 1.26**	8.58 ± 0.50	11.88 ± 0.54	13.82 ± 0.86	9.55 ± 1.16	11.71 ± 0.45	10.91 ± 1.85	10.88 ± 0.47	10.94 ± 0.56	21.95 ± 1.27
PickupGestureWiimoteZ	50.80 ± 15.01	68.40 ± 2.61	**73.60 ± 2.19**	51.60 ± 1.67	54.40 ± 6.84	56.67 ± 3.06	26.67 ± 1.15	54.67 ± 1.15	42.00 ± 5.10	60.67 ± 1.15	54.40 ± 6.84	70.40 ± 2.61
PigAirwayPressure	26.35 ± 2.05	12.88 ± 0.53	**50.58 ± 0.92**	4.81 ± 0.00	4.13 ± 0.43	4.81 ± 0.48	2.08 ± 1.00	6.25 ± 1.27	4.62 ± 0.87	6.73 ± 0.48	5.00 ± 0.65	9.62 ± 0.48
PigArtPressure	54.71 ± 1.57	25.19 ± 5.85	**69.81 ± 3.29**	8.27 ± 0.40	6.06 ± 2.62	8.97 ± 0.28	6.41 ± 1.00	9.78 ± 0.73	6.35 ± 0.92	8.65 ± 1.27	18.08 ± 2.06	31.73 ± 1.92
PigCVP	28.85 ± 2.68	19.62 ± 1.84	**61.35 ± 2.42**	7.12 ± 0.40	3.27 ± 1.61	7.37 ± 0.73	7.05 ± 1.21	8.33 ± 1.69	4.13 ± 0.80	5.93 ± 1.11	11.63 ± 1.24	19.52 ± 1.34
Plane	**100.00 ± 0.00**	99.43 ± 1.28	**100.00 ± 0.00**	96.57 ± 1.28	96.95 ± 1.24	98.10 ± 0.00	98.73 ± 0.55	97.14 ± 0.95	98.48 ± 0.52	97.14 ± 0.95	49.71 ± 5.36	99.24 ± 1.24
PowerCons	84.00 ± 3.74	85.67 ± 2.68	**88.11 ± 1.91**	73.22 ± 7.34	83.67 ± 3.78	85.93 ± 2.85	53.33 ± 0.56	86.30 ± 2.25	59.67 ± 5.39	84.63 ± 0.85	74.33 ± 14.06	82.33 ± 5.46
ProximalPhalanxOutlineAgeGroup	69.07 ± 5.10	80.49 ± 4.77	80.98 ± 2.76	80.98 ± 2.95	75.90 ± 8.78	81.30 ± 5.55	81.14 ± 3.66	**81.63 ± 4.89**	76.88 ± 9.31	77.56 ± 6.40	68.78 ± 16.36	79.80 ± 3.30
ProximalPhalanxOutlineCorrect	63.71 ± 11.74	64.60 ± 2.91	63.85 ± 3.21	56.01 ± 10.53	61.86 ± 4.47	64.60 ± 0.69	**66.55 ± 3.51**	64.26 ± 0.69	61.86 ± 6.47	65.98 ± 3.31	51.34 ± 15.85	60.07 ± 4.20
ProximalPhalanxTW	60.78 ± 11.91	70.05 ± 3.85	**72.39 ± 1.88**	51.12 ± 7.39	70.73 ± 1.42	64.07 ± 2.20	60.16 ± 2.69	65.53 ± 2.25	57.95 ± 5.18	59.84 ± 3.91	25.07 ± 15.12	69.66 ± 3.47
RefrigerationDevices	42.67 ± 7.00	44.27 ± 6.42	43.41 ± 5.31	34.72 ± 1.59	37.71 ± 6.85	44.00 ± 2.97	34.40 ± 1.33	34.76 ± 2.56	35.31 ± 4.05	**44.62 ± 2.22**	32.37 ± 1.67	41.23 ± 4.90
Rock	41.60 ± 6.84	34.40 ± 4.56	36.00 ± 1.41	81.60 ± 0.89	57.20 ± 9.01	82.67 ± 1.15	84.67 ± 6.11	**88.00 ± 0.00**	62.00 ± 6.16	86.67 ± 3.06	30.80 ± 4.82	76.40 ± 2.97
ScreenType	35.84 ± 5.22	36.48 ± 5.07	37.23 ± 5.64	36.37 ± 2.19	39.15 ± 5.32	**40.53 ± 1.87**	35.73 ± 2.12	36.09 ± 2.74	33.12 ± 2.51	39.82 ± 2.27	36.32 ± 3.91	37.12 ± 3.93
SemgHandGenderCh2	63.47 ± 5.58	**74.93 ± 6.24**	65.43 ± 6.18	67.83 ± 4.45	66.37 ± 2.41	62.89 ± 2.12	51.67 ± 5.43	65.78 ± 6.45	55.43 ± 2.94	60.72 ± 4.24	58.77 ± 7.13	63.80 ± 10.73
SemgHandMovementCh2	37.69 ± 5.41	36.27 ± 1.15	37.82 ± 1.58	37.42 ± 2.10	32.22 ± 2.74	30.67 ± 2.78	25.04 ± 0.56	34.52 ± 3.16	25.69 ± 1.85	33.04 ± 3.33	33.64 ± 4.19	**45.56 ± 2.93**

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Table 9: Per-dataset 10-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	TimeMorph
SemgHandSubjectCh2	43.56 ± 2.78	45.20 ± 5.06	40.22 ± 5.31	62.80 ± 4.54	54.13 ± 4.47	60.37 ± 0.71	34.22 ± 3.27	**68.07 ± 2.65**	39.96 ± 2.48	60.96 ± 1.10	38.18 ± 2.42	66.71 ± 4.45
ShakeGestureWiimoteZ	82.00 ± 6.16	91.20 ± 2.28	88.80 ± 1.10	55.60 ± 2.97	68.40 ± 7.67	60.00 ± 4.00	33.33 ± 3.06	62.67 ± 3.06	45.20 ± 7.29	56.67 ± 1.15	68.40 ± 7.67	**91.60 ± 2.97**
ShapeletSim	86.33 ± 6.80	**100.00 ± 0.00**	**100.00 ± 0.00**	48.78 ± 0.82	95.22 ± 4.26	59.07 ± 5.56	48.70 ± 1.40	50.19 ± 2.80	51.00 ± 1.98	49.81 ± 1.95	52.33 ± 4.64	96.67 ± 1.96
ShapesAll	68.60 ± 2.31	66.63 ± 1.82	68.60 ± 2.31	62.47 ± 0.34	65.47 ± 2.49	72.11 ± 0.69	12.89 ± 4.16	73.28 ± 0.38	31.00 ± 5.80	65.56 ± 1.08	70.00 ± 1.20	**83.13 ± 0.32**
SmallKitchenAppliances	**67.20 ± 8.07**	61.87 ± 4.21	64.69 ± 7.08	36.59 ± 1.57	60.00 ± 3.85	55.11 ± 4.59	41.51 ± 2.40	38.76 ± 2.56	47.25 ± 3.08	51.02 ± 1.54	41.33 ± 5.03	57.60 ± 5.56
SmoothSubspace	**92.00 ± 1.56**	87.07 ± 3.67	**92.00 ± 3.02**	38.13 ± 6.59	67.73 ± 10.31	79.56 ± 2.78	80.67 ± 0.67	60.00 ± 11.02	86.13 ± 3.38	90.67 ± 4.00	43.60 ± 9.96	89.47 ± 2.60
SonyAIBORobotSurface1	95.11 ± 2.42	**96.01 ± 0.87**	92.45 ± 1.40	42.96 ± 0.07	87.95 ± 1.92	64.28 ± 1.83	67.05 ± 1.01	76.43 ± 0.69	64.86 ± 0.43	67.83 ± 1.06	45.09 ± 4.84	86.09 ± 3.97
SonyAIBORobotSurface2	**96.73 ± 3.11**	92.82 ± 3.91	94.67 ± 2.40	77.50 ± 0.71	90.95 ± 2.30	81.25 ± 2.70	83.07 ± 2.21	86.08 ± 2.22	82.71 ± 1.50	85.73 ± 2.18	61.15 ± 12.88	84.11 ± 2.29
StarLightCurves	71.47 ± 14.56	82.56 ± 3.87	75.10 ± 5.06	74.16 ± 3.22	77.51 ± 4.09	75.29 ± 4.53	29.60 ± 9.11	76.31 ± 3.24	30.20 ± 4.95	74.55 ± 5.89	70.71 ± 14.11	**85.26 ± 3.54**
Strawberry	79.73 ± 8.80	76.70 ± 3.26	76.70 ± 4.55	60.81 ± 1.94	79.89 ± 6.30	61.98 ± 1.58	**80.63 ± 1.13**	59.10 ± 1.09	77.51 ± 4.36	68.11 ± 2.15	50.54 ± 10.66	75.84 ± 0.99
SwedishLeaf	**88.03 ± 1.39**	83.68 ± 1.26	**88.03 ± 1.39**	66.11 ± 1.36	73.47 ± 2.54	75.95 ± 1.36	60.69 ± 4.66	81.39 ± 2.49	68.80 ± 1.87	74.72 ± 1.82	28.96 ± 3.20	85.89 ± 1.27
Symbols	**93.83 ± 2.55**	89.89 ± 7.08	92.52 ± 1.13	75.94 ± 0.93	85.87 ± 1.87	83.32 ± 0.52	34.77 ± 0.30	85.96 ± 1.03	61.21 ± 7.54	82.51 ± 0.70	21.11 ± 2.93	89.33 ± 0.82
SyntheticControl	95.60 ± 1.14	**97.87 ± 0.61**	97.60 ± 0.43	59.07 ± 3.98	94.20 ± 1.64	90.78 ± 1.64	59.44 ± 1.95	79.44 ± 4.86	68.40 ± 3.43	90.11 ± 1.35	22.60 ± 7.62	97.33 ± 0.85
ToeSegmentation1	**93.25 ± 1.60**	89.56 ± 3.33	91.58 ± 1.68	55.53 ± 3.14	68.95 ± 9.85	56.73 ± 5.97	50.29 ± 6.71	55.85 ± 3.55	49.39 ± 2.18	55.70 ± 4.98	50.53 ± 1.40	80.53 ± 2.75
ToeSegmentation2	81.38 ± 10.25	**87.08 ± 4.53**	85.85 ± 2.64	50.15 ± 3.62	73.38 ± 4.70	68.21 ± 10.44	56.67 ± 7.86	58.46 ± 6.30	46.77 ± 3.41	61.54 ± 3.35	43.54 ± 14.75	80.31 ± 4.70
Trace	92.20 ± 4.60	**100.00 ± 0.00**	95.20 ± 5.07	55.80 ± 5.72	86.20 ± 10.35	72.67 ± 1.53	55.67 ± 4.51	61.33 ± 4.62	74.80 ± 3.27	78.00 ± 4.36	46.80 ± 8.17	**100.00 ± 0.00**
TwoLeadECG	**99.86 ± 0.08**	98.79 ± 1.25	97.56 ± 2.37	58.05 ± 1.68	91.89 ± 5.00	70.85 ± 3.05	75.62 ± 1.63	65.26 ± 1.91	78.86 ± 1.56	78.99 ± 2.23	54.01 ± 3.73	99.68 ± 0.24
TwoPatterns	70.89 ± 4.77	77.56 ± 2.96	**78.86 ± 3.59**	42.28 ± 1.08	74.73 ± 6.48	42.55 ± 1.37	26.22 ± 0.85	42.35 ± 1.82	29.82 ± 3.35	37.95 ± 1.72	25.60 ± 4.32	70.06 ± 0.70
UMD	92.78 ± 3.35	98.33 ± 1.05	**98.75 ± 0.58**	64.17 ± 3.05	91.53 ± 4.77	76.62 ± 2.44	75.93 ± 7.23	89.58 ± 4.22	93.06 ± 1.47	87.27 ± 3.21	47.50 ± 8.96	98.67 ± 7.42
UWaveGestureLibraryAll	48.45 ± 3.31	48.15 ± 2.07	49.06 ± 3.81	80.57 ± 3.71	68.35 ± 4.39	84.19 ± 2.90	27.54 ± 1.68	**84.63 ± 3.19**	42.77 ± 5.45	83.32 ± 3.39	81.20 ± 2.71	52.41 ± 1.70
UWaveGestureLibraryX	51.98 ± 1.26	57.08 ± 2.26	52.99 ± 2.93	60.99 ± 2.04	57.69 ± 1.76	**62.37 ± 1.85**	17.30 ± 0.19	62.30 ± 0.55	36.29 ± 8.88	61.26 ± 1.95	56.81 ± 2.90	53.50 ± 1.91
UWaveGestureLibraryY	45.58 ± 1.39	47.42 ± 0.67	46.02 ± 2.85	55.44 ± 1.29	53.52 ± 3.23	**56.83 ± 0.99**	23.04 ± 3.42	56.65 ± 1.12	32.16 ± 4.98	55.22 ± 1.56	46.61 ± 1.40	52.03 ± 0.97
UWaveGestureLibraryZ	48.34 ± 2.78	51.60 ± 2.75	44.60 ± 1.67	53.18 ± 1.58	53.45 ± 2.14	54.99 ± 0.25	20.01 ± 2.19	54.36 ± 2.47	34.40 ± 4.42	51.61 ± 2.72	46.80 ± 1.81	**60.65 ± 2.84**
Wafer	84.60 ± 6.19	78.03 ± 11.28	81.53 ± 4.78	69.44 ± 4.07	**84.77 ± 11.10**	82.09 ± 7.52	76.30 ± 12.19	83.57 ± 7.70	81.81 ± 10.33	81.67 ± 9.12	78.96 ± 10.53	82.53 ± 8.58
Wine	**68.15 ± 6.33**	52.59 ± 5.80	64.07 ± 8.03	50.74 ± 1.66	50.00 ± 0.00	58.64 ± 5.35	50.62 ± 1.07	50.00 ± 3.70	50.00 ± 0.00	62.96 ± 7.41	47.78 ± 4.97	49.26 ± 2.11
WordSynonyms	35.14 ± 2.03	30.72 ± 3.50	35.14 ± 2.03	35.86 ± 2.07	39.44 ± 4.12	51.78 ± 3.11	15.15 ± 0.48	51.10 ± 4.21	25.64 ± 4.61	49.01 ± 4.47	33.92 ± 1.82	**58.90 ± 2.04**
Worms	50.39 ± 8.29	**51.95 ± 5.35**	45.71 ± 7.71	23.38 ± 5.11	39.74 ± 4.27	23.38 ± 6.49	26.84 ± 3.97	27.27 ± 9.09	25.45 ± 3.74	25.97 ± 9.09	20.78 ± 5.35	48.57 ± 4.91
WormsTwoClass	**64.94 ± 5.51**	63.12 ± 4.37	61.30 ± 7.20	49.09 ± 4.63	47.27 ± 1.97	50.65 ± 6.49	48.05 ± 4.50	48.05 ± 4.68	48.05 ± 3.56	51.95 ± 4.50	46.49 ± 5.15	56.36 ± 3.85
Yoga	59.91 ± 2.90	**59.97 ± 4.98**	59.88 ± 3.11	55.72 ± 1.71	57.07 ± 1.49	58.79 ± 3.68	53.03 ± 3.90	59.02 ± 1.75	51.43 ± 2.65	59.87 ± 3.99	51.89 ± 2.29	57.73 ± 2.05
Avg. Acc.	66.24	64.93	67.18	55.43	62.72	62.79	46.04	61.94	51.50	63.13	46.88	**69.16**
Avg. Rank	4.83	5.03	4.55	8.00	6.23	5.75	9.87	5.73	8.87	5.64	9.58	**3.91**
#Best	25	15	28	6	5	7	3	5	1	8	1	**40**

Table 10: Key few-shot adaptation settings for TimeMorph.

Component	Setting
Base LLM	`meta-llama/Llama-3.2-1B`
Vision backbone	`facebook/dinov2-base`
Classification prompt suffix / answer	`Class:` / one class token, e.g., $\langle c_y \rangle$ followed by EOS
Morphology-map construction	Overlapping-window grid transform with per-sequence median/IQR normalization
Temporal patch length / stride	16 / 8
Temporal hidden size / heads	128 / 8
Branch projectors	LayerNorm–Linear–GELU–Dropout MLPs when dimensions differ
Language projector	MLP projector into the LLM hidden dimension followed by output LayerNorm
Default downstream checkpoint	Stage II (TSQA transfer)
Warm-up / joint epochs	5 / 55
Batch size / eval batch size	8 / 8
Gradient accumulation	4
LR (encoder / projector / LoRA)	2×10^{-4} / 1×10^{-4} / 1×10^{-4}
Language-side adaptation	LoRA on `q/k/v/o_proj`, `gate_proj`, `up_proj`, and `down_proj` ($r = 16$, $\alpha = 32$)
Tokenizer training mode	`class_rows`
Anonymous class-token initialization	pretrained vocabulary-row mean plus small random perturbation
Prediction rule	anonymous class-token scoring; semantic-prior class-token scoring when label text is meaningful
Freeze vision backbone	true
Downstream data augmentation	none

B Additional Experimental Details

We summarize the key implementation choices of TimeMorph for the few-shot benchmark in Table 10. These settings are fixed across the main benchmark reported in Section 4.

Morphology-map transform. The operator Γ_{img} in Eq. 4 applies per-sample min–max normalization over the two-dimensional grid and a fixed contrast adjustment $u \mapsto u^{0.8}$. In the default morphology mode, the effective window length is $\max(1, \lfloor \sqrt{T} \rfloor)$, the stride length is $\max(1, \lfloor 0.1q \rfloor)$, incomplete windows are completed by replicate padding on the left, and the resulting grid is resized to the DINOv2 image resolution with nearest-neighbor interpolation before RGB replication. We use DINOv2 patch tokens from the configured intermediate layer and exclude the class token.

Stage I pretraining hyperparameters. Stage I uses mask ratio 0.4, branch-dropout probability 0.1, jitter standard deviation 0.03, scaling range $[0.8, 1.2]$, and time-mask ratio 0.1 for the two augmented views. The Stage I loss weights are $\lambda_{\mathrm{rec}} = 1.0$, $\lambda_t = 0.25$, $\lambda_v = 0.25$, and $\lambda_f = 0.5$. These values are fixed before the downstream few-shot runs.

B.1 Pretraining Stages and Data Sources

Table 11 summarizes the three training stages discussed in the main paper and clarifies which stage provides the default downstream checkpoint.

After stage-wise pretraining, downstream few-shot adaptation uses only the sampled support labels of the target dataset. For UCR targets contained in the Stage I pool, unlabeled TRAIN sequences may already have contributed to the encoder pretraining.

B.2 Code Release and Assets

We include an anonymized code package with the submission. The package includes the TimeMorph implementation, the few-shot adaptation and evaluation scripts, baseline launchers, reporting scripts for the tables in this paper, environment files, and README instructions for preparing the public datasets used in our experiments. The code is released under the MIT license. The package does not redistribute raw third-party datasets or gated pretrained model weights; users should obtain those assets from the original providers and comply with their licenses and terms of use. The included scripts document which experiments are directly reproducible from the package and which require separately downloaded datasets or pretrained checkpoints.

B.3 Asset and License Table

Table 12 summarizes the main existing assets used in this paper. We list the original source, role, version when available, license or stated terms, and whether the asset is redistributed in our release.

Table 11: Stage-wise data sources and roles for TimeMorph.

Stage	Training data	Validation / monitoring data	Role in the main UCR few-shot benchmark
Stage I	Raw TSQA-train series, raw M4-train series, and official TRAIN-split raw series from a 98-dataset UCR training pool.	Raw TSQA validation/test series and raw M4 validation/test series. No UCR official TEST split is used.	Learns dual-view representations before the LLM interface; corresponds to `stage0_encoder_ssl` in the implementation.
Stage II	TSQA train split only, with standard language-model training on time-series question answering.	TSQA validation/test splits.	Provides the default downstream checkpoint used by all main few-shot results in Section 4; corresponds to `stage1_tsqa_transfer`.
Stage III	Optional mixture of TSQA, M4 captioning, and synthetic attribute supervision derived from TSQA/M4 raw series.	Corresponding validation/test splits of the same mixture.	Used only for additional semantic enrichment or caption specialization; not used for the default main benchmark initialization unless explicitly stated. This corresponds to `stage2_semantic_bridge` and later caption specialization in code.

For assets whose official source does not state a separate dataset license, we cite the official repository or archive page and do not redistribute the raw data.

Table 12: Existing assets used in this paper and their licenses or stated terms. URLs were checked during manuscript preparation.

Asset	Role	Source / version	License or stated terms	Redistribution in our release
UCR Archive 2018	Main few-shot benchmark and unlabeled Stage I source pool	TSC website; Zenodo v1	Zenodo record lists Creative Commons Attribution 4.0 International (CC BY 4.0); archive should be cited via Dau et al.	Not redistributed; scripts expect users to download from the original archive or cited mirror.
TSQA / Time-MQA	Stage I raw series and Stage II time-series question answering	Hugging Face dataset card	Apache-2.0 according to the Hugging Face dataset metadata	Not redistributed; scripts load from the original dataset source.
M4 Competition dataset	Stage I raw series and optional captioning / semantic enrichment data	Official M4 repository	The official repository publicly provides the dataset and asks users to cite the M4 competition resources; no separate dataset LICENSE file was found in the repository.	Not redistributed; users download from the official repository.
MIT-BIH Arrhythmia Database	External semantic-label evaluation	PhysioNet v1.0.0	Open Data Commons Attribution License v1.0; PhysioNet citation required	Not redistributed; preprocessing scripts require a local PhysioNet download.
PhysioNet/CinC 2017 AF dataset	External semantic-label evaluation	PhysioNet v1.0.0	Open Data Commons Attribution License v1.0; PhysioNet and challenge citations required	Not redistributed; preprocessing scripts require a local PhysioNet download.
Llama 3.2 1B	Pretrained causal LLM backbone	Hugging Face model card	Llama 3.2 Community License and Acceptable Use Policy	Not redistributed; users must obtain access from Meta / Hugging Face.
DINOv2-base	Frozen visual backbone	Hugging Face model card	Apache-2.0	Not redistributed; loaded from the original model repository.

Table 13: Compute-resource budget for the reported experiment groups. Downstream jobs use single RTX 3090 workers unless noted otherwise.

Experiment group	Scope	Compute workers	Mean per run	Total GPU-hours	Notes
Stage I–II pretraining	Curriculum pretraining checkpoint used to initialize downstream adaptation	1 RTX 3090 worker	–	19.81	Stage I encoder SSL used 9.57 GPU-hours; Stage I–II together used 19.81 GPU-hours.
Optional Stage III enrichment	Semantic-enrichment and caption-specialization training used outside the default UCR results	1 RTX 3090 worker	–	29.03	Not used for the default UCR benchmark.
TimeMorph main UCR benchmark	128 UCR datasets, 1/2/5/10-shot, five runs per setting; 2,560 completed downstream runs	6 single-GPU workers	0.67 GPU-h/run	1,715	Batch size 8, gradient accumulation 4, 60 downstream epochs.
UCR baseline runs	ResNet, TapNet, COSCO-ResNet, PatchTST, GPT4TS, and TSLib-family baselines under the same few-shot protocol	6 single-GPU workers	0.40–0.49 GPU-h/run	10,937	Includes the retained main UCR baseline logs; GPT4TS covers all 128 UCR datasets.
External-dataset TimeMorph runs	MIT-BIH and CinC2017 under 10/20/30-shot protocols	single-GPU workers	1.69 GPU-h/run	102	Same downstream adaptation recipe as the external few-shot experiments.
External-dataset baseline runs	Reported baseline families on MIT-BIH and CinC2017	single-GPU workers	0.68 GPU-h/job	15	Launcher jobs cover both external datasets and baseline families.
Top-level ablations	No-pretraining, temporal-only, visual-only, class-token decoding, and no-LLM backbone ablations	6 single-GPU workers	1.67 GPU-h/run	3,895	Includes the two retained no-LLM linear runs added together.
Semantic-prior ablation	External semantic-label interface runs	single-GPU workers	1.69 GPU-h/run	31	Uses the same 10/20/30-shot external evaluation protocol.
Efficiency probe	Representative UCR runtime subset with parameter, memory, latency, and throughput measurements	one RTX 3090	–	–	Detailed runtime and memory values are reported in Tables 4, 16, and 18.

Table 12: Existing assets used in this paper and their licenses or stated terms (continued).

Asset	Role	Source / version	License or stated terms	Redistribution in our release
OpenTSLM base code	Third-party time-series language-model components used as a starting point	OpenTSLM [Langer et al., 2025] and its official project repository	MIT license in the third-party repository license file	Modified code is released in anonymized form under MIT, preserving applicable notices.
Python ML libraries, including PyTorch, Transformers, PEFT, NumPy, pandas, scikit-learn, and related dependencies	Training, adaptation, evaluation, and reporting	Version constraints are listed in the released environment files	Governed by their respective open-source licenses	Not vendored; installed by the environment files.

B.4 Computational Resource Reporting

All reported downstream experiments were run on a local server with six NVIDIA GeForce RTX 3090 GPUs (24GB each), two Intel Xeon Silver 4314 CPUs, and 251GiB system memory. Unless otherwise stated, each training job used one GPU, and the batch launchers executed up to six single-GPU workers in parallel. Table 13 summarizes the compute used by the main experiment groups. GPU-hours are summed over single-GPU jobs; wall-clock time can be lower when multiple workers run concurrently. The efficiency measurements in Appendix B.7 additionally report peak memory, train-step latency, inference latency, throughput, total parameters, and downstream-updated parameters on the representative runtime subset.

Retained launcher summaries also include failed or exploratory jobs. These runs account for less than 2 additional GPU-hours in the preserved logs, and dry-run checks add a negligible amount of compute.

Table 14: Few-shot classification accuracy (%) on two external datasets. Each cell reports mean ± standard deviation over 10 runs under the same few-shot protocol. Best is bold and second-best is underlined within each dataset-shot column.

Model	MIT-BIH			CinC2017		
	10-shot	20-shot	30-shot	10-shot	20-shot	30-shot
TimeMorph	55.03 ± 9.40	**63.95 ± 13.80**	**64.15 ± 5.32**	**32.80 ± 9.08**	**36.98 ± 4.27**	34.13 ± 9.73
ResNet	37.97 ± 14.13	55.66 ± 8.58	57.07 ± 9.05	30.95 ± 12.41	33.73 ± 7.43	**36.20 ± 4.84**
TapNet	31.94 ± 7.18	40.15 ± 10.28	44.11 ± 11.14	30.05 ± 6.73	32.80 ± 8.81	31.01 ± 11.54
COSCO-ResNet	32.33 ± 14.25	35.08 ± 13.75	41.17 ± 8.94	31.41 ± 4.48	31.04 ± 6.97	31.98 ± 8.37
PatchTST	38.94 ± 14.35	55.36 ± 8.84	57.08 ± 9.40	26.07 ± 6.24	30.35 ± 7.51	31.21 ± 3.36
GPT4TS	32.92 ± 16.63	42.48 ± 10.82	54.87 ± 10.04	30.80 ± 6.07	27.56 ± 4.33	29.23 ± 3.03
TimesNet	51.01 ± 5.84	54.21 ± 7.81	56.15 ± 8.77	27.25 ± 2.81	28.41 ± 4.24	29.43 ± 2.44
DLinear	55.57 ± 9.85	57.45 ± 7.95	58.84 ± 8.69	24.92 ± 1.16	25.17 ± 1.22	24.53 ± 1.70
Autoformer	38.05 ± 5.65	42.49 ± 8.58	44.39 ± 7.25	28.37 ± 5.44	27.67 ± 2.76	26.30 ± 4.36
Crossformer	53.48 ± 9.36	58.55 ± 10.43	58.24 ± 9.71	24.02 ± 6.10	25.50 ± 2.86	27.35 ± 2.35
FEDformer	**58.88 ± 5.77**	56.80 ± 9.98	55.12 ± 10.09	20.35 ± 8.32	26.90 ± 3.54	27.91 ± 4.87
Informer	53.19 ± 6.03	54.20 ± 7.61	52.35 ± 10.80	26.93 ± 9.30	28.14 ± 4.01	30.12 ± 5.95

Table 15: Semantic-prior class-token ablation on two external datasets (accuracy, %). Each cell reports mean ± standard deviation over 10 runs under the same few-shot protocol. The better variant within each dataset-shot column is bold. Δ denotes Semantic prior minus Anonymous labels.

Variant	MIT-BIH			CinC2017		
	10-shot	20-shot	30-shot	10-shot	20-shot	30-shot
Anonymous labels	55.03 ± 9.40	63.95 ± 13.80	64.15 ± 5.32	32.80 ± 9.08	**36.98 ± 4.27**	34.13 ± 9.73
Semantic prior	**57.05 ± 10.96**	**66.62 ± 6.54**	**68.87 ± 11.23**	**34.72 ± 3.15**	35.24 ± 1.24	**38.86 ± 7.00**
Semantic prior − Anonymous	+2.02	+2.67	+4.72	+1.92	-1.74	+4.73

B.5 External Dataset Results

Table 14 reports the full 10-, 20-, and 30-shot results on MIT-BIH and CinC2017. All methods use the same external few-shot settings and reported run count; we use these datasets as transfer checks beyond the primary UCR benchmark rather than as a claim of uniform dominance across external domains.

B.6 Semantic-Prior Decision Ablation

Table 15 reports the full semantic-prior comparison. Both variants use one-pass class-token scoring; the semantic-prior variant additionally initializes and regularizes class-token rows from label-verbalizer prototypes. The semantic prior improves the averaged performance on the reported semantic-label datasets, while individual shot-level deltas remain dataset-dependent.

B.7 Efficiency Summary

Table 16 reports computational efficiency and adaptation cost under the 10-shot UCR protocol. Runtime and memory are measured on the predefined representative UCR subset with batch size 8 on a single RTX 3090, while parameter counts are reported separately for total deployed parameters and downstream-updated parameters.

B.8 Baseline Training Recipes and Hyperparameters

Table 17 summarizes the training rules used by the reported baseline families. The goal of this table is to make the few-shot comparison protocol explicit: all methods share the same sampled support/query split, but they do not all use the same checkpoint-selection rule or the same epoch schedule.

B.9 Computational Efficiency Probe Details

The efficiency probe is implemented by the repository's `efficiency_probe.py` script. It uses the same model families as the main benchmark and omits TimeMorph-only ablations. The default runtime subset contains COFFEE, GUNPOINT, ECG5000, ELECTRICDEVICES, CROP, ACSF1,

Table 16: Computational efficiency and adaptation cost under the 10-shot UCR protocol. Total parameters count the full deployed model, while updated parameters count only the parameters optimized during downstream adaptation. Adapter size estimates the per-dataset trainable-state footprint. Runtime is measured on representative UCR datasets with batch size 8 on a single RTX 3090.

Model	Total params	Updated params	Updated %	Adapter MB	Peak GB	Train ms/step	Infer ms/ex.	Infer ex./s
TimeMorph	1.28B	12.3M	0.96	46.7	13.75	1207.8	156.85	6.4
ResNet	629.1K	629.1K	100.00	2.4	0.33	13.2	0.26	4273.3
TapNet	912.1K	912.1K	100.00	3.5	0.22	15.0	0.47	3345.0
COSCO-ResNet	662.1K	662.1K	100.00	2.5	0.58	50.8	0.25	4315.1
PatchTST	665.4K	598.3K	90.80	2.3	1.54	32.6	0.75	2149.8
DLinear	1.2M	1.2M	100.00	4.6	0.04	2.7	0.07	14389.5
TimesNet	2.5M	2.5M	100.00	9.7	0.14	66.7	2.46	614.5
Autoformer	1.2M	1.2M	100.00	4.5	0.40	23.1	11.10	91.3
Crossformer	2.8M	2.8M	100.00	10.8	0.06	24.9	0.59	1709.6
FEDformer	1.8M	1.8M	100.00	6.7	0.26	244.1	5.17	201.9
Informer	1.4M	1.4M	100.00	5.3	0.22	18.0	0.62	1690.1
GPT4TS	82.2M	1.1M	1.34	4.2	0.56	22.6	0.86	1181.2

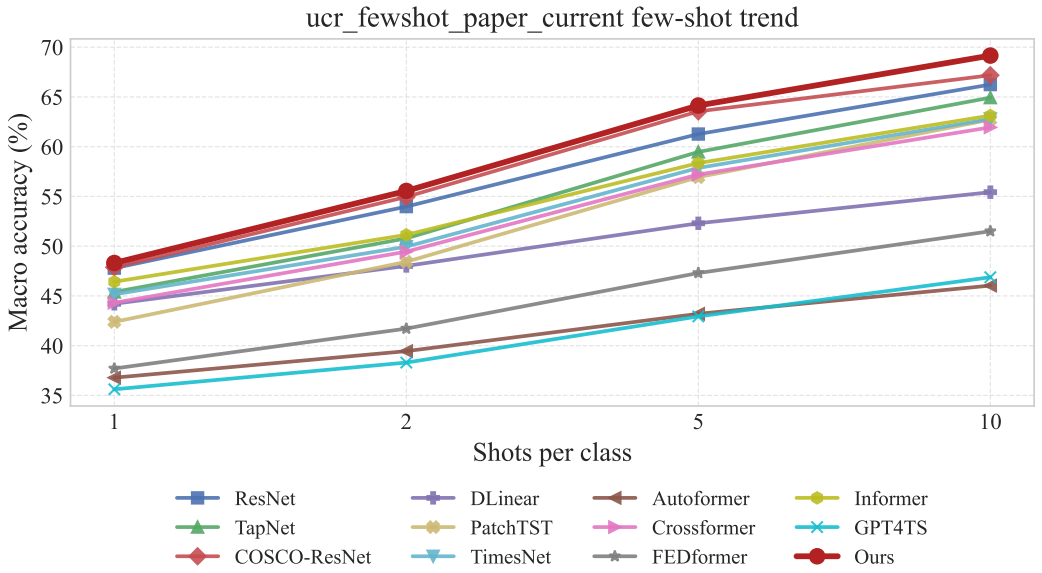

Figure 3: Shot-scaling trend on the 128 UCR datasets. Each point reports the mean accuracy over five runs. TimeMorph has the highest macro-averaged accuracy across all four shot settings.

PHONEME, and NONINVASIVEFETALECGTHORAX1, covering short, medium, and long UCR series as well as different class counts. The script writes the LaTeX tables, CSV summary, and accuracy-versus-updated-parameters plot to the efficiency-report directory.

C Additional Plot

Figure 3 visualizes the shot-scaling trend over the 128 UCR datasets. TimeMorph has the highest macro-averaged accuracy across the entire few-shot range, with the largest absolute advantage appearing at 10-shot.

Table 17: Few-shot training recipes for TimeMorph and the reported baseline families. All methods use the same sampled support sets within each run and evaluate on all official TEST examples from the target dataset.

Method / family	Optimizer	Epochs	Batch / LR	Model selection rule	Val	ES	Few-shot notes
TimeMorph	AdamW	5 warm-up + 55 joint	8 / 2×10^{-4} (encoder), 1×10^{-4} (projector, LoRA)	Report the last Phase II checkpoint	No	No	Class-token rows remain trainable; language-side adaptation uses LoRA in Phase II.
ResNet, TapNet	AdamW	Fixed schedule in the few-shot runner (60 in the benchmark configs)	model-specific / benchmark configs use 10^{-4}	Best support-set accuracy, tie-broken by support loss	No	No	Trained only on the sampled support set; the support set doubles as the model-selection set.
COSCO-ResNet	SAM with SGD base optimizer + prototypical loss	Fixed schedule (100 in the benchmark configs)	128 / 10^{-2} by default	Final trained model; support centroids computed after training	No	No	Query predictions are made by distances to support centroids rather than a learned classifier head.
PatchTST	AdamW + linear warmup	Two-phase fixed schedule (head warm-up + joint fine-tuning; 60 total epochs in the benchmark configs)	16 / separate backbone-head LRs	Report the last Phase II checkpoint	No	No	Uses the same support/query split and full dataset class set as the main protocol.
TSLib family (DLINEAR, AUTOFORMER, CROSSFORMER, FEDFORMER, INFORMER, TIMESNET)	RAdam + cosine schedule	Fixed model-specific schedule from the TSLib profile or benchmark override	usually 16 / 10^{-3}	Report the last checkpoint	No	No	Single-stage training; evaluation masks logits to the target dataset's class set. TimesNet uses a shorter profile than the other TSLib baselines.
GPT4TS	RAdam (default runner setting)	Fixed schedule (60 in the benchmark configs)	64 / 10^{-4} in the benchmark configs	Report the last checkpoint	No	No	Implemented with the One-Fits-All classification runner under the same sampled support/query protocol.

Table 18: Per-dataset runtime details for ACSF1 from the representative UCR runtime subset. Aggregate runtime and memory statistics over the representative subset are reported in Tables 4 and 16.

Dataset	Model	Peak GB	Train ms/step	Infer ms/ex.	Infer ex./s
ACSF1	TimeMorph	13.75	1262.4	173.87	5.8
ACSF1	ResNet	1.01	26.0	0.32	3167.0
ACSF1	TapNet	0.59	33.7	1.45	688.2
ACSF1	COSCO-ResNet	1.64	119.2	0.43	2303.7
ACSF1	PatchTST	6.50	107.6	2.23	447.7
ACSF1	DLinear	0.10	4.5	0.07	13898.9
ACSF1	TimesNet	0.27	121.8	4.54	220.5
ACSF1	Autoformer	1.19	40.5	12.53	79.8
ACSF1	Crossformer	0.12	25.0	0.58	1723.7
ACSF1	FEDformer	0.61	277.0	7.03	142.3
ACSF1	Informer	0.56	23.4	1.09	917.6
ACSF1	GPT4TS	1.00	29.1	0.89	1121.4

Table 19: Ablation of aggregate curriculum pretraining on the 128 UCR datasets under the same five-run full-label-space few-shot protocol as the main results. Each value is mean accuracy (%) over five runs, and Drop denotes the average accuracy decrease relative to the full TimeMorph model.

Variant	1-shot	2-shot	5-shot	10-shot	Avg	Drop
w/o Curriculum Pretraining	44.70	51.81	60.65	66.60	55.94	3.35
TimeMorph	**48.31**	**55.55**	**64.14**	**69.16**	**59.29**	–

D Ablation Tables

Table 5 in the main paper reports the compact ablation summary. Tables 19–21 expand that summary into three focused views under the same few-shot protocol. Table 22 summarizes the LLM-backbone ablation, and Table 23 reports its full per-dataset comparison on all 128 UCR datasets. Because UCR class identifiers are anonymous, the label-space ablation should be interpreted as testing closed-set class-token scoring; the semantic-prior class-token ablation is reported on external datasets with meaningful class names in Appendix B.6.

Table 20: Ablation of dual-view time-series-to-language modeling on the 128 UCR datasets under the same five-run full-label-space few-shot protocol as the main results. These comparisons assess the effect of removing one top-level view at a time. Each value is mean accuracy (%) over five runs, and Drop denotes the average accuracy decrease relative to the full TimeMorph model.

Variant	1-shot	2-shot	5-shot	10-shot	Avg	Drop
Temporal only	43.51	50.55	58.40	64.39	54.21	5.08
Visual only	45.88	50.02	60.86	63.58	55.09	4.20
TimeMorph	**48.31**	**55.55**	**64.14**	**69.16**	**59.29**	–

Table 21: Ablation of closed-set class-token label-space scoring on the 128 UCR datasets under the same five-run full-label-space few-shot protocol as the main results. Each value is mean accuracy (%) over five runs, and Drop denotes the average accuracy decrease relative to the full TimeMorph model. Since UCR labels are anonymous, this ablation does not evaluate semantic label priors.

Variant	1-shot	2-shot	5-shot	10-shot	Avg	Drop
w/o Class-Token Label-Space Scoring	44.46	50.44	60.08	65.59	55.14	4.15
TimeMorph	**48.31**	**55.55**	**64.14**	**69.16**	**59.29**	–

Table 22: Ablation of the LLM backbone on all 128 UCR datasets under the same five-run full-label-space few-shot protocol as the main results. Each value is benchmark-level mean accuracy (%). Drop denotes the average accuracy decrease relative to full TimeMorph, and W/T/L counts per-dataset, per-shot wins/ties/losses of the ablated variant against TimeMorph.

Variant	1-shot	2-shot	5-shot	10-shot	Avg	Drop	W/T/L
TimeMorph	**48.31**	**55.55**	**64.14**	**69.16**	**59.29**	–	–
w/o LLM Backbone	44.89	48.29	54.22	58.43	51.46	7.83	82/2/428

Table 23: Per-dataset LLM-backbone ablation on all 128 UCR datasets. Each value is mean ± standard deviation accuracy (%) over five runs.

Dataset	TimeMorph				w/o LLM Backbone			
	1-shot	2-shot	5-shot	10-shot	1-shot	2-shot	5-shot	10-shot
ACSF1	44.00 ± 5.20	54.33 ± 4.16	68.33 ± 2.89	81.33 ± 1.53	36.00 ± 10.37	36.99 ± 3.96	51.60 ± 5.46	56.83 ± 3.44
Adiac	45.17 ± 5.28	56.01 ± 1.25	69.51 ± 2.42	76.42 ± 1.68	23.01 ± 1.68	21.53 ± 2.11	36.37 ± 4.24	29.79 ± 2.69
AllGestureWiimoteX	25.11 ± 4.65	26.51 ± 4.36	39.40 ± 2.86	52.31 ± 2.26	20.23 ± 4.10	23.34 ± 2.88	25.33 ± 1.71	29.65 ± 1.51
AllGestureWiimoteY	25.54 ± 4.42	29.43 ± 1.69	48.11 ± 2.89	56.34 ± 3.54	23.11 ± 3.11	26.46 ± 2.53	35.37 ± 1.87	40.62 ± 2.17
AllGestureWiimoteZ	22.26 ± 3.96	27.91 ± 5.21	36.14 ± 0.97	44.43 ± 2.91	22.26 ± 4.37	21.82 ± 5.60	31.91 ± 3.85	31.91 ± 1.75
ArrowHead	52.57 ± 3.43	57.03 ± 2.81	73.60 ± 3.71	81.49 ± 3.22	44.57 ± 6.72	47.20 ± 4.58	56.26 ± 6.78	53.91 ± 3.82
BME	66.40 ± 3.88	79.47 ± 11.40	97.73 ± 1.53	99.20 ± 0.30	60.54 ± 2.92	74.53 ± 11.00	93.47 ± 5.34	87.33 ± 3.81
Beef	34.00 ± 2.79	43.33 ± 5.27	51.33 ± 6.91	68.00 ± 1.83	23.60 ± 8.69	31.00 ± 6.06	36.67 ± 7.07	48.00 ± 3.80
BeetleFly	70.00 ± 11.73	75.00 ± 18.03	91.67 ± 2.89	96.67 ± 2.89	56.00 ± 5.48	58.00 ± 4.47	43.26 ± 9.62	51.02 ± 4.18
BirdChicken	68.00 ± 14.83	69.00 ± 22.19	85.00 ± 11.73	89.00 ± 5.48	70.63 ± 12.94	72.61 ± 9.62	86.00 ± 8.94	88.00 ± 5.70
CBF	72.48 ± 4.85	85.91 ± 4.23	93.76 ± 3.90	97.71 ± 0.69	62.60 ± 4.62	71.61 ± 11.07	87.40 ± 4.28	90.04 ± 1.83
Car	55.67 ± 8.94	63.33 ± 7.45	69.67 ± 3.98	76.33 ± 5.45	43.33 ± 12.86	53.00 ± 7.21	60.33 ± 8.50	63.30 ± 3.80
Chinatown	93.59 ± 3.92	91.84 ± 4.85	93.46 ± 10.56	94.56 ± 3.09	49.64 ± 10.98	72.36 ± 12.11	78.02 ± 7.25	67.53 ± 3.06
ChlorineConcentration	32.61 ± 3.30	32.47 ± 3.03	39.58 ± 3.20	34.77 ± 6.93	26.41 ± 3.06	34.99 ± 5.96	28.12 ± 1.89	36.44 ± 3.56
CinCECGTorso	48.74 ± 6.47	62.95 ± 6.66	82.56 ± 7.22	94.71 ± 4.65	26.12 ± 5.37	35.79 ± 7.99	37.54 ± 6.84	49.80 ± 4.72
Coffee	75.71 ± 13.69	88.57 ± 4.66	94.29 ± 1.96	95.71 ± 2.99	68.57 ± 11.68	79.14 ± 15.15	72.86 ± 20.45	84.63 ± 9.65
Computers	56.40 ± 5.77	57.73 ± 17.44	62.67 ± 4.74	65.60 ± 6.66	49.94 ± 7.38	58.96 ± 7.32	54.83 ± 7.81	64.00 ± 4.03
CricketX	26.26 ± 5.29	35.98 ± 3.13	49.06 ± 3.21	66.92 ± 2.71	22.54 ± 5.37	31.90 ± 2.09	38.97 ± 3.38	44.63 ± 2.29
CricketY	23.33 ± 5.19	30.43 ± 5.23	49.66 ± 1.41	57.95 ± 1.33	25.13 ± 3.56	33.28 ± 2.21	31.57 ± 1.65	37.21 ± 1.59
CricketZ	29.28 ± 4.82	33.33 ± 3.02	53.68 ± 2.06	65.30 ± 1.16	28.36 ± 1.29	22.47 ± 4.90	42.87 ± 1.47	39.57 ± 1.96
Crop	36.48 ± 3.38	43.94 ± 2.11	52.18 ± 2.24	57.97 ± 1.25	24.11 ± 1.75	27.36 ± 1.59	33.24 ± 2.17	40.94 ± 1.89
DiatomSizeReduction	85.82 ± 8.55	90.39 ± 5.62	93.66 ± 1.97	92.61 ± 2.00	80.97 ± 8.56	79.48 ± 7.19	77.78 ± 3.00	75.23 ± 1.32
DistalPhalanxOutlineAgeGroup	42.01 ± 16.44	67.77 ± 3.39	69.35 ± 1.20	70.65 ± 2.18	40.07 ± 18.16	70.22 ± 4.60	70.65 ± 1.56	68.22 ± 1.79
DistalPhalanxOutlineCorrect	60.58 ± 2.98	55.80 ± 14.95	63.77 ± 11.63	60.51 ± 11.85	60.58 ± 6.43	43.93 ± 12.26	64.49 ± 7.76	51.57 ± 13.65
DistalPhalanxTW	54.53 ± 1.72	51.37 ± 8.97	56.40 ± 4.87	54.10 ± 3.11	51.51 ± 3.83	42.41 ± 13.90	52.23 ± 3.58	55.11 ± 1.40
DodgerLoopDay	34.50 ± 4.38	35.50 ± 8.78	45.75 ± 3.26	48.75 ± 1.77	35.25 ± 4.79	30.50 ± 9.91	34.54 ± 2.98	42.00 ± 3.89
DodgerLoopGame	56.96 ± 11.69	61.01 ± 13.03	72.61 ± 10.25	83.62 ± 0.65	50.29 ± 8.64	50.89 ± 10.24	56.19 ± 4.46	73.19 ± 1.36
DodgerLoopWeekend	85.36 ± 4.30	84.64 ± 2.73	93.62 ± 2.37	97.83 ± 0.00	87.68 ± 3.80	81.22 ± 2.83	85.03 ± 2.14	92.32 ± 2.38
ECG200	60.60 ± 6.91	73.00 ± 3.46	67.40 ± 8.91	77.40 ± 4.88	55.12 ± 12.30	67.80 ± 5.22	60.91 ± 2.97	77.60 ± 2.70
ECG5000	67.18 ± 10.57	67.68 ± 9.76	73.89 ± 8.21	73.78 ± 6.18	59.38 ± 6.12	57.59 ± 16.45	70.09 ± 6.73	81.51 ± 1.30
ECGFiveDays	61.77 ± 6.61	69.12 ± 9.82	70.96 ± 3.91	75.28 ± 1.71	56.83 ± 8.58	55.49 ± 14.64	68.63 ± 5.73	69.31 ± 6.09
EOGHorizontalSignal	29.06 ± 4.16	37.35 ± 4.07	46.35 ± 2.27	49.17 ± 1.62	22.53 ± 3.83	35.36 ± 2.46	36.71 ± 2.23	46.13 ± 2.16
EOGVerticalSignal	27.35 ± 1.90	31.82 ± 3.47	36.74 ± 2.24	42.21 ± 2.26	23.82 ± 4.88	30.29 ± 2.15	35.64 ± 2.16	40.42 ± 2.51
Earthquakes	55.54 ± 11.57	55.25 ± 14.33	54.68 ± 9.31	54.68 ± 7.63	51.25 ± 10.58	56.83 ± 12.51	39.18 ± 13.02	49.50 ± 9.78
ElectricDevices	33.14 ± 10.96	47.14 ± 3.99	47.06 ± 4.18	49.42 ± 5.02	29.52 ± 15.31	46.51 ± 2.37	38.78 ± 5.04	46.90 ± 1.68
EthanolLevel	26.68 ± 1.86	28.00 ± 3.11	25.92 ± 1.40	28.24 ± 0.50	25.40 ± 10.20	27.07 ± 2.35	24.23 ± 1.59	22.78 ± 1.87
FaceAll	32.84 ± 6.86	39.09 ± 1.28	54.54 ± 1.26	63.01 ± 1.34	22.95 ± 5.55	28.36 ± 2.34	39.13 ± 1.42	45.90 ± 2.02
FaceFour	54.77 ± 8.25	68.55 ± 4.91	75.68 ± 4.37	79.09 ± 1.90	47.00 ± 4.73	59.09 ± 9.02	64.19 ± 1.74	70.83 ± 2.37
FacesUCR	50.37 ± 2.98	55.08 ± 2.32	68.58 ± 2.57	79.08 ± 1.41	38.49 ± 2.57	38.70 ± 1.46	53.71 ± 1.53	57.12 ± 2.59
FiftyWords	36.48 ± 2.66	47.30 ± 2.89	61.58 ± 2.55	71.34 ± 2.10	34.07 ± 2.53	28.74 ± 1.89	41.35 ± 3.11	54.65 ± 2.54
Fish	34.74 ± 4.91	49.91 ± 10.25	64.57 ± 5.92	78.29 ± 3.92	28.02 ± 1.84	48.95 ± 8.27	47.60 ± 3.15	57.71 ± 3.96
FordA	50.71 ± 0.98	53.35 ± 2.84	54.00 ± 2.22	57.15 ± 1.85	50.58 ± 1.08	48.20 ± 2.46	52.30 ± 1.37	47.08 ± 2.81
FordB	49.75 ± 2.56	50.42 ± 2.72	52.32 ± 2.78	55.60 ± 1.91	49.09 ± 2.24	46.11 ± 1.63	51.15 ± 1.17	52.30 ± 2.63

Table 23: Per-dataset LLM-backbone ablation on all 128 UCR datasets (continued).

Dataset	TimeMorph				w/o LLM Backbone			
	1-shot	2-shot	5-shot	10-shot	1-shot	2-shot	5-shot	10-shot
FreezerRegularTrain	61.90 ± 22.25	79.14 ± 7.94	85.11 ± 3.83	89.20 ± 5.65	64.18 ± 29.51	60.75 ± 22.33	66.06 ± 4.26	71.86 ± 2.07
FreezerSmallTrain	66.53 ± 10.64	64.30 ± 13.28	75.05 ± 3.10	82.86 ± 3.49	70.05 ± 6.00	68.60 ± 6.15	70.24 ± 4.62	72.81 ± 5.88
Fungi	79.35 ± 5.30	81.97 ± 3.24	81.18 ± 3.16	79.57 ± 1.20	80.29 ± 2.96	75.78 ± 1.73	70.51 ± 1.42	76.88 ± 5.18
GestureMidAirD1	32.00 ± 3.94	44.92 ± 6.31	60.31 ± 2.35	67.08 ± 1.00	24.60 ± 2.91	33.59 ± 1.60	37.55 ± 3.20	45.38 ± 4.07
GestureMidAirD2	33.85 ± 3.03	43.85 ± 4.03	54.62 ± 1.96	55.69 ± 1.85	34.62 ± 2.04	31.06 ± 4.07	36.00 ± 4.24	45.90 ± 1.94
GestureMidAirD3	18.46 ± 2.61	21.08 ± 1.85	32.15 ± 3.33	33.69 ± 2.63	10.43 ± 4.50	20.26 ± 1.18	16.34 ± 1.60	21.54 ± 1.54
GesturePebbleZ1	35.81 ± 4.48	37.91 ± 6.35	63.84 ± 8.99	75.12 ± 5.19	38.95 ± 6.29	35.56 ± 2.93	52.31 ± 6.43	62.21 ± 2.53
GesturePebbleZ2	33.29 ± 5.85	41.39 ± 4.24	60.89 ± 8.06	75.57 ± 4.64	31.84 ± 9.52	42.41 ± 4.15	54.01 ± 5.38	63.63 ± 1.59
GunPoint	58.00 ± 7.79	79.33 ± 16.25	90.13 ± 10.49	97.47 ± 2.23	53.73 ± 4.44	56.07 ± 1.33	78.44 ± 10.42	86.00 ± 1.76
GunPointAgeSpan	60.51 ± 10.32	64.49 ± 7.90	81.52 ± 9.45	92.72 ± 5.80	59.81 ± 3.85	53.38 ± 2.69	69.47 ± 5.35	69.61 ± 7.46
GunPointMaleVersusFemale	81.01 ± 10.12	87.66 ± 5.66	94.62 ± 6.24	98.54 ± 0.94	86.71 ± 15.43	84.36 ± 8.65	96.31 ± 7.99	93.86 ± 4.95
GunPointOldVersusYoung	57.97 ± 10.78	70.92 ± 5.05	80.44 ± 8.63	86.79 ± 4.71	59.58 ± 4.78	59.30 ± 6.18	68.57 ± 16.69	79.86 ± 4.31
Ham	52.76 ± 9.30	50.86 ± 8.07	60.57 ± 9.91	62.86 ± 3.37	46.79 ± 8.46	55.56 ± 11.12	50.97 ± 8.21	59.68 ± 4.50
HandOutlines	61.73 ± 6.31	64.59 ± 10.14	68.54 ± 6.21	71.51 ± 5.35	54.54 ± 3.64	56.13 ± 17.33	65.68 ± 1.62	64.64 ± 6.36
Haptics	25.65 ± 4.99	26.62 ± 4.64	26.95 ± 4.88	39.22 ± 4.29	22.51 ± 3.93	20.82 ± 4.72	25.61 ± 4.50	30.41 ± 5.05
Herring	50.00 ± 10.00	55.00 ± 4.34	45.00 ± 4.87	55.31 ± 5.01	48.44 ± 10.94	52.08 ± 11.93	38.36 ± 6.31	47.81 ± 5.02
HouseTwenty	62.18 ± 8.04	71.93 ± 10.30	84.37 ± 7.51	89.24 ± 2.81	59.67 ± 2.95	55.06 ± 5.25	71.71 ± 6.19	82.91 ± 3.40
InlineSkate	17.13 ± 2.24	18.98 ± 2.54	28.07 ± 1.19	33.49 ± 2.77	10.94 ± 2.73	18.67 ± 1.77	20.69 ± 1.77	24.79 ± 1.21
InsectEPGRegularTrain	55.34 ± 14.47	68.76 ± 9.60	80.24 ± 3.80	86.59 ± 4.06	47.68 ± 8.08	55.50 ± 7.77	74.16 ± 3.42	78.58 ± 3.07
InsectEPGSmallTrain	50.76 ± 7.86	67.79 ± 2.45	83.05 ± 2.80	85.06 ± 2.08	47.39 ± 13.33	54.47 ± 9.84	71.04 ± 3.82	79.25 ± 1.83
InsectWingbeatSound	25.27 ± 2.20	31.49 ± 3.72	41.44 ± 2.25	51.36 ± 2.05	24.23 ± 2.98	28.79 ± 12.26	27.48 ± 3.11	32.75 ± 1.72
ItalyPowerDemand	58.52 ± 15.27	65.03 ± 13.70	84.26 ± 5.94	92.19 ± 2.62	74.70 ± 22.36	70.00 ± 20.81	82.59 ± 2.87	85.77 ± 2.91
LargeKitchenAppliances	45.44 ± 5.60	42.56 ± 5.00	50.13 ± 3.07	55.09 ± 5.30	45.33 ± 5.47	37.16 ± 2.59	47.94 ± 10.07	51.73 ± 2.91
Lightning2	59.67 ± 8.87	63.28 ± 8.25	69.51 ± 10.47	77.05 ± 4.18	45.81 ± 2.50	51.35 ± 9.46	70.49 ± 5.91	70.49 ± 2.84
Lightning7	41.64 ± 7.60	46.03 ± 6.17	66.30 ± 3.83	66.85 ± 7.34	45.66 ± 5.19	51.13 ± 13.07	60.45 ± 4.81	69.41 ± 1.58
Mallat	68.24 ± 7.54	80.67 ± 4.06	90.88 ± 1.15	89.72 ± 1.50	55.80 ± 1.73	68.88 ± 4.23	62.67 ± 1.52	56.53 ± 2.14
Meat	55.67 ± 9.62	57.67 ± 17.30	64.00 ± 11.82	64.00 ± 6.08	38.44 ± 10.72	65.56 ± 9.62	54.06 ± 1.69	60.00 ± 11.67
MedicalImages	27.66 ± 4.41	31.63 ± 5.59	42.03 ± 2.75	54.68 ± 2.10	27.63 ± 4.85	22.27 ± 5.68	31.38 ± 2.14	44.21 ± 8.13
MelbournePedestrian	39.16 ± 5.22	49.78 ± 4.68	65.81 ± 3.77	70.88 ± 1.49	44.25 ± 10.53	39.50 ± 2.46	56.47 ± 3.42	56.95 ± 3.78
MiddlePhalanxOutlineAgeGroup	39.48 ± 7.40	42.99 ± 12.95	49.74 ± 4.72	51.04 ± 7.11	38.31 ± 10.31	37.01 ± 6.87	49.02 ± 9.83	55.01 ± 3.10
MiddlePhalanxOutlineCorrect	51.68 ± 5.00	48.87 ± 3.56	48.45 ± 7.93	59.18 ± 5.65	52.58 ± 9.24	37.18 ± 8.48	42.13 ± 13.83	59.91 ± 2.41
MiddlePhalanxTW	37.14 ± 16.84	44.81 ± 7.78	43.64 ± 5.71	44.42 ± 6.27	31.16 ± 9.77	45.89 ± 7.24	38.14 ± 10.50	47.19 ± 2.62
MixedShapesRegularTrain	56.06 ± 5.34	67.98 ± 4.06	76.68 ± 5.11	79.60 ± 1.74	48.40 ± 5.50	64.48 ± 7.28	63.75 ± 1.60	68.34 ± 2.44
MixedShapesSmallTrain	63.30 ± 3.34	68.35 ± 3.26	78.71 ± 1.70	81.61 ± 2.47	59.07 ± 4.89	56.71 ± 4.10	71.13 ± 2.34	70.18 ± 1.20
MoteStrain	54.36 ± 12.77	58.56 ± 15.12	80.37 ± 4.50	89.23 ± 1.53	57.45 ± 18.70	56.23 ± 15.93	76.82 ± 2.70	82.96 ± 2.06
NonInvasiveFetalECGThorax1	39.16 ± 2.28	50.58 ± 2.13	67.42 ± 0.56	76.84 ± 1.40	19.86 ± 11.16	36.49 ± 5.57	41.83 ± 2.75	51.30 ± 2.44
NonInvasiveFetalECGThorax2	47.48 ± 2.06	59.72 ± 2.76	75.31 ± 0.82	82.53 ± 1.33	34.34 ± 18.53	41.64 ± 4.16	55.73 ± 4.11	66.77 ± 2.82
OSULeaf	29.83 ± 5.82	36.28 ± 6.97	49.92 ± 6.46	59.42 ± 3.58	29.06 ± 5.33	27.82 ± 3.53	40.22 ± 5.80	45.14 ± 4.16
OliveOil	28.00 ± 14.26	23.33 ± 9.72	28.67 ± 11.69	40.00 ± 0.00	28.81 ± 8.82	33.33 ± 11.55	28.89 ± 11.71	32.18 ± 3.12
PLAID	32.10 ± 5.72	40.19 ± 5.06	52.96 ± 5.39	59.89 ± 3.28	30.60 ± 5.25	40.04 ± 6.05	44.30 ± 2.95	50.48 ± 2.42
PhalangesOutlinesCorrect	51.93 ± 9.54	50.84 ± 2.93	53.61 ± 8.69	55.73 ± 8.07	49.64 ± 6.76	49.30 ± 5.83	52.45 ± 7.84	52.41 ± 12.20
Phoneme	6.87 ± 1.15	10.02 ± 0.59	15.63 ± 1.56	21.95 ± 1.27	7.12 ± 7.01	10.50 ± 5.15	12.54 ± 2.38	13.83 ± 4.92
PickupGestureWiimoteZ	47.20 ± 7.29	58.80 ± 3.03	68.00 ± 2.45	70.40 ± 2.61	36.79 ± 7.02	54.30 ± 1.15	61.33 ± 2.31	63.33 ± 4.62
PigAirwayPressure	5.48 ± 2.19	9.81 ± 0.43	9.81 ± 0.73	9.62 ± 0.48	5.77 ± 5.21	1.98 ± 6.09	2.13 ± 3.02	4.81 ± 4.75

Table 23: Per-dataset LLM-backbone ablation on all 128 UCR datasets (continued).

Dataset	TimeMorph				w/o LLM Backbone			
	1-shot	2-shot	5-shot	10-shot	1-shot	2-shot	5-shot	10-shot
PigArtPressure	15.38 ± 4.07	30.67 ± 1.38	33.08 ± 1.33	31.73 ± 1.92	10.84 ± 6.22	17.77 ± 6.36	23.56 ± 6.58	24.52 ± 4.99
PigCVP	11.44 ± 2.08	19.33 ± 0.99	19.52 ± 0.55	19.52 ± 1.34	2.87 ± 6.86	16.83 ± 4.61	8.34 ± 5.63	18.27 ± 4.70
Plane	93.90 ± 6.92	99.24 ± 0.80	99.24 ± 1.24	99.24 ± 1.24	86.43 ± 6.38	90.48 ± 7.48	95.35 ± 4.00	99.05 ± 4.73
PowerCons	60.89 ± 11.09	66.78 ± 15.52	76.11 ± 6.55	82.33 ± 5.46	74.54 ± 5.79	48.89 ± 2.76	48.89 ± 2.53	78.33 ± 5.30
ProximalPhalanxOutlineAgeGroup	48.98 ± 23.27	66.24 ± 17.89	76.88 ± 5.79	79.80 ± 3.30	78.54 ± 5.81	77.14 ± 1.53	83.90 ± 5.10	71.55 ± 5.03
ProximalPhalanxOutlineCorrect	52.23 ± 12.03	59.93 ± 12.82	65.84 ± 4.70	60.07 ± 4.20	53.30 ± 6.90	64.26 ± 6.47	62.89 ± 6.88	56.41 ± 4.70
ProximalPhalanxTW	55.22 ± 6.63	57.76 ± 7.59	66.24 ± 3.19	69.66 ± 3.47	60.98 ± 6.42	32.61 ± 2.52	66.56 ± 2.70	45.37 ± 5.25
RefrigerationDevices	29.76 ± 11.02	33.39 ± 7.13	40.96 ± 5.59	41.23 ± 4.90	31.20 ± 6.47	22.70 ± 9.56	41.60 ± 4.41	35.99 ± 5.28
Rock	39.20 ± 13.75	64.00 ± 7.07	76.00 ± 3.16	76.40 ± 2.97	15.38 ± 5.99	66.00 ± 1.33	56.00 ± 2.49	59.26 ± 5.03
ScreenType	32.53 ± 4.84	35.63 ± 4.40	34.51 ± 4.14	37.12 ± 3.93	33.60 ± 5.59	33.60 ± 10.13	29.46 ± 6.08	27.02 ± 4.70
SemgHandGenderCh2	51.90 ± 14.92	59.73 ± 10.87	66.80 ± 7.25	63.80 ± 10.73	42.00 ± 5.38	62.60 ± 6.87	60.33 ± 6.50	65.10 ± 4.64
SemgHandMovementCh2	28.13 ± 4.95	31.16 ± 4.70	36.31 ± 5.92	45.56 ± 2.93	24.23 ± 6.12	26.00 ± 6.35	32.00 ± 5.84	33.02 ± 4.67
SemgHandSubjectCh2	37.42 ± 8.82	43.20 ± 6.72	54.31 ± 3.37	66.71 ± 4.45	34.53 ± 5.61	44.22 ± 3.78	37.33 ± 2.46	50.07 ± 4.87
ShakeGestureWiimoteZ	60.40 ± 7.92	81.60 ± 2.61	92.00 ± 0.00	91.60 ± 2.97	65.41 ± 5.86	67.10 ± 9.20	82.00 ± 4.14	84.00 ± 5.13
ShapeletSim	62.44 ± 7.44	67.44 ± 10.92	92.00 ± 3.88	96.67 ± 1.96	52.78 ± 6.14	54.20 ± 3.69	86.67 ± 4.69	89.91 ± 4.88
ShapesAll	50.20 ± 3.03	63.03 ± 2.22	76.20 ± 1.83	83.13 ± 0.32	39.56 ± 5.66	52.67 ± 4.73	64.00 ± 3.93	64.79 ± 5.11
SmallKitchenAppliances	42.88 ± 15.27	55.68 ± 8.74	51.63 ± 12.25	57.60 ± 5.56	51.68 ± 6.21	56.00 ± 7.17	45.07 ± 3.10	50.83 ± 5.17
SmoothSubspace	68.53 ± 8.95	85.07 ± 5.20	88.00 ± 2.87	89.47 ± 2.60	58.67 ± 6.21	70.07 ± 9.78	68.82 ± 6.33	87.33 ± 5.00
SonyAIBORobotSurface1	57.20 ± 8.62	71.85 ± 8.28	84.86 ± 2.67	86.09 ± 3.97	60.07 ± 6.82	69.72 ± 7.23	63.35 ± 6.13	78.04 ± 4.96
SonyAIBORobotSurface2	66.09 ± 10.78	68.21 ± 5.28	77.21 ± 3.67	84.11 ± 2.29	62.12 ± 6.33	61.26 ± 3.67	67.66 ± 3.79	75.87 ± 4.74
StarLightCurves	63.03 ± 14.23	74.47 ± 15.08	81.86 ± 7.93	85.26 ± 3.54	57.02 ± 5.60	77.25 ± 4.74	78.64 ± 3.60	82.39 ± 4.84
Strawberry	51.46 ± 5.21	59.41 ± 6.38	58.05 ± 7.04	75.84 ± 0.99	49.59 ± 6.56	50.26 ± 3.79	52.97 ± 3.13	59.73 ± 5.33
SwedishLeaf	57.98 ± 2.72	64.99 ± 3.83	79.78 ± 2.56	85.89 ± 1.27	52.96 ± 5.38	49.52 ± 7.99	72.64 ± 5.75	67.14 ± 4.64
Symbols	80.94 ± 7.54	82.69 ± 3.43	88.04 ± 1.30	89.33 ± 0.82	81.28 ± 6.48	75.18 ± 8.14	91.16 ± 6.71	76.98 ± 4.74
SyntheticControl	84.20 ± 6.54	87.93 ± 3.95	95.07 ± 1.62	97.33 ± 0.85	75.00 ± 5.85	76.53 ± 3.66	90.33 ± 5.01	86.88 ± 5.09
ToeSegmentation1	58.16 ± 3.70	60.26 ± 8.54	70.53 ± 10.39	80.53 ± 2.75	67.11 ± 6.03	67.98 ± 4.47	60.35 ± 4.09	69.71 ± 4.78
ToeSegmentation2	75.23 ± 12.13	76.15 ± 12.44	78.00 ± 6.36	80.31 ± 4.70	69.23 ± 5.52	64.64 ± 3.10	61.40 ± 7.00	70.77 ± 5.33
Trace	99.20 ± 1.79	98.40 ± 3.05	100.00 ± 0.00	100.00 ± 0.00	91.94 ± 6.69	95.02 ± 4.02	100.00 ± 5.50	100.00 ± 4.75
TwoLeadECG	64.88 ± 11.28	74.63 ± 8.09	96.68 ± 5.58	99.68 ± 0.24	62.30 ± 5.15	57.67 ± 10.34	63.90 ± 7.16	79.33 ± 5.36
TwoPatterns	45.74 ± 2.64	78.23 ± 0.41	74.22 ± 2.81	70.06 ± 0.70	38.08 ± 3.18	35.76 ± 4.48	51.19 ± 4.48	62.35 ± 7.74
UMD	45.47 ± 8.73	78.86 ± 14.11	95.78 ± 2.33	98.67 ± 7.42	67.92 ± 11.10	74.03 ± 5.05	75.30 ± 6.79	77.01 ± 2.84
UWaveGestureLibraryAll	33.37 ± 0.26	42.54 ± 0.28	48.36 ± 0.30	52.41 ± 1.70	24.30 ± 3.96	31.46 ± 2.66	46.43 ± 3.46	51.65 ± 2.85
UWaveGestureLibraryX	30.33 ± 0.23	42.54 ± 0.28	47.36 ± 0.30	53.50 ± 1.91	29.70 ± 5.14	40.57 ± 4.12	52.13 ± 1.78	50.30 ± 1.99
UWaveGestureLibraryY	40.36 ± 0.25	42.54 ± 0.28	51.36 ± 0.31	52.03 ± 0.97	33.46 ± 3.28	36.52 ± 4.08	37.16 ± 2.40	42.03 ± 2.25
UWaveGestureLibraryZ	47.44 ± 2.61	59.88 ± 2.03	55.54 ± 2.61	60.65 ± 2.84	30.72 ± 3.79	34.57 ± 4.09	51.20 ± 2.23	56.31 ± 2.07
Wafer	46.27 ± 7.11	55.01 ± 15.62	71.65 ± 9.58	82.53 ± 8.58	68.14 ± 4.41	61.14 ± 4.11	86.02 ± 2.12	74.41 ± 2.23
Wine	51.11 ± 7.48	48.89 ± 5.34	51.48 ± 2.03	49.26 ± 2.11	61.11 ± 3.61	54.41 ± 4.09	50.00 ± 1.85	44.98 ± 2.10
WordSynonyms	20.91 ± 2.73	34.92 ± 2.64	47.05 ± 3.87	58.90 ± 2.04	13.64 ± 4.08	15.54 ± 4.11	25.24 ± 2.13	24.01 ± 2.14
Worms	24.68 ± 8.95	38.70 ± 4.63	44.42 ± 6.26	48.57 ± 4.91	42.86 ± 4.63	27.93 ± 4.11	42.86 ± 1.81	37.41 ± 2.24
WormsTwoClass	47.53 ± 10.69	54.03 ± 7.72	54.81 ± 7.59	56.36 ± 3.85	48.22 ± 4.09	45.45 ± 4.11	45.60 ± 2.17	55.84 ± 2.14
Yoga	50.42 ± 4.51	50.64 ± 4.90	51.31 ± 4.34	57.73 ± 2.05	51.50 ± 3.76	50.15 ± 4.11	47.98 ± 2.40	53.63 ± 2.20
Average	48.31	55.55	64.14	69.16	44.89	48.29	54.22	58.43

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